

Matrix-associated extracellular vesicles modulate smooth muscle cell adhesion and directionality by presenting collagen VI

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eLife Assessment

This paper explores the role of extracellular vesicles in providing extracellular matrix signals for migration of vascular smooth muscle cells. The evidence, based on cell culture experiments and supporting imaging of human samples, is mostly **convincing**. The paper will be **valuable** for researchers investigating cell migration during vessel repair and atherogenesis.

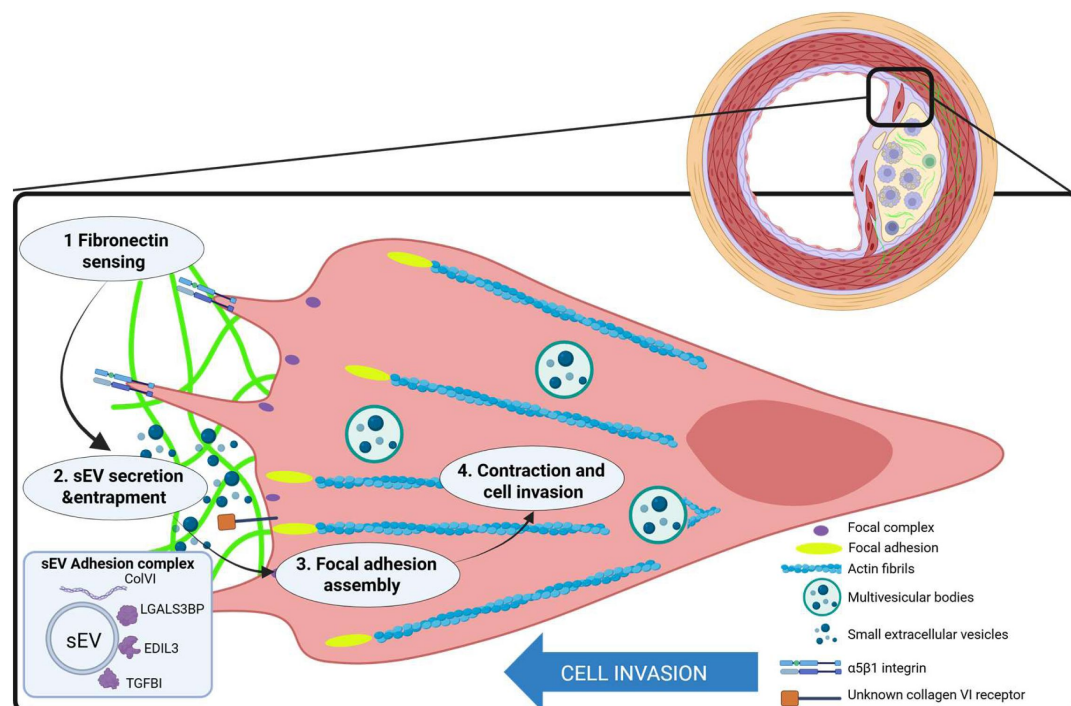
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Abstract

The extracellular matrix (ECM) supports blood vessel architecture and functionality and undergoes active remodelling during vascular repair and atherogenesis. Vascular smooth muscle cells (VSMCs) are essential for vessel repair and, via their secretome, can invade from the vessel media into the intima to mediate ECM remodelling. Accumulation of fibronectin (FN) is a hallmark of early vascular repair and atherosclerosis. Here we show that FN stimulates VSMCs to secrete small extracellular vesicles (sEVs) by activating the $\beta 1$ integrin/FAK/Src pathway as well as Arp2/3-dependent branching of the actin cytoskeleton. We found that sEVs are trapped by the ECM in vitro and colocalise with FN in symptomatic atherosclerotic plaques in vivo. Functionally, ECM-trapped sEVs induced the formation of focal adhesions (FA) with enhanced pulling forces at the cellular periphery preventing cellular spreading and adhesion. Proteomic and GO pathway analysis revealed that VSMC-derived sEVs display a cell adhesion signature and are specifically enriched with collagen VI on the sEV surface. In vitro assays identified collagen VI as playing a key role in cell adhesion and invasion directionality. Taken together our data suggests that the accumulation of FN is a key early event in vessel repair acting to promote secretion of collagen VI enriched sEVs by VSMCs. These sEVs stimulate directional invasion, most likely by triggering peripheral focal adhesion formation and actomyosin contraction to exert sufficient traction force to enable VSMC movement within the complex vascular ECM network.

Figure Abstract

Vascular smooth muscle cells sense fibronectin via $\beta 1$ integrin and secrete small extracellular vesicles loaded with collagen VI. These extracellular vesicles are entrapped in the extracellular matrix and induce formation of peripheral focal adhesions presenting adhesion complex ECM proteins including collagen VI, LGALS3BP, EDIL3 and TGFBI. Focal adhesions anchor the extracellular matrix to actin fibrils in the cell. Contraction of the actin fibrils generates the mechanical force for directional cell invasion through the matrix. Created with [BioRender.com](https://www.biorender.com).



Introduction

The healthy arterial vasculature is dominated by a highly organised medial extracellular matrix (ECM) containing organised layers of contractile vascular smooth muscle cells (VSMCs). Vascular pathologies such as atherosclerosis are associated with dramatic remodelling of the ECM ultimately leading to plaque rupture and myocardial infarction or stroke¹. Progenitor medial VSMCs are essential for vessel repair and must invade through the ECM to form the protective intimal fibrous cap in the plaque. This process of VSMC invasion actively contributes to ECM remodelling²⁻⁴. The accumulation of liver-derived fibronectin (FN) in the vasculature is an early biomarker of atherosclerotic plaque formation and conditional FN knockout in the liver blocked VSMC invasion and fibrous cap formation in the ApoE mouse model. This suggests that FN is an essential signal for VSMC recruitment and invasion during vascular repair, yet its exact role remains unknown⁵⁻⁷.

FN has been shown to play key signalling roles by modulating cellular spreading, adhesion, invasion, differentiation and viability during both developmental and pathological processes⁸. Cell adhesion to FN is primarily mediated by $\alpha 5 \beta 1$ integrins and proteoglycans acting together to activate small GTP-binding proteins, Cdc42, Rac1, and Rho which in turn induce branched actin cytoskeletal rearrangements to form cellular membrane protrusions, filopodia and lamellipodia, which are attached to the ECM via transient peripheral focal complexes⁸⁻¹¹. In turn, maturation of focal complexes into focal adhesions (FAs) anchors cytoplasmic actin stress fibers to the ECM and actin polymerisation and actomyosin-mediated contractility generate the traction forces required for cell body locomotion^{12, 13}.

Exosome-like small extracellular vesicles (sEVs) are novel, cell-matrix crosstalk entities that decorate the ECM and form “migration paths” for tumour cells by enhancing cell adhesion, motility and directionality¹⁴⁻¹⁷. Mechanistically, sEVs are secreted at the cellular leading edge and promote FA formation by presenting fibronectin (FN)^{14, 15, 18, 19}. In addition, sEVs can stimulate motility of immune cells and Dictyostelium discoideum by presenting cytokines

and/or generating chemoattractants^{20–22}. We recently showed that FN is also enriched in VSMC-derived sEVs but the exact role of sEVs in VSMC migration and invasion within the ECM meshwork environment of the vasculature remains unexplored²³.

Here we report that FN in the ECM induces sEV secretion by activating $\beta 1$ integrin/FAK/Src and Arp2/3-dependent actin cytoskeleton assembly. In turn, sEVs are trapped to ECM and these ECM-associated sEVs regulate VSMC invasion directionality in a 3D model. Mechanistically ECM-associated sEVs induced earlier formation of focal adhesions (FA) with enhanced pulling forces at the cell periphery thus switching the leading edge from protrusion activity into contractile mode to enable cell body propulsion. Importantly, we found that the sEV cargo, collagen VI, is indispensable for triggering FA formation and directed cell invasion. We hypothesize that sEV-triggered peripheral FA formation anchors the cellular leading edge and activates cell contraction to exert sufficient force to allow VSMC invasion into the complex ECM fibre meshwork. This novel mechanism opens a unique therapeutic opportunity to target specifically VSMC invasion activity during vascular repair and restenosis.

Results

1. ECM components involved in vessel injury repair stimulate sEV release via $\beta 1$ integrins

FN is detectable in the vasculature as early as the fatty streak stage and preceding major atherogenic alterations such as neointima formation and it is also a novel marker for symptomatic carotid plaques^{5–7}. Given its presence during early stages of vessel injury we hypothesized that FN may modulate sEV secretion to enable vessel repair. FN is secreted as a monomeric protein forming fibrils upon cellular binding so we compared the effects of monomeric FN with the effects of FN immobilised on tissue culture plates to mimic FN fibrils⁸. Plating VSMCs on immobilised FN increased the release of CD63+/CD81+ sEVs 3.5 $\pm$ 0.6 fold whilst addition of soluble FN had no effect on EV secretion as detected by CD63-bead capture assay (**Fig 1A**). Fibrillar collagen I but not a non-fibrillar laminin also stimulated secretion of CD63+/CD81+ sEVs by VSMCs to the same extent as FN (**Fig S1A**).

We next tested if the native VSMC-derived 3D matrix that contains FN and collagen could modulate sEV secretion²⁴. VSMCs were induced to produce ECM, cells were removed and fresh VSMCs plated onto these 3D matrices. VSMCs acquired an elongated shape (**Figs 1B** and **1C**) which is typical for mesenchymal cells in a 3D environment²⁴ and increased secretion of CD63+/CD81+ sEVs compared to VSMCs plated onto the non-coated plastic (**Fig 1D**). We observed no changes in the size distribution of sEVs secreted by VSMCs plated either on plastic or FN matrix as detected using Nanoparticle Tracking Analysis (NTA) (**Fig 1E**). To further characterise the EV populations released western blotting confirmed that sEVs isolated in both conditions were loaded with similar levels of FN and the sEV-specific markers CD63 and Syntenin-1 and lacked the EV-specific marker, α -actinin-4 (**Fig 1F**)²⁵. In addition, treatment of the cells plated onto the matrix with sphingomyelin phosphodiesterase 3 inhibitor (3-O-Methyl-Sphingomyelin, 3-OMS) revealed that CD63+/CD81+ sEV secretion in response to collagen and FN was regulated by sphingomyelin phosphodiesterase 3 (**Fig S1B**) which regulates sEV generation in multivesicular bodies²⁶. Hence, FN triggers secretion of CD63+/CD81+ sEVs, most likely with a late endosomal origin.

Both collagen I and FN are ECM ligands that bind and transduce intracellular signalling via $\beta 1$ integrin. Therefore, we explored the effect of $\beta 1$ integrin activating (12G10) or inhibiting antibodies (4B4) on FN-stimulated secretion of CD63+/CD81+ sEVs. Activation of $\beta 1$ integrin using 12G10 antibody in VSMCs enhanced the effect of FN on sEV secretion (**Fig 1G**). Inhibition of $\beta 1$ integrin using the 4B4 antibody blocked CD63+/CD81+ sEV secretion by VSMCs plated on FN (**Fig**

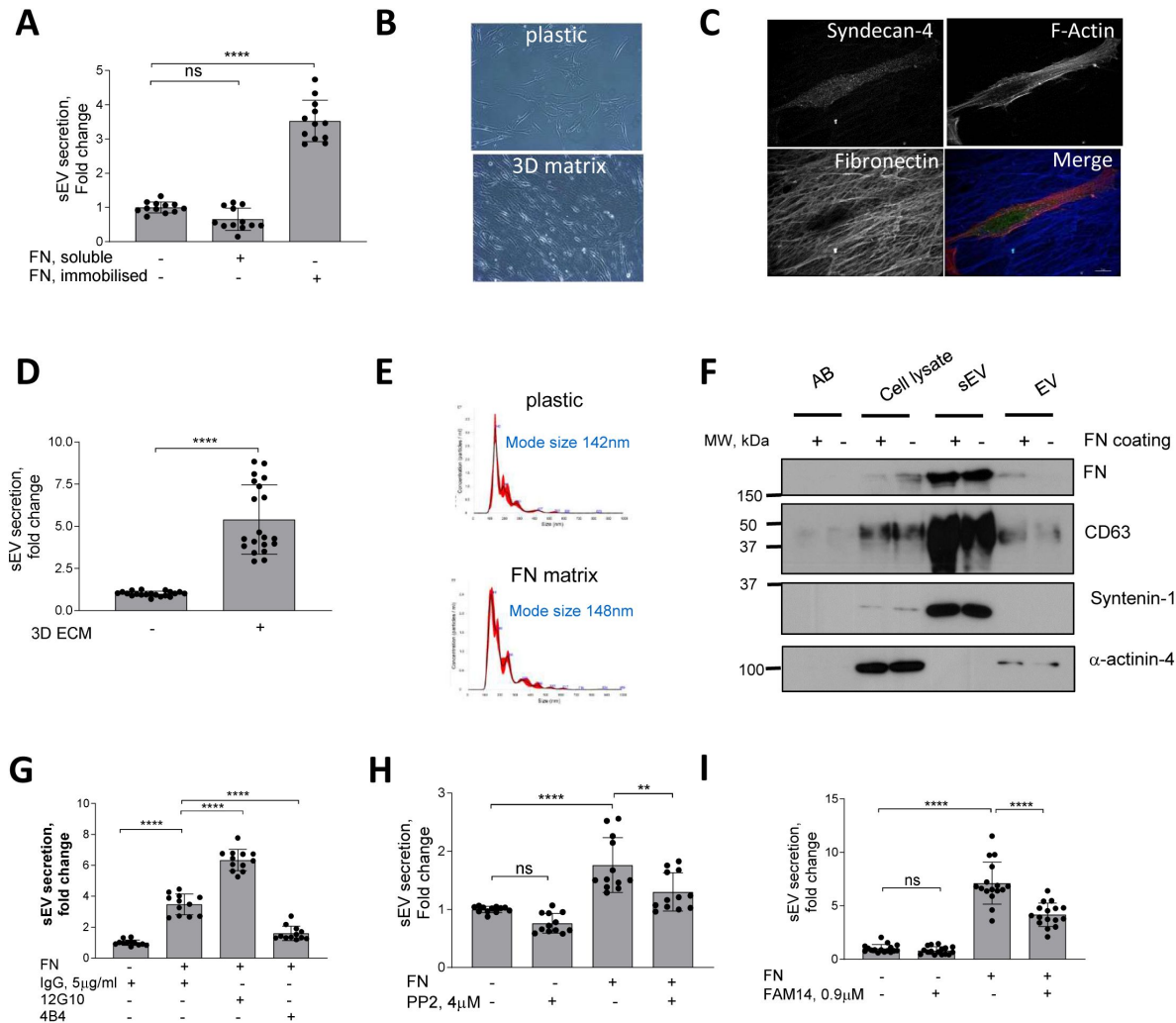


Figure 1.

FN matrix stimulates sEV secretion by VSMCs *in vitro*.

A, Immobilised FN but not a soluble FN promotes sEV secretion. Cells were cultured for 24h and sEVs in conditioned media were measured by CD63-beads assay. N=3 biological replicates, n=4 technical replicates, ANOVA, ****p<0.0001 **B**, Micrograph showing VSMC plated onto plastic or 3D matrix. **C**, VSMCs were plated on the 3D matrix for 24h, fixed and labelled for Syndecan-4 (green), F-actin (phalloidin, red) and fibronectin (blue) Size bar, 10µm. **D**, 3D matrix promote sEV secretion. Cells were cultured for 24h and conditioned media was collected and sEV secretion was measured by CD63-beads assay. N=5, biological replicates, n=4 technical replicates, t-test **E**, FN matrix does not affect sEV mode size. VSMCs were plated on non-coated or FN-coated flasks and incubated for 24h. Isolated sEVs were analysed by Nanoparticle Tracking Analysis. Representative from N=3 biological replicates. **F**, sEV and EV markers distribution is not altered by FN matrix. Cells were plated on non-coated or FN-coated flasks and apoptotic bodies (AB, 1.2K pellet), extracellular vesicles (EV, 10K pellet) and small extracellular vesicles (sEVs, 100K pellet) were isolated by differential ultracentrifugation and analysed by western blotting. Representative image from N=3 biological replicates. **G**, FN induces secretion of sEVs by activating $\beta 1$ integrin. VSMCs were plated on non-coated or FN-coated plates in the absence or presence of integrin activating (12G10) or inhibiting (4B4) antibodies for 24h and conditioned media was analysed by CD63-bead assay. N=3 biological replicates, n=4 technical replicates, ANOVA, ****p<0.0001 **H**, Src is required for the sEV secretion. Cells were plated and sEV secretion was measured as in 2A. N=3, biological replicates, n=4, technical replicates, ANOVA, **p<0.0001 **I**, Inhibition of FAK blocks FN-induced sEV secretion. Cells were plated and sEV secretion was measured as in 1A. N=3 biological replicates, n=4 technical replicates, ANOVA.

1G [↗](#)). Next, we tested the role of the $\beta 1$ integrin downstream signalling mediators, FAK and Src²⁷ [↗](#). Blocking these pathways with small inhibitors (FAM14 and PP2, respectively) reduced sEV secretion by cells plated on FN but not on plastic (**Figs 1H** [↗](#) and **1I** [↗](#)). Taken together these data suggest that FN matrices stimulate secretion of CD63+/CD81+ sEVs via the $\beta 1$ integrin signalling pathway.

2. FN-induced sEV secretion is modulated by Arp2/3 and formin-dependent actin cytoskeleton remodelling

ECM-induced focal FAs initiate dynamic remodelling of the actin cytoskeleton, a process regulated by the Arp2/3 complex and formins, which have recently been implicated in exosome secretion in lymphocytes and tumor cells^{28–30} [↗](#). So next we tested the contribution of Arp2/3 and formins by using the small molecule inhibitors, CK666 and SMIFH2, respectively^{31, 32} [↗](#). CK666 reduced sEV secretion both in VSMCs plated on plastic and FN matrix while inhibition of formins using SMIFH2 was only effective on the FN matrix (**Figs 2A** [↗](#) and **2B** [↗](#)) indicating that formins may be involved in FN-dependent sEV secretion.

As regulators of branched actin assembly, the Arp2/3 complex and cortactin are thought to contribute to sEV secretion in tumour cells by mediating MVB intracellular transport and plasma membrane docking^{28, 33} [↗](#). Therefore, we overexpressed the Arp2/3 subunit, ARPC2-GFP and the F-actin marker, F-tractin-RFP in VSMCs and performed live-cell imaging. As expected, Arp2/3 and F-actin bundles formed a distinct lamellipodia scaffold in the cellular cortex (**Fig S2A** [↗](#) and Supplementary Video S2). Unexpectedly, we also observed numerous Arp2/3/F-actin positive spots moving through the VSMC cytoplasm that resembled previously described endosome actin tails observed in *Xenopus* eggs³³ [↗](#) and parasite infected cells where actin comet tails propel parasites via filopodia to neighbouring cells^{34, 35} [↗](#) (**Fig S2A** [↗](#), arrow, and Supplementary Video S2). Analysis of the intracellular distribution of Arp2/3 and CD63-positive endosomes in VSMCs showed CD63-MVB propulsion by the F-actin tail in live cells (**Fig 2C** [↗](#) and Supplementary Video S4). Although these were rare events, possibly due to the short observational time in a single microscopic plane (**Fig 2C** [↗](#)) we observed numerous F-actin spots in fixed VSMCs that were positive both for F-actin and cortactin indicating that these are branched-actin tails (**Fig 2D** [↗](#)). Moreover, cortactin/F-actin spots colocalised with CD63+ endosomes and addition of the SMPD3 inhibitor, 3-OMS, induced the appearance of enlarged doughnut-like cortactin/F-actin/CD63 complexes resembling invadopodia-like structures similar to those observed in tumour cells (**Fig 2D** [↗](#), arrowheads)¹⁸ [↗](#). To quantify CD63 overlap with the actin tail-like structures, we extracted round-shaped actin structures and calculated the thresholded Manders colocalization coefficient (**Fig S2B** [↗](#)). We observed overlap between F-actin tails and CD63 as well as close proximity of these markers in fixed VSMCs (**Fig S2B** [↗](#)). Approximately 50% of the F-actin tails were associated with ~13% of all endosomes ($tM1=0.44\pm0.23$ and $tM2=0.13\pm0.06$, respectively, $N=3$). Addition of 3-OMS enhanced this overlap further ($tM1=0.75\pm0.18$ and $tM2=0.25\pm0.09$) suggesting that Arp2/3-driven branched F-actin tails are involved in CD63+ MVB intracellular transport in VSMCs.

Formins and the Arp2/3 complex play a crucial role in the formation of filopodia, a cellular protrusion required for sensing the extracellular environment and cell-ECM interactions³⁶ [↗](#). To test whether MVBs can be delivered to filopodia, we stained VSMCs for Myosin-10 (Myo10)³⁷ [↗](#). We observed no difference between total filopodia number per cell on plastic or FN matrices ($n=18\pm 8$ and $n=14\pm 3$, respectively) however the presence of endogenous CD63+ MVBs along the Myo10-positive filopodia were observed in both conditions (**Fig 2E** [↗](#), arrows). Filopodia have been implicated in sEV capture and delivery to endocytosis “hot-spots”³⁸ [↗](#), so next we examined the directionality of CD63+ MVB movement in filopodia by overexpressing Myo10-GFP and CD63-RFP in live VSMCs. Importantly, we observed anterograde MVB transport toward the filopodia tip (**Fig 2F** [↗](#) and Supplementary Video S2) indicative of MVB secretion.

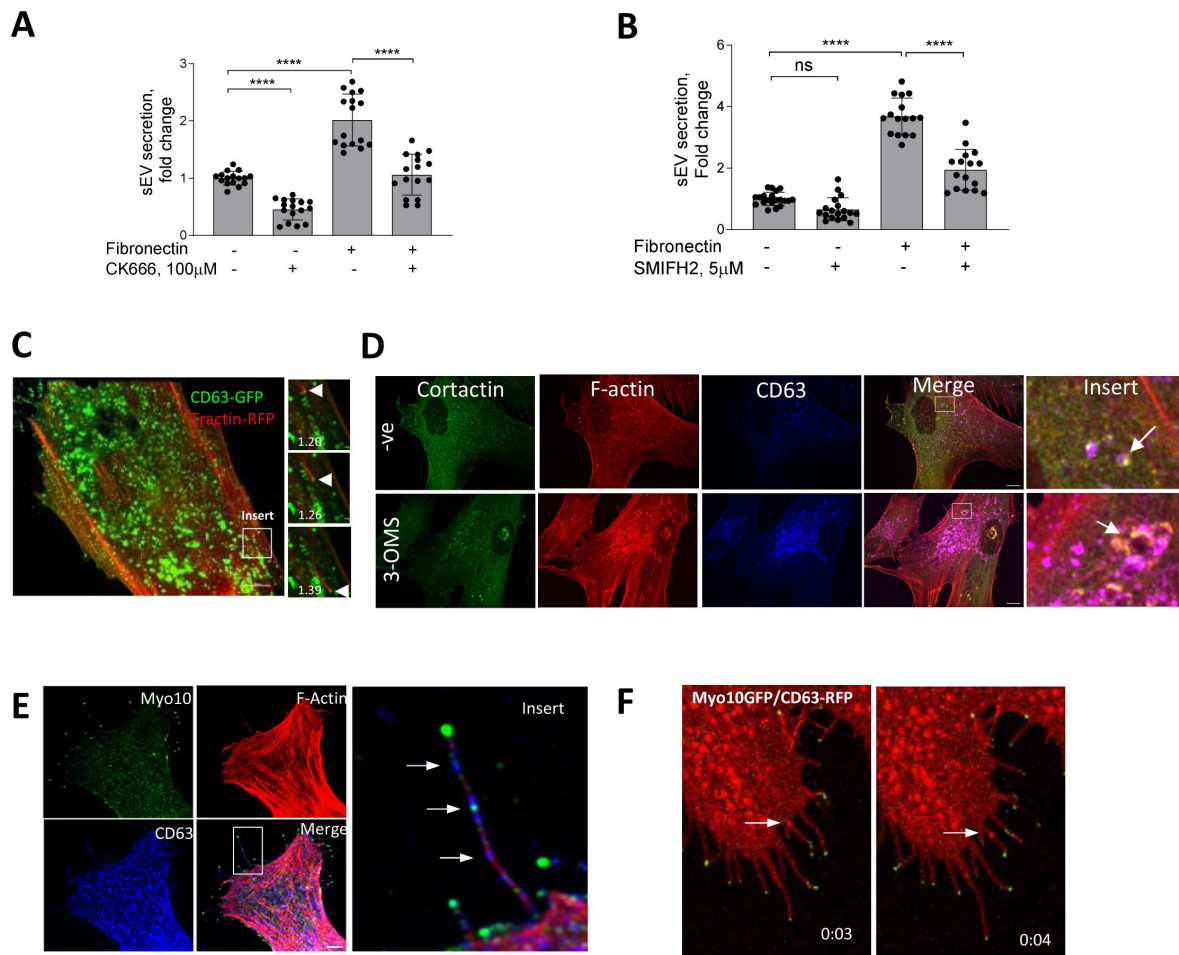


Figure 2.

sEV secretion is regulated by Arp2/3 and formin-dependent actin cytoskeleton remodelling.

A, Arp2/3 inhibition with CK666 reduces sEV secretion in VSMC. Cells were plated onto non-coated or FN-coated plates for 24h and conditioned media was analyzed by CD63-bead assay. N=4 biological replicates, n=4 technical replicates, ANOVA. **B**, Formin inhibitor, SMIFH2 blocking filopodia formation reduces sEV secretion from FN-plated cells only. Cells were plated onto non-coated or FN-coated plates for 24h and conditioned media was analyzed by CD63-bead assay. N=4-5 biological replicates, n=2-4 technical replicates, ANOVA. **C**, Still images of a time-lapse showing that transported MVBs are attached to F-actin tails. VSMCs were co-transfected with CD63-GFP and F-tractin-RFP and time-lapse was captured using confocal spinning-disk microscopy. Arrow head – position of CD63 MVB across time. Time, min. Size bar, 10μm. **D**, VSMC were plated for 24h in the absence (top, -ve) or presence of sEV secretion inhibitor (3-OMS, 10μM). Cells were fixed and stained for cortactin, F-actin and CD63. Note an overlap between CD63 endosomes and branched actin proteins (F-actin and cortactin), which is enhanced by 3-OMS treatment (arrow). Size bar, 10μm. **E**, CD63 MVBs (arrows) are detected in filopodia-like structures. VSMCs were plated on FN-coated plates for 24h and cells were stained for Myo10 (green), CD63 (blue), F-actin (phalloidin, red). Size bar, 10μm. **F**, Still images of a time-lapse showing that MVBs are transported to the filopodia tip in the live VSMC. Cells were co-transfected by CD63-RFP and Myo10-GFP, cultured for 24h and time-lapse video was captured using confocal spinning disk microscopy. Snapshots were taken at T=3s and T=4s after start of the video.

We also attempted to visualise sEV release in filopodia using CD63-pHluorin where fluorescence is only observed upon the fusion of MVBs with the plasma membrane³⁹. Using total internal reflection fluorescence microscopy (TIRF) we observed the typical “burst”-like appearance of sEV secretion at the cell-ECM interface in full agreement with an earlier report showing MVB recruitment to invadopodia-like structures in tumor cells¹⁸ (**Fig S2B** and Supplementary Video S1). Although we also observed an intense CD63-pHluorin staining along filopodia-like structures we were not able to detect typical “burst”-like events to confirm sEV secretion in filopodia. (**Fig S2C** and Supplemental Video S1).

Taken together these data show that sEV secretion by VSMCs is regulated via Arp2/3 and formin-dependent actin cytoskeleton remodelling. Branched actin tails can potentially contribute to MVB intracellular transport and secretion at the VSMC-ECM interface. Interestingly, CD63+ MVBs can be observed in filopodia-like structures suggesting that sEV secretion can also occur spatially via cellular protrusion-like filopodia but more studies are needed to confirm this hypothesis.

3. sEVs are trapped in the ECM in vitro and in atherosclerotic plaque

EV trapping in the ECM is a prominent feature of the blood vessel media and intima and EVs become more abundant in the ECM with ageing and disease^{40, 41}. Therefore, we next set out to determine if sEVs that are secreted toward the ECM can be trapped in the native matrix in vitro. VSMCs were plated on plastic or FN for 24 h and sEVs were visualised by CD63 immunofluorescence. We observed CD63 puncta in proximity to protrusions at the cell periphery (**Fig 3A**). These CD63+ sEVs were observed both on the plastic and FN-coated plates suggesting that sEVs can bind to the ECM secreted by VSMCs over the 24h incubation. In agreement with our previous data we also observed CD63+ sEV trapping in the native ECM produced by VSMCs over a 7 day period and their reduction/absence from the matrix when VSMCs were treated with SMPD3 siRNA (**Fig 3B**)⁴¹.

To quantify ECM-trapped sEVs we applied a modified protocol for the sequential extraction of extracellular proteins using salt buffer (0.5M NaCl) to release sEVs which are loosely-attached to ECM via ionic interactions, followed by 4M guanidine HCl buffer (GuHCl) treatment to solubilize strongly-bound sEVs (**Fig S3A**)⁴². We quantified total sEV and characterised the sEV tetraspanin profile in conditioned media, and the 0.5M NaCl and GuHCl fractions using ExoView. The total particle count showed that EVs are both loosely bound and strongly trapped within the ECM. sEV tetraspanin profiling showed differences between these 3 EV populations. While there was close similarity between the conditioned media and the 0.5M NaCl fraction with high abundance of CD63+/CD81+ sEVs as well as CD63+/CD81+/CD9+ in both fractions (**Fig S3A**). In contrast, the GuHCl fraction was particularly enriched with CD63+ and CD63+/CD81+ sEVs with very low abundance of CD9+ EVs (**Fig S3A**). The abundance of CD63+/CD81+ sEVs was confirmed independently by a CD63+ bead capture assay in the media and loosely bound fractions (**Fig S3B**).

We previously found that the serum protein prothrombin binds to the sEV surface both in the media and MVB lumen showing it is recycled in sEVs and catalyses thrombogenesis being on the sEV surface⁴³. So we investigated whether FN can also be associated with sEV surface where it can be directly involved in sEV-cell cross-talk⁴³. We treated serum-deprived primary human aortic VSMCs with FN-Alexa568 and found that it was endocytosed and subsequently delivered to early and late endosomes together with fetuin A, another abundant serum protein that is a recycled sEV cargo and elevated in plaques (Figs S3C and S3D). CD63 visualisation with a different fluorophore (Alexa488) confirmed FN colocalization with CD63+ MVBs (**Fig S3E**). Next, we stained non-serum deprived VSMC cultured in normal growth media (RPMI supplemented with 20% FBS) with an anti-FN antibody and observed colocalization of CD63 and serum-derived FN. Co-localisation was reduced likely due to competitive bulk protein uptake by non-deprived cells

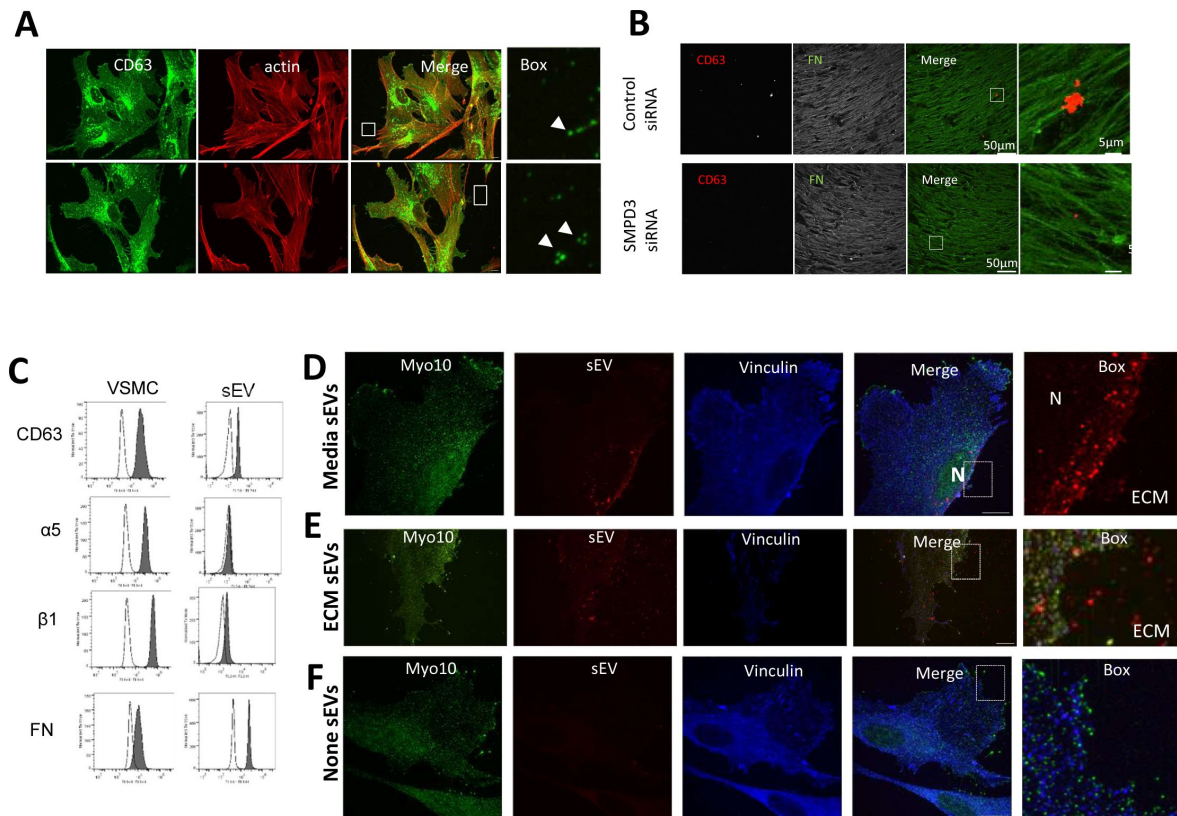


Figure 3.

Endogenous and exogenous sEVs are trapped by ECM *in vitro*.

A, VSMC were plated onto non-coated (top) or FN-coated (bottom) plastic for 24h and stained for the membrane sEV marker, CD63 and F-actin. Note accumulation of CD63 puncta in close proximity to filopodia-like projections. Size bar, 10 μ m. **B**, VSMCs were plated onto gelatin-coated plates and treated with control or SMPD3 siRNA for 72h. 3D matrices were generated and stained for CD63 and FN. Images were acquired using Nikon AX inverted confocal microscope. Note the decrease in CD63-positive sEVs associated with the FN fibrils. **C**, FN is presented on the surface of the VSMC-derived sEV along with $\alpha 5\beta 1$ integrin. VSMC sEVs were immobilised on the 4 μ m beads. sEV-beads and VSMCs were stained with the antibodies (filled graphs) in non-permeabilised conditions and analysed by flow cytometry. **D**, VSMC were plated on FN-coated dishes and Alexa568-labelled sEV were added to the cell media for 3h. Cells were fixed and stained for filopodia marker Myo10 (green) and vinculin (blue). Note perinuclear localisation of internalised sEVs. Size bar, 10 μ m. Representative image from N=3 biological replicates. **E**, VSMC were plated on FN-coated dishes pre-coated with Alexa568-labelled sEV and incubated for 24h. Cell staining as in **Fig 3D**. Note even distribution of sEVs across the matrix and cell area. **F**, VSMC were plated on the FN-coated dishes in the absence of Alexa568-labelled sEV and incubated for 24h. Cell staining as in **Fig 3D**. Note the absence of signal in sEV channel.

(Fig S3F [↗](#)). Notably, when we compared FN distribution in sparsely growing VSMCs versus confluent cells we found that FN intracellular spots, as well as colocalization with CD63, completely disappeared in the confluent state (Fig S3F [↗](#) and S3G [↗](#)).

This correlated with nearly complete loss of CD63+/CD81+ sEV secretion by the confluent cells indicating that confluence abrogates intracellular FN trafficking as well as sEV secretion by VSMCs (Fig S3H [↗](#)). Finally, FN could be co-purified with sEVs from VSMC conditioned media (Fig S3I [↗](#)) and detected on the surface of sEVs by flow cytometry confirming its loading and secretion via sEVs (Fig 3C [↗](#)).

To understand how VSMCs interact with sEVs in the ECM versus in the media we isolated and fluorescently-labelled sEVs and added them either to the ECM or the media and tracked sEV distribution. Super-resolution microscopy (iSIM) revealed that the addition of sEVs to the cell media resulted in fast uptake by VSMCs and delivery to the nuclear periphery (Fig 3D [↗](#) and Fig S3J [↗](#)). This was particularly obvious in 3D projections (Figs S3J). In contrast, purified sEVs added to FN-coated plates resulting in their even distribution across the ECM were not internalised even 24h after cell addition with no perinuclear localisation of ECM-associated sEVs observed (Fig 3E [↗](#) and Fig S3K [↗](#)). No red fluorescence was observed in the absence of added sEVs (Fig 3F [↗](#)).

To spatially map the accumulation of sEV markers in the atherosclerotic plaque in vivo we collected 12 carotid atherosclerotic plaques and adjacent intact vascular segments excised during carotid endarterectomy. Unbiased proteomics analysis confirmed considerable differences between these 2 sample groups, revealing 213 plaque-specific and 111 intact arterial-specific proteins (Fig 4A [↗](#) and Fig 4B [↗](#), Tables S1-S5). Differential expression analysis identified 46 proteins significantly overexpressed (fold change ≥ 2 and FDR-adjusted P value ≤ 0.05) in plaques and 13 proteins significantly upregulated in the intact arterial segments (Fig 4C [↗](#), Table S3). Among the top proteins differentially expressed in plaques were catalytic lipid-associated bioscavenger paraoxonase 1, atherogenic apolipoprotein B-100, HDL-associated apolipoprotein D, iron-associated protein haptoglobin, and inflammation-related matrix metalloproteinase-9 (Table S3). These proteins have previously been implicated in lipid metabolism alterations as well as inflammation in the plaque hence indicating the advanced atherosclerotic plaque signature of the analysed samples. Comparison of the differentially expressed proteins also revealed an accumulation in atherosclerotic regions of fetuin-A (Alpha-2-HS-glycoprotein, P02765, 2.7 fold increase, $p=0.003869659$), an abundant sEV cargo protein which is recycled by VSMCs²³[↗](#). Likewise, the level of another sEV cargo, Apolipoprotein B-100 (P04114) was also significantly elevated (2.8 fold increase, $p=0.000124446$) in atherosclerotic plaque (Table S2)²³[↗](#). Notably, a negative regulator of sEV secretion, Coronin 1C (Q9ULV4) was downregulated in the plaque (0.6 fold decrease, $p=0.000805372$, Table S2) consistent with previous studies that suggested that sEV secretion is upregulated during plaque progression⁴⁰[↗](#), ⁴⁴[↗](#).

To test if FN associates with sEV markers in atherosclerosis, we investigated the spatial association of FN with sEV markers using the sEV-specific marker CD81. Staining of atherosclerotic plaques with haematoxylin and eosin revealed well-defined regions with the neointima as well as tunica media layers formed by phenotypically transitioned or contractile VSMCs, respectively (Fig S4A [↗](#)). Masson's trichrome staining of atherosclerotic plaques showed abundant haemorrhages in the neointima, and sporadic haemorrhages in the tunica media (Fig S4B [↗](#)). Staining of atherosclerotic plaques with orcein indicated weak connective tissue staining in the atheroma with a confluent extracellular lipid core, and strong specific staining at the tunica media containing elastic fibres which correlated well with the intact elastin fibrils in the tunica media (Figs S4C [↗](#) and S4D [↗](#)). Using this clear morphological demarcation, we found that FN accumulated both in the neointima and the tunica media where it was significantly colocalised with the sEV marker, CD81 (Fig. 4D [↗](#), 4E [↗](#) and 4F [↗](#)). Notably CD81 and FN colocalization was particularly prominent in cell-free, matrix-rich plaque regions (Figs. 4E [↗](#) and 4F [↗](#)).

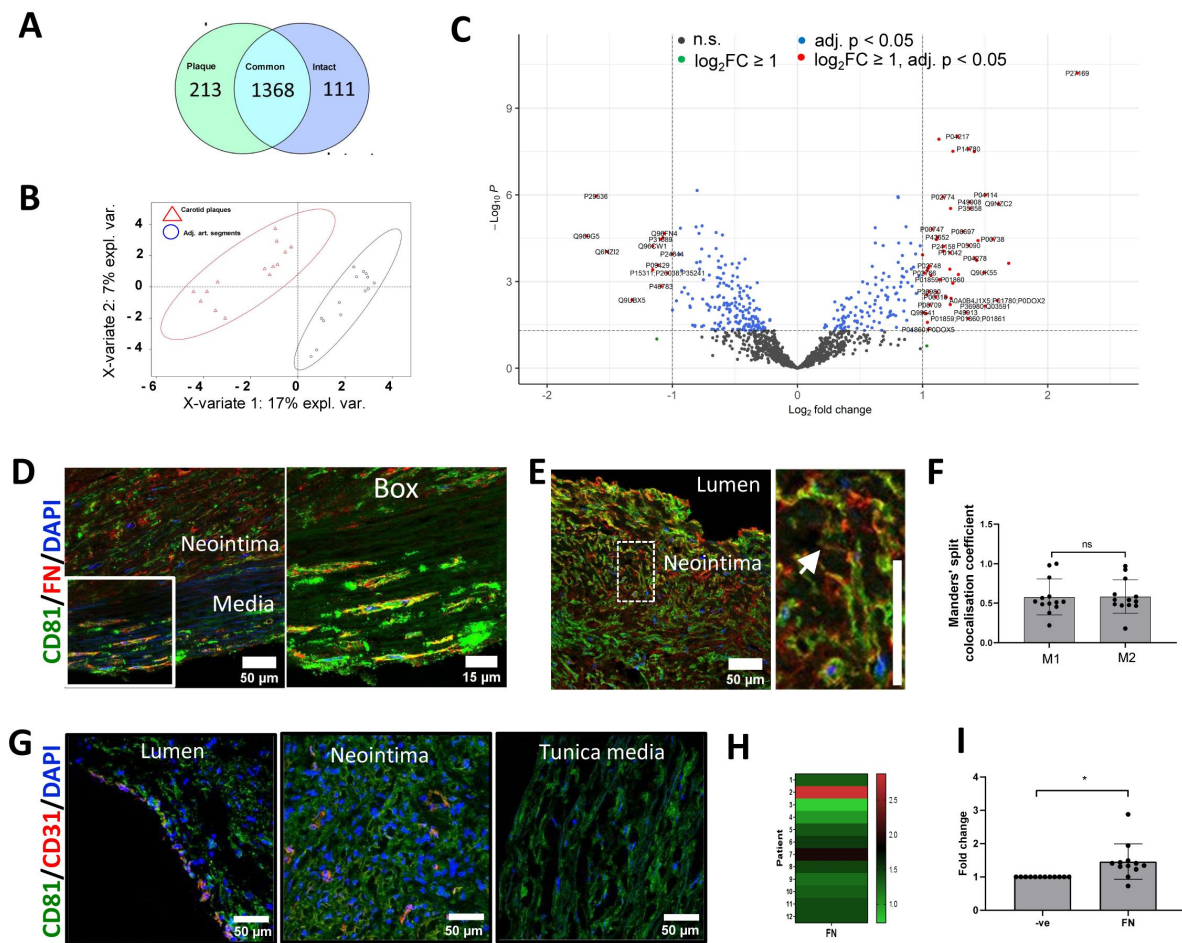


Figure 4.

Fibronectin deposition in the atherosclerotic plaque is spatially associated with sEV markers.

A, Proteomic profiling of pairwise collected carotid atherosclerotic plaques and adjacent intact arterial segments. Venn diagram shows the number of plaque-specific ($n = 213$) and intact-specific ($n = 111$) proteins as well as the number of proteins which are common for both vascular regions (1,368). **B**, Partial least-squares discriminant analysis indicates clear classification pattern between the carotid plaques (indicated by red triangles) and adjacent intact arterial segments (indicated by blue circles). $N = 14$ patients. **C**, Volcano plot illustrates that 46 proteins are significantly overexpressed in plaques whilst 13 proteins are significantly upregulated in adjacent intact arterial segments. **D**, **E**, Atherosclerotic plaques were co-stained for fibronectin (FN) and sEV marker, CD81. Cell nuclei were counterstained with DAPI. Main figure: $\times 200$ magnification, size bar, $50\ \mu\text{m}$. Box: $\times 400$ magnification, size bar, $15\ \mu\text{m}$. Note an accumulation of FN in the neointima. **F**, Spatial distribution of FN and CD81 in the neointima. Note high overlap between FN and CD81 in the extracellular matrix. $\times 200$ magnification, size bar, $50\ \mu\text{m}$. Manders' split colocalization coefficient for the overlap of FN with CD81 (M1) and CD81 with FN (M2). Neointima region as in [Fig. 4E](#). **G**, Co-localisation of endothelial cells (stained for CD31/PECAM1, red colour) and sEVs (stained for CD81, green colour). Nuclei are counterstained with 4',6-diamidino-2-phenylindole (DAPI, blue colour). Magnification: $\times 200$, scale bar: $50\ \mu\text{m}$. Note endothelialised (i.e., those covered with CD31-positive cells) segments at the luminal side and abundant CD81 expression both in the neointima and in the tunica media. Endothelial cells (CD31, red colour) are largely co-localised with CD81-positive extracellular vesicles (green colour). Note CD31-positive capillaries in the neointima. **H**, **I**, Quantification of FN content in atherosclerotic plaques. Samples were analysed by western blot and bands intensity was quantified in ImageJ. Fold change was calculated as ratio of band intensity in the atherosclerotic plaque to band intensity in the adjacent intact arterial segments normalised to GAPDH. Note that FN content is elevated in atherosclerotic plaques relative to the adjacent intact arterial segments. Paired t-test.

Interestingly, colocalization analysis of CD81 and FN distribution showed that there was a trend of higher colocalization levels of CD81 with FN in the neointima as compared to tunica media which was not significant (Thresholded Mander's split colocalisation coefficient for CD81 colocalization with FN 0.58 ± 0.21 and 0.49 ± 0.14 , respectively, $N=13$, $p=0.052$, paired T test). An enhanced expression of CD81 by endothelial cells in early atheroma has been previously reported so to study the contribution of CD81+ sEVs derived from endothelial cells we investigated the localisation of CD31 and CD81⁴⁵. In agreement with a previous study, we found that the majority of CD31 colocalises with CD81 (Thresholded Mander's split colocalization coefficient 0.54 ± 0.11 , $N=6$) indicating that endothelial cells express CD81 (Fig 4G)⁴⁵. However, only a minor fraction of total CD81 colocalised with CD31 (Thresholded Mander's split colocalization coefficient 0.24 ± 0.06 , $N=6$) confirming that the majority of CD81 in the neointima is originating from the most abundant VSMCs. Finally, western blot analysis confirmed FN accumulation in the plaque region (Figs 4H, 4I and 54E). To understand the origin of plaque FN we performed RT-PCR. FN expression could neither be detected in plaques nor in intact vessels, although it was abundant in the liver (data not shown), suggesting that the circulation is a key source of FN in atherosclerotic plaques.

4. sEVs stimulate VSMC directional invasion and induce early peripheral focal adhesion assembly with an enhanced pulling force

FN as a cargo in sEVs promotes FA formation in tumour cells and increases cell speed^{14, 15}. As we found that FN is loaded into VSMC-derived sEVs we hypothesized that ECM-entrapped sEVs can enhance cell migration by increasing cell adhesion and FA formation in the context of a FN-rich ECM. Therefore, we tested the effect of sEV deposition onto the FN matrix on VSMC migration in 2D and 3D models. We found that FN coating promoted VSMC velocity and inhibition of bulk sEV secretion with 3-OMS reduced VSMC speed in a 2D single-cell migration model (Figs. 5A, 5B) in agreement with previous studies using tumour cells^{14, 15}. However, addition of sEVs to the ECM had no effect on VSMC speed at baseline but rescued cell speed and distance in the presence of the sEV secretion inhibitor, 3-OMS suggesting the EVs are not primarily regulating cell speed (Figs 5A and 5B).

To assess the effect of sEVs on cell directionality we exploited a 3D model developed by Sung et al¹⁵ and examined VSMC invasion using a μ -Slide Chemotaxis assay where cells were embedded into FN-enriched 3D Matrigel matrices in the absence or presence of exogenously-added sEVs (Fig. 5C). We developed a script to automatically measure cell invasion parameters (track length, cell speed, straightness, accumulated distance, lateral and vertical displacement and parallel forward motion index (FMI)). The addition of sEVs had no effect on cell speed, accumulated distance or straightness (Figs. 5D, 5E, 5F). However, VSMCs invading in the presence of embedded sEVs migrated more aligned to the FBS gradient compared to cells plated in the absence of sEVs. To test whether this was a feature of all EV populations we also performed the experiment in the presence of larger EVs pelleted at the 10,000xg centrifugation step and observed no effect suggesting this mechanism was specific to sEVs (Figs. 5G and 5H). Hence, ECM-associated sEVs have modest influence on VSMC speed but influence VSMC invasion directionality.

We wondered whether VSMC migration and invasion were affected by the FN cargo in sEVs that was shown previously to promote tumour cell migration by enhancing FA assembly and cellular adhesion to FN¹⁴. Therefore, we measured VSMC adhesion using an adhesion assay where sEVs were added to FN-coated plates and cells were plated onto these matrices for 1h and firmly attached cells counted. Surprisingly we observed a marked reduction of VSMC adhesion to FN in the presence of sEVs (Fig 6A) suggesting that ECM-associated sEV can impair VSMC adhesion. Mesenchymal cell attachment to the matrix is mediated by transient focal complexes present in the cellular protrusion as well as centripetal FAs that link the cellular cytoskeleton to the matrix and mediate the pulling force to move the cell body^{12, 13}. Consistently we observed focal complexes in cellular protrusions as well as more centripetal FAs which were associated with mature actin stress fibres in VSMCs plated onto FN matrix (Fig S5A). Therefore, we used a live

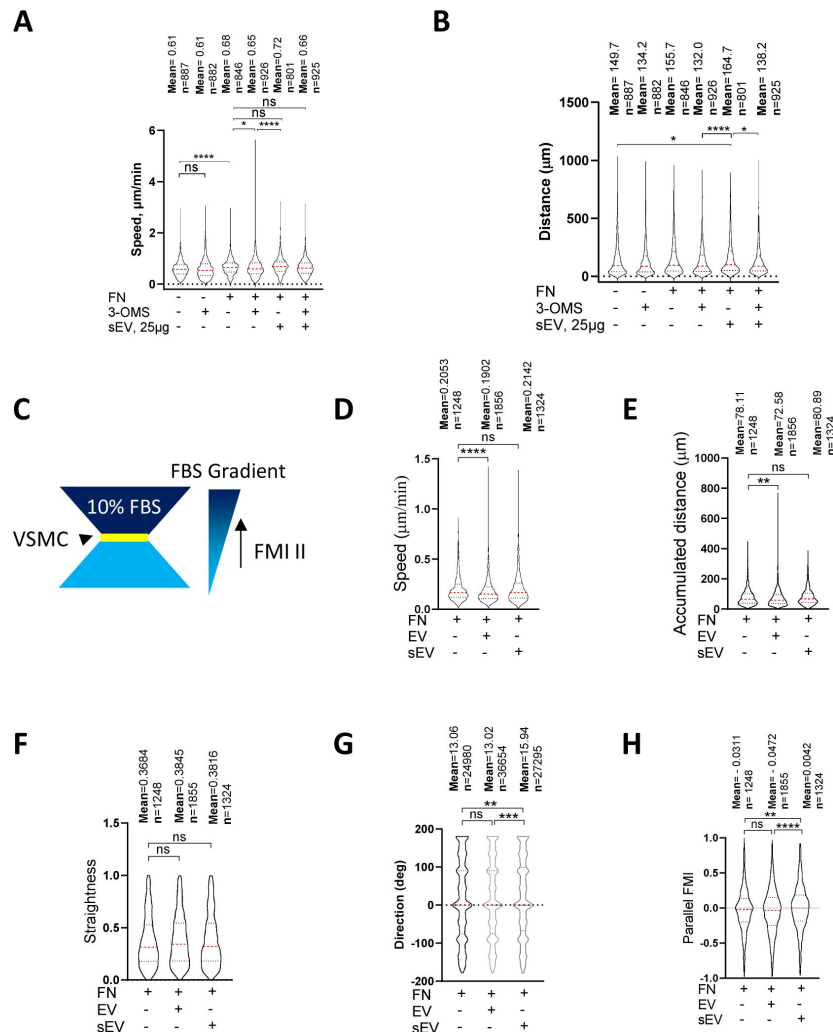


Figure 5.

sEV induces directional VSMC invasion.

A, B, VSMC migration in 2D assay. VSMC were plated onto FN in the absence or presence of SMPD3 inhibitor (3-OMS) and/or sEV (25μg). Cells were tracked for 8h. Kruskal-Wallis with Dunn's multiple comparison test, N=2-3 biological replicates, n indicates the total number of tracked cells per condition, **, p<0.01, ****, p<0.0001. **C**, Chemotaxis μ-slide diagram. Yellow, cell chamber, blue chemoattractant-free medium chamber, dark blue – chemoattractant medium chamber. **D-H**, sEV promote directional VSMC invasion. Cells were seeded to the FN-enriched Matrigel matrix in μ-Slide Chemotaxis assay and stained with DraQ5. Cell tracking was conducted by OperaPhenix microscope for 12h and cell invasion parameters were quantified using Columbus. Kruskal-Wallis with Dunn's multiple comparison test, N=4, biological replicates, n indicates the total number of tracked cells per condition, **, p<0.01, ***, p<0.001, ****, p<0.0001.

cell spreading assay to monitor cellular adhesion over time using ACEA's xCELLigence. As expected, FN alone stimulated VSMC spreading and adhesion (**Figs. 6B** and **6C**). Addition of sEVs did not impact cellular adhesion over the first ~15 min but then the adhesion was stalled in the presence of sEVs (**Fig 6B** and **6C**) suggesting a defect in further cellular spreading. Again, to test whether this was specific for the sEV subset, we also tested the effect of larger EVs pelleted at 10,000x g and found that these EVs had no effect on VSMC adhesion and spreading onto the FN matrix (**Fig 6B** and **6C**). To interrogate the stalling event induced by sEVs in VSMCs we plated cells and counted the number of FAs, average size as well as distance from the cell periphery after 30 mins using TIRF microscopy. Once again plating VSMCs onto FN increased the cell size, number of FAs as well as the number of centripetal FAs as compared to cells spreading over non-coated plastic (**Figs 6E-I**). Importantly, addition of sEVs to FN significantly reduced the number of FAs and the newly formed FAs were formed in close proximity to the cell periphery (**Figs. 6E** and **6F**). Interestingly, the average FA size was similar between the various conditions (**Fig 5I**).

Cellular traction force is generated by the FA gradient upon FA turnover - formation at the leading edge and disassembly at the tail^{12, 13, 46}. To test if sEVs can influence FA turnover in migrating VSMCs we examined FA turnover in the presence of sEVs. Adhesion sites were visualised by paxillin-RFP reporter expression and the FA turnover index was calculated by counting FA overlap over time using a previously developed algorithm to track individual FAs (**Fig S5B**)⁴⁷. Interestingly, the FA turnover index remains the same in the presence of sEVs indicating that FA stability was not altered (6J). These data correlate well with no changes observed in FA average size across various conditions (**Fig 6I**) and indicate that sEVs are not influencing FA assembly or “life-cycle” directly. Rho-dependent activation of actomyosin contractility stabilises FAs by mechanical forces halting fibroblast spreading onto FN^{48, 49}. We hypothesized that sEVs trigger the appearance of premature, peripheral FAs by activating Rho-dependent cellular contractility. To test whether sEVs can modulate mature FA contractility we measured individual traction force which is generated by mature adhesion sites in the absence or presence of sEVs (**Fig. S5C**). VSMCs transfected with paxillin-RFP were plated on FN and sEV-covered PDMS pillars and pillar displacements were calculated as previously described⁵⁰⁻⁵². Importantly, the traction force was increased in the presence of sEVs (**Fig 6K**). Altogether these data suggest that ECM-associated sEVs can trigger the formation of FAs with enhanced pulling force at the cell periphery.

5. The sEV cargo Collagen VI regulates focal adhesion formation in VSMCs and cell invasion

To identify the components triggering peripheral FA formation we compared the proteomic composition of sEVs with the larger EV fraction, which had no effect on FA formation. We identified 257 proteins in sEVs and 168 proteins in EVs with 142 proteins common between both datasets (**Fig 7A**, Tables S6-S8, **Fig S6A**). Functional enrichment analysis revealed that the top 5 clusters were related to cell migration - ECM organization, movement of cell or subcellular component, cell adhesion, leukocyte migration and cell-cell adhesion (**Fig 7B**, **Fig S6B**, Table S8). The cell adhesion cluster included collagen VI (chains COL6A1, COL6A2 and COL6A3), extracellular matrix glycoproteins (FN and thrombospondin, THBS2, THBS1) as well as EGF-like repeat and discoidin I-like domain-containing protein (EDIL3) and transforming growth factor-beta-induced protein ig-h3 (TGFBI) (**Fig 7B**, **Fig S6B**, Table S8). Notably, these sEV proteins are predominantly involved in cell-matrix interactions and cellular adhesion. The lectin galactoside-binding soluble 3 binding protein (LGALS3BP) which regulates cell adhesion and motility was also presented exclusively in sEVs (Table S6). These cell adhesion modulating proteins including Collagen VI, TGFBI, EDIL and LGALS3BP were either uniquely or highly enriched in sEVs compared to larger EVs as detected by western blotting (**Fig 7C**).

Collagen VI was the most abundant protein in VSMC-derived sEVs (**Fig 7B**, Table S7) and was previously implicated in the interaction with the proteoglycan NG2⁵³ and suppression of cell spreading on FN⁵⁴. To confirm the presence of collagen VI in ECM-associated sEVs we analysed

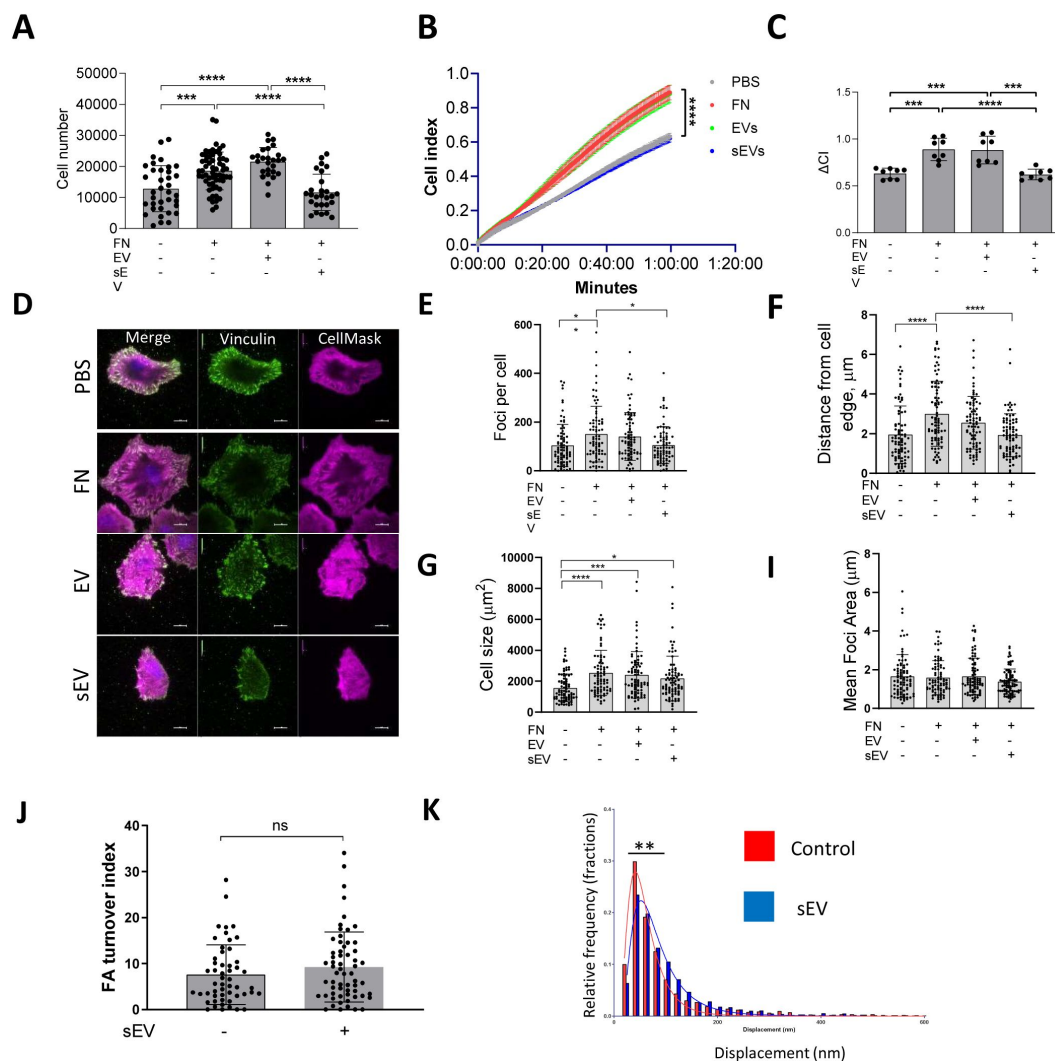


Figure 6.

sEVs induce formation of peripheral FAs.

A, VSMC were plated on FN matrix for 30min and adhered cells were counted by crystal violet staining. N=6 biological replicates, n=4-10 technical replicates, ANOVA, *** $p < 0.001$, ****, $p < 0.0001$. **B**, **C**, VSMC spreading onto FN was tracked by using ACEA's xCELLigence Real-Time Cell Analysis. Note that FN matrix promoted VSMC adhesion but addition of sEVs inhibited cell spreading. N=3 biological replicates, n=2-3 technical replicates, ANOVA. **D**, VSMCs were spread onto FN for 30 min and cells were fixed and stained with CellMask (magenta) and vinculin (green). Size bar, 10 μm. **E**, **F**, **G**, **I** Quantification of FA number, distance from plasma membrane, cell size and mean FA size per cell, respectively. FA were stained as in 5D and quantified. Representative data from N=3 biological replicates, n=80-82 total number of tracked cells per condition, ANOVA, * $p < 0.05$, ** $p < 0.01$. **J**, Focal adhesion turnover is not affected by sEVs. VSMC were transfected with Paxillin-RFP and plated on the FN in the absence or presence of immobilised sEVs. Images were captured for 30min using confocal spinning disk microscopy and FA turnover was quantified using extracted images analysis, N=4, total n=53-61 cells per condition, Unpaired T-test. **K**, sEV induces formation of strong-pulling FAs. VSMC transfected with Paxillin-RFP were plated on the PDMS pillars which were covered with FN and sEVs and pillar displacements were quantified. ** $p < 0.01$, Unpaired t-test, Data represent 2,170-2322 measurements per condition pooled from two independent biological replicates.

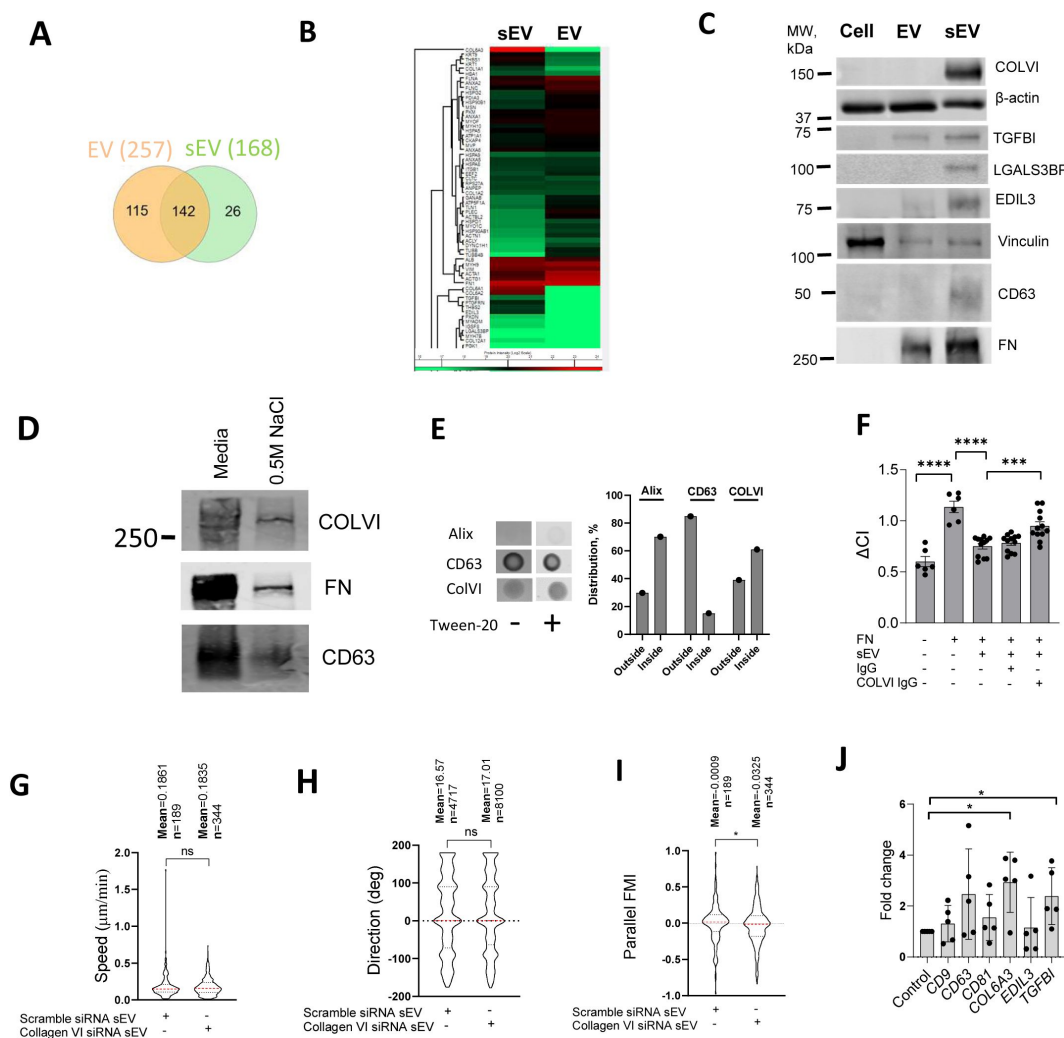


Figure 7.

sEVs block focal adhesion formation by presenting collagen VI.

A, Proteomic analysis of VSMC-derived sEVs and EVs. Venn diagram. N=3 biological replicates. **B**, Protein enrichment in the EV and sEV proteome. Heat Map. N=3 biological replicates. **C**, Western blot validation of sEV cargos. EVs and sEVs were isolated from VSMC conditioned media by differential ultracentrifugation and analysed by western blotting. Representative image from N=3 biological replicates. **D**, Collagen VI was presented in sEVs from conditioned media and in ECM-associated sEVs. sEVs were isolated from conditioned media and 0.5M NaCl ECM fraction by differential ultracentrifugation and analysed by western blotting. **E**, Collagen VI is presented on the surface of sEVs. sEVs were analysed by dot-blot in the non-permeabilised or permeabilised (PBS-0.2%-tween20) conditions. Staining intensity was quantified for non-permeabilised (outside) and permeabilised (inside) conditions. Representative from N=2 biological replicates. **F**, VSMC adhesion is regulated by collagen VI loaded on sEV. FN matrices were incubated with sEV and anti-collagen VI antibody (COLVI IgG) or control IgG. Cell adhesion was tracked by using ACEA's xCELLigence Real-Time Cell Analysis. ANOVA, N=3 biological replicates, ***, $p < 0.001$, ****, $p < 0.0001$. **G**, **H**, **I**, sEVs promote directional VSMC invasion. VSMCs were treated with control siRNA (Scramble) or collagen VI-specific siRNA pools for 24h and were seeded to the FN-enriched Matrigel matrix in a μ -Slide Chemotaxis assay and stained with Draq5. Cell tracking was conducted by OperaPhenix microscope for 12h and cell invasion parameters were quantified using Columbus, n indicates the total number of tracked cells per condition, Kolmogorov-Smirnov test, *, $p < 0.05$. **I**, Real-time PCR analysis of expression of CD9, CD63, CD81, COL6A3, EDIL3 and TGFBI in atherosclerotic plaque. *, $p < 0.05$, Paired t-test, N=5 patients.

sEVs extracted from the 3D matrix using 0.5M NaCl treatment and showed that both collagen VI and FN are present (**Fig 7D**). Next, we analysed the distribution of collagen VI using dot-blot. Alix staining was bright only upon permeabilization of sEV indicating that it is preferentially a luminal protein (**Fig 7E**). On the contrary, CD63 staining was similar in both conditions showing that it is surface protein (**Fig 7E**). Interestingly, collagen VI staining revealed that 40% of the protein is located on the outside surface with 60% in the sEV lumen (**Fig 7E**). To test the role of collagen VI in sEV-induced changes in VSMC spreading we incubated FN-deposited sEVs with an anti-collagen VI antibody to block its interaction with the proteoglycan NG2⁵⁵. Addition of the anti-collagen VI antibody restored VSMC spreading on the FN matrix as compared to non-specific IgG treatment (**Fig. 7F**). Moreover, sEVs isolated from VSMCs after collagen VI knockdown using siRNA had no effect in the 3D invasion model on cell speed (**Fig 7G**) or direction (**Fig. 7H**) but reduced cell alignment to the gradient as compared to VSMCs treated with a scrambled siRNA control (**Fig. 7I**). Finally, we compared the expression of collagen VI, EDIL3 and TGFBI between the plaque and control aorta tissues (Table S1). Importantly, collagen VI and TGFBI expression were significantly elevated in the plaque (**Fig 7J**) consistent with increased deposition of sEVs during disease progression. Altogether these data indicate that sEVs can induce early formation of mature peripheral FAs by presenting collagen VI thus acting to modulate cell directionality upon the invasion (Graphical Abstract).

Discussion

VSMC migration and invasion to the site of vascular injury is crucial for vessel repair as well as the pathogenic development of atherosclerotic plaque, however the mechanisms of effective invasion through the complex vascular ECM meshwork have been poorly studied. Here we show that FN, a novel marker of unstable plaques⁶, enables VSMC migration and invasion by stimulating secretion of collagen VI-loaded sEVs which decorate the ECM and stimulate FA maturation. Notably, our data are consistent and extend previous reports showing that EV secretion by tumour cells enhances nascent adhesion formation hence facilitating tumor cell migration and invasion¹⁴⁻¹⁶. In particular, we found that sEVs induce formation of FAs with an enhanced pulling force at the cell periphery thus favouring actomyosin-mediated contractility which is essential for cell body propulsion and locomotion.

Mesenchymal cell migration begins with protrusive activity at the leading edge followed by the forward movement of the cell body^{13, 49, 56}. Cell directionality is guided via the integrin $\beta 1$ “probe” on the tip of cellular protrusions, filopodia and lamellipodia⁵⁷. Cellular protrusions are attached to the ECM via focal complexes and extended by the physical force generated by the branched network of actin filaments beneath the plasma membrane⁵⁸. Focal complexes either disassemble or mature into the elongated centripetally located FAs⁴⁸. In turn, these mature FAs anchor the ECM to actin stress fibres and the traction force generated by actomyosin-mediated contractility pulls the FAs rearward and the cell body forward^{12, 13}. Here we report that $\beta 1$ integrin activation triggers sEV release followed by sEV entrapment by the ECM. Curiously we observed CD63+ MVB transport toward the filopodia tips as well as inhibition of sEV-secretion with filopodia formation inhibitors suggesting that sEV secretion can be directly linked to filopodia but further studies are needed to define the contribution of this pathway to the overall sEV secretion by cells. To gain insight into how sEVs can stimulate cell invasiveness we explored VSMC protrusive activity using live cell spreading assays and TIRF imaging. FN stimulated VSMC spreading by increasing the number of FAs, which were formed centripetally from the cell plasma membrane. FN-associated sEV dramatically ceased VSMC spreading by inducing early formation of mature FAs at the cell periphery. Cell spreading and adhesion are orchestrated by the Rho family of small GTPases, Cdc42, Rac1 and Rho with Cdc42 and Rac1 modulating filopodia and lamellipodia and focal complex formation and Rho controlling FA maturation and actomyosin-mediated contractility^{11, 59, 60}. Interestingly, in fibroblasts Rho is specifically degraded in cellular protrusions and cellular spreading on FN is accompanied by transient Rho suppression during the

fast cell spreading phase^{49,61,62}. This phase is followed by Rho activation leading to formation of actin stress fibres, tension increase on the focal complexes and their maturation into elongated centripetally located FAs^{48,49,61,63}. Excessive Rho activation leads to premature FA assembly and stress fiber formation resulting in reduced cellular protrusions, inhibition of cellular spreading and motility on the FN matrix^{49,61,64-66}. Altogether these data indicate that Rho activity, in mesenchymal cells with extensive FA networks such as fibroblasts or VSMCs, can slow down cell migration by immobilising cells¹³. However, we found that sEVs were not influencing mature FA stability. On the contrary, sEV-induced FAs were spatially restricted to the cell periphery near cellular protrusions and were characterised by an enhanced pulling force activity. Interestingly, fibroblast polarization is driven by Smurf1-dependent RhoA ubiquitinylation and localised degradation in cellular protrusions⁶². A recent study indicated that Rho-dependent contractility is essential for cell invasion in a 3D model and it is tempting to speculate that local Rho activity can enable VSMC invasion by stabilizing FAs at the leading edge and stimulating actomyosin-mediated contractility and cell body movement⁶⁷. On the other hand Rho-ROCK activity is critical for the protease-independent rounded motility of tumour cells and cell types with few adhesion contacts (ameboid) when Rho-dependent contractile forces generate hydrostatic pressure forming multiple membrane blebs to invade the ECM^{68,69}. In fact, we observed that an extensive secretion of sEVs effectively ceased protrusion activity; also VSMCs acquired a rounded morphology when “hovering” over the FN matrix decorated with sEVs (data not shown). Hence, it will be interesting in future studies to investigate whether sEVs can stimulate Rho activity by presenting adhesion modulators— particularly collagen VI—on their surface, thereby guiding cell directionality during invasion.

Collagen VI is a nonfibrillar collagen that assembles into beaded microfilaments upon secretion and it plays both structural and signalling roles⁷⁰. Interestingly, type VI collagen deposition by interstitial fibroblasts is increased in the infarcted myocardium and collagen VI knockout in mice improves cardiac function, structure and remodelling after myocardial infarction via unknown pathways^{71,72}. Immunohistochemical analysis showed that in the healthy vasculature collagen VI is detected in the endothelial basement membrane in the intima and it also forms fibrillar structures between smooth muscle cells in the media. In the fibrous plaque collagen VI is diffusely distributed throughout both the fibrous cap and atheroma⁷³. Curiously, treatment of ApoE^{-/-} mice with antibodies to collagen VI reduced atherosclerosis but its exact role has remained unknown⁷⁴. We and others detected Collagen VI in sEVs but its functional significance remained unknown^{23,75}. Here we showed that collagen VI is essential for early FA formation at the VSMC periphery as well as VSMC invasion in a 3D model. We propose that collagen VI modulates FA formation either by activating cellular signalling or acting as a structural component changing local ECM stiffness¹². Identified collagen VI receptors include $\alpha 3 \beta 1$ integrin, the cell surface proteoglycan chondroitin sulfate proteoglycan-4 (CSPG4; also known as NG2), and the anthrax toxin receptors 1 and 2⁷⁶⁻⁷⁹. Interestingly, a novel, recently identified Collagen VI signalling receptor (CMG2/ANTXR2) mediates localised Rho activation upon collagen VI binding^{79,80}. NG2 also activates the Rho/ROCK pathway leading to effective amoeboid invasiveness of tumour cells which is characterised by excessive blebbing and enhanced actomyosin contractility^{68,81}. Hence, binding of collagen VI to these receptors can potentially locally activate Rho and trigger FA formation and actomyosin contractility thus increasing VSMC motility and directional invasiveness. Of note, it was also reported that NG2 expression is elevated in VSMCs in the atherosclerotic plaque and NG2 knockout prevents plaque formation^{53,82}. Curiously, endotrophin, a collagen VI $\alpha 3$ chain-derived profibrotic fragment has recently been associated with a higher risk of arterial stiffness and cardiovascular and all-cause death in patients with diabetes type 1 and atherosclerosis, respectively^{83,84}.

In addition to collagen VI the unique adhesion cluster in VSMC-derived sEVs also includes EGF-like repeat and discoidin I-like domain-containing protein (EDIL3), transforming growth factor-beta-induced protein ig-h3 (TGFB1) and the lectin galactoside-binding soluble 3 binding protein (LGALS3BP) and these proteins are also directly implicated in activation of integrin signalling and

cellular invasiveness^{85, 87}. Although we found that collagen VI plays the key role in sEV-induced early formation of FAs in VSMCs, it is tempting to speculate that the high sEV efficacy in stimulating FA formation is driven by cooperative action of this unique adhesion complex on the sEVs surface and targeting this novel sEV-dependent mechanism of VSMC invasion may open-up new therapeutic opportunities to modulate atherosclerotic plaque development or even to prevent undesired VSMC motility in restenosis.

We also identified a positive feedback loop and showed that FN stimulates sEV secretion by VSMCs. sEV secretion is a tightly regulated process which is governed by activation of signalling pathways including activation of G-protein coupled receptors³⁹ and alterations in cytosolic calcium⁸⁸, small GTPases Rab7, Rab27a, Rab27b and Rab35^{89, 90}, vesicular trafficking scaffold proteins including syntenin⁹¹, sortilin⁹² and CD63⁹³ or accumulation of ceramide²⁶. We found that the fibrillar ECM proteins, FN and collagen I induce sEV secretion by activating the $\beta 1$ integrin/FAK/Src and Arp2/3-dependent pathways. Our in vivo FN staining data are in good agreement with previous studies showing accumulation of FN in late-stage plaques^{6, 7}. Moreover, we observed close co-distribution of FN and CD81 in the plaque suggesting that accumulating FN matrices can stimulate sEV secretion in vivo. Notably, expression of the major FN receptor, $\alpha 5 \beta 1$ integrin is re-activated upon VSMC de-differentiation following vascular injury⁹⁴. It is thought the $\alpha 5 \beta 1$ integrin mediates FN matrix assembly, whilst $\beta 3$ integrins are important for cell-matrix interactions^{94, 95}. Our data sheds new light on the functional role of $\alpha 5 \beta 1$ as an ECM sensor inducing production of collagen VI-loaded sEVs which in turn enhance cell invasion in the complex ECM meshwork. Although our small inhibitors and integrin modulating antibody data clearly indicate that $\beta 1$ activation triggers sEV secretion via activation of actin assembly we cannot fully rule out that FN may also be modulating growth factor activity which in turn contributes to sEV secretion by VSMCs²³. Excessive collagen and elastin matrix breakdown in atheroma has been tightly linked to acute coronary events hence it will be interesting to study the possible link between sEV secretion and plaque stability as sEV-dependent invasion is also likely to influence the necessary ECM degradation induced by invading cells⁹⁶.

In summary, cooperative activation of integrin signalling and F-actin cytoskeleton pathways results in the secretion of sEVs which associate with the ECM and play a signalling role by controlling FA formation and cell-ECM crosstalk. Further studies are needed to test these mechanisms across various cell types and ECM matrices.

Materials and Methods

Materials

Proteins were fibronectin (Cell Guidance Systems, AP-23), collagen I (Gibco, #A1048301), laminin (Roche, 11243217001), Matrigel (Corning, #356237), Phalloidin-rhodamin (ThermoFisherScientific, R415). Peptides were Gly-Arg-Gly-Asp-Ser-Pro (GRGDSP, Merck, SCP0157) and scramble control Gly-Arg-Ala-Asp-Ser-Pro (GRADSP, Merck, SCP0156). All chemical inhibitors were diluted in DMSO and were: 3-O-Methyl-Sphingomyelin (SMPD3 inhibitor, Enzo Life technologies, BML-SL225-0005), FAM14 inhibitor (FAK inhibitor, Abcam, ab146739), PP2 (Src inhibitor, Life technologies, PHZ1223), CK666 (Arp2/3 inhibitor, Abcam ab141231), SMIFH2 (Formin inhibitor, Sigma, S4826). Control siRNA pool (ON-TARGETplus siRNA, Dharmacon, D-001810-10-05), collagen VI siRNA ON-TARGETplus siRNA (COL6A3, Human), SMARTPool, Horizon, L-003646-00-0005). Antibodies were CD9 (SA35-08 clone, NBP2-67310, Novus Biologicals), CD63 (BD Pharmingen, 556019), CD81 (BD Pharmingen™, 555676, B-11, SantaCruz, sc-166029 and M38 clone, NBP1-44861, Novus Biologicals), Syntenin-1 (Abcam, ab133267), Syndecan-4 (Abcam, ab24511), α -Actinin-4 (Abcam, ab108198), fibronectin (Abcam, ab2413, ab6328 [IST-9] (3D matrix staining) and F14 clone, ab45688 (clinical samples analysis)), $\beta 1$ activating (12G10) antibody was previously described⁹⁷, 4B4 integrin inhibiting antibody (Beckman Coulter, 41116015), vinculin (Sigma, V9264), $\alpha 5$ integrin (P1D6,

Abcam, ab78614), Myo10 (Sigma, HPA024223), gelatin-3BP/MAC-2BP (R&D systems, AF2226), EDIL3 antibody (R&D systems, MAB6046), TGFBI (Sigma, SAB2501486), IgG mouse (Sigma PP54), Anti-collagen Type VI antibody, clone 3C4 (Sigma, MAB1944) (functional blocking), Anti-collagen Type VI antibody (Abcam, AB6588) (western blot) Anti-collagen Type VI antibody (Abcam, AB182744) (dot blot) p34-Arc/ARPC2 antibody (Millipore, #07-227), Cortactin, LGALS3BP (R&D, AF2226), GAPDH (ab139416, Abcam), Alix (Clone 3A9, ThermoFisher Scientific, MA1-83977) Donkey Anti-mouse Secondary IRDye@680 LT (Li-COR, 926-68022) and Donkey Anti-Rabbit Secondary IRDye@800 CW (Li-COR, 926-32213) DNA plasmids were: CD63-GFP was kindly provided by Dr Aviva Tolkovsky⁹⁸, CD63-pHluorin was previously described³⁹, CD63-RFP was previously published⁹⁹, Paxillin-RFP was previously described⁹⁷, ARPC2-GFP DNA vector was kind gift from Professor Michael Way laboratory³⁴, F-tractin-RFP was kindly provided by Dr Thomas S. Randall (King's College London, UK), Myo10-GFP (Addgene Plasmid#135403) was previously described¹⁰⁰.

Cell culture

Human VSMCs were isolated as previously described¹⁰¹ and were cultured in Dulbecco's modified Eagle medium (Sigma) supplemented with 20% fetal calf serum, 100 U/ml penicillin, 100 µg/ml streptomycin and 2 mmol/L L-glutamine (Gibco) and used between passages 4 and 12.

VSMC adhesion

24 well plate (Corning Costar) was incubated with 5 µg/ml fibronectin in PBS overnight at +4°C. Next, sEVs (10 µg/ml) diluted in PBS were added to the wells and incubated overnight at +4°C. Wells were washed with PBS, blocked with PBS-1% BSA for 30min at 37°C and washed 3 times with PBS again. VSMCs were incubated in serum free media, M199 supplemented with 0.5% BSA, 100 U/ml penicillin and 100 µg/ml streptomycin overnight and removed by brief trypsin treatment. Next, 20,000 cells were added to each well and incubated for 30min at +37°C. Unattached cells were washed away with PBS and attached cells were fixed with 3.7% PFA for 15 min at 37°C. Cells were washed with H₂O 3 times and stained with 0.1% crystal violet for 30min at 10% CH₃COOH for 5min at room temperature. Samples were transferred to 96 well plate and absorbance was measured at 570nm using the spectrophotometer (Tecan GENios Pro).

iCELLigence adhesion protocol

E-Plate L8 (Acea) was coated with FN (5 µg/mL) in PBS at 4°C overnight. Next, E-Plate L8 was gently washed once with cold PBS followed by the incubation with EV or sEV (10 µg/ml) at 4°C overnight. The following day, E-Plate L8 was incubated with 0.1% BSA in PBS for 30min at 37°C to block non-specific binding sites and washed with PBS before adding 250 µL of M199 media supplemented with 20% exosome-free FBS. VSMCs were serum-deprived by incubation in M199 media supplemented with 0.5% BSA overnight, were passaged with trypsin and resuspended in media M199 supplemented with 20% exosome-free FBS to a final concentration 80,000 cells/mL. Cell aliquot (250 µL) was added to E8-plate L8 which was then transferred to iCELLigence device for the adhesion assay. Cell adhesion was measured at 37°C with time intervals 20sec for 1h.

Focal adhesion turnover assay

iBIDI 35mm dishes were incubated with FN (5 µg/ml) in PBS overnight at +41°C and treated with exosomes (10 µg/ml) diluted in PBS overnight at +41°C. 500,000 cells were transfected with Paxillin-RFP/Myo10-GFP by using electroporation (see below) and plated on iBIDI 35mm dishes coated with FN and sEVs in DMEM supplemented with 20% exosome-free FBS, 100 U/ml penicillin, and 100 µg/ml streptomycin. Next, cells were transferred to DMEM supplemented with 20% exosome-free FBS, 10mM HEPES, 100 U/ml penicillin, and 100 µg/ml streptomycin and images were acquired (Nikon Spinning Disk) every 1min for 30min at 37°C. FA images were extracted from timelapses and FA turnover was quantified using Mathematica by producing the adhesion map as previously described⁴⁷.

Confocal spinning disk microscopy and live cell imaging

VSMC were transfected using electroporation (see below) and plated onto FN-coated 35mm iBIDI glass bottom dish and incubated for 48h. Then cells were transferred to DMEM supplemented with 20% exosome-free FBS, 10mM HEPES, 100 U/ml penicillin, and 100 µg/ml streptomycin and images were acquired every 1min for 20min at 37°C using a Nikon Ti-E (inverted) microscope equipped with a Yokogawa spinning disk and a Neo 5.5 sCMOS camera (Andor) and 60x or 100x/1.40 NA Plan Apo λ oil objectives (Nikon) were used. Images were acquired using NIS Elements AR 4.2 software. Cells were maintained at 37°C, 5% CO₂ throughout the experiment via a CO₂ chamber and a temperature-regulated Perspex box which housed the microscope stage and turret.

iSIM

Slides was imaged and super-resolved images collected using a Visitech-iSIM module coupled to a Nikon Ti-E microscope using a Nikon 100x 1.49NA TIRF oil immersion lens. Blue fluorescence was excited with a 405nm laser and emission filtered through a 460/50 filter. Green fluorescence was excited with a 488nm laser and emission filtered through a 525/50 filter. Red Fluorescence was excited with a 561nm laser and emission filtered through a 630/30 filter. Far Red fluorescence was excited with a 640nm laser and emission filtered through a 710/60 filter. Images were collected at focal planes spaced apart by 0.08µm. Data was deconvolved using a Richardson-Lucy algorithm utilizing 20 iterations using the Nikon deconvolution software module (Nikon Elements). 65 stack images were taken with Z step 0.08µm. ImageJ (153t) or NIS Elements (5.21.00, built1483, 64bit) software were used for image analysis and assembly.

Cell electroporation

iBidi dishes were coated with FN (5µg/ml) in PBS for 1 hour at 37°C or overnight at 4°C. VSMCs were grown to 70% confluence, washed twice with EBSS and detached by trypsin treatment for 5min at +37°C. Cells (500,000) were mixed with plasmids (2.5µg each) in electroporation buffer (100µl, Lonza) and transfected using either Nucleofector II (program U25) or Nucleofector III (program CM-137).

Isolation of apoptotic bodies, extracellular vesicles and small extracellular vesicles

Flasks (T150) were incubated with PBS or FN (5µg/ml) in PBS overnight at +4°C and VSMCs (≈10⁶ cells) were plated and incubated for 16h at +37°C. Next, cells were washed with EBSS 3 times and incubated with DMEM supplemented with 0,1% BSA, 100 U/ml penicillin, and 100 µg/ml streptomycin. Conditioned media was collected and centrifuged at 2,500 rpm (Thermo Scientific Heraeus Multifuge 3SR+ centrifuge, rotor Sorvall 75006445) for 5min to remove apoptotic bodies (AB, 1.2K pellet). Supernatant was transferred to centrifugation tube (Nalgene™ Oak Ridge High-Speed Polycarbonate Centrifuge Tubes, Thermo Fisher Scientific, 3138-0050) and centrifuged 10,000 xg for 30min. 10K pellet (EVs) was collected, washed in PBS once and kept at -80°C until further analysis. 10K supernatant was transferred to ultracentrifugation tubes (Polycarbonate tubes for ultracentrifugation, BeckmanCoulter, 355647) and centrifuged at 35,000 rpm (100,000xg) for 40min at 4°C (Fixed-angle rotor, Beckman Coulter Optima Max Ultracentrifuge). 100K pellet (sEVs) was washed with PBS twice and re-suspended in PBS. EV and sEV pellets were kept at -80°C for the proteomic analysis and freshly isolated sEVs were used for all other assays.

EV and sEV proteomic analysis

EV and sEV samples were submitted to the CEMS Proteomics Facility in the James Black Centre, King's College London for mass spectrometric analysis. The data were processed by Proteome Discover software for protein identification and quantification. Scaffold 5 proteome software was utilized to visualize differential protein expression. Individual proteins intensities were acquired

and transformed to log₂ scale before Hierarchical clustering analysis based on Euclidean distance and k-means processing. Differential genes defined by multiple groups comparison were applied to perform the gene ontology functional assay through using the DAVID Gene Ontology website (<https://david.ncifcrf.gov/>). Selected genes which showed abundant in each condition were applied to generate Venn diagram through Interacti website (<http://www.interactivenn.net/>).

Cell lysis

Cells were washed with PBS and removed by cell-scraper in 1ml of PBS. The cells were pelleted by centrifugation at 1000xg for 5min and cell pellets were kept at -80°C until further analysis. To prepare cell lysates, pellets were incubated with lysis buffer (0.1M TrisHCl(pH8.1), 150 mM NaCl, 1% Triton X-100 and protease inhibitors cocktail (Sigma, 1:100)) for 15min on ice. Cell lysates were subjected to ultrasound (Branson Sonifier 150), centrifuged at 16,363xg for 15 minutes (Eppendorf) at 4°C and supernatants were collected and analysed by western blotting.

CD63-bead capturing assay

CD63-bead capturing assay was conducted as previously described²³ with some modifications. In brief, CD63 antibody (35µg) was immobilised on 1x10⁸ 4µm aldehyde-sulfate beads (Invitrogen) in 30mM MES buffer (pH 6.0) overnight at room temperature and spun down by centrifugation (3,000xg, 5min). Next, beads were washed 3 times with PBS containing 4%BSA and kept in PBS containing 0,1% Glycine and 0,1% NaN₃ at +4°C. VSMCs were plated onto 24 well plate (10,000 cells per well) and incubated in the complete media overnight. Next, cells were washed 3 times with EBSS and incubated in M199 supplemented with 2.5% exosome-free FBS in the presence of absence of inhibitors for 2h. Then, conditioned media was replaced once to the fresh aliquot of 2.5% exosome-free FBS in the presence or absence of inhibitors and cells were incubated for 18-24h. VSMC conditioned media was collected and centrifuged and centrifuged at 2,500xg for 5 min. VSMCs were detached from the plate by trypsin treatment and viable cells were quantified using NC3000 (ChemoMetec A/S). The conditioned media supernatants were mixed with 1µL of anti-CD63-coated beads and incubated on a shaker overnight at +4°C. Beads were washed with PBS-2%BSA and incubated with anti-CD81-PE antibody (1:50 in PBS containing 2% BSA) for 1h at room temperature. Next, beads were washed with PBS-2%BSA and PBS and analysed by flow cytometry (BD Accuri™ C6). sEV secretion (fold change) was calculated as ratio of Arbitrary Units (fluorescence units x percentage of positive beads and normalized to the number of viable VSMCs) in the treatment and control conditions.

Exosome labelling with Alexa Fluor® 568 C5 Maleimide

Exosomes (10µg/ml) were incubated with 200µg/mL Alexa Fluor® 568 C5 Maleimide (ThermoFisherScientific, #A20341) in PBS for 1h at room temperature. An excessive dye was removed by using Exosome Spin Columns (MW 3000, ThermoFisherScientific, #4484449) according to the manufacturer protocol.

Fetuin-A and Fibronectin labelling and uptake studies

Bovine fetuin-A (Sigma) and Fibronectin were labelled using an Alexa488 (A10235) and Alex568 (A10238) labelling kits in accordance with the manufacturer's protocol (Invitrogen). VSMCs were serum-starved for 16h and then incubated with Alexa488-labeled fetuin-A (10µg/mL) and Alexa555-labeled fibronectin (10µg/mL) from 30 to 180 minutes at 37°C.

VSMC and beads-coupled sEVs flow cytometry

VSMC were removed from the plate by brief trypsin treatment, washed with M199 media supplemented with 20% FBS and resuspended in HBSS supplemented with 5%FBS. Cells were kept on ice throughout the protocol. Next, cells (200,000) were incubated with primary antibody for

30min on ice, washed with HBSS-5%FBS and incubated with secondary fluorescently-labelled antibody 30min on ice. Then cells were washed twice with HBSS-5%FBS and once with PBS and analysed by flow cytometry (BD FACScalibur, BD Bioscience).

Flow cytometry analysis of beads-coupled sEVs was conducted as described before¹⁰²[\[102\]](#). In brief, exosomes (10µg) isolated by differential centrifugation were coupled to 4µm surfactant-free aldehyde/sulfate latex beads (Invitrogen) and were incubated with primary antibody and fluorescently labeled secondary antibody and analysed by flow cytometry on BD FACScalibur (BD Bioscience). Data were analysed using the Cell Quest Software (BD). Cells and beads stained with isotype-control antibodies were used as a negative control in all experiments.

Nanoparticle Tracking Analysis

Exosomes were diluted 1:150 and analysed using LM-10 using the light scattering mode of the NanoSight LM10 with sCMOS camera (NanoSight Ltd, Amesbury, United Kingdom). 5 frames (30 s each) were captured for each sample with camera level 10 and background detection level 11. Captured video was analysed using NTA software (NTA 3.2 Dev Build 3.2.16).

Western blotting and dot blotting

For western blotting, cell lysates or apoptotic bodies, microvesicles and exosomes in Laemmli loading buffer were separated by 10% SDS-PAGE. Next, separated proteins were transferred to ECL immobilon-P membranes (Millipore) using semi-dry transfer (BioRad). Membranes were incubated in the blocking buffer (TBST supplemented with 5% milk and 0.05% tween-20) for 1h at room temperature, then incubated with primary antibody overnight at +4°C. Next, membranes were washed with blocking buffer and incubated with secondary HRP-conjugated antibody and washed again in PBS supplemented with 0.05% tween-20 and PBS. Protein bands were detected using ECL plus (Pierce ECL Western Blotting Substrate, #32109). Alternately, membranes were incubated with the secondary fluorescent antibody (Alexa fluor antibody) and washed again with TBST or PBST. Blots were detected using Odyssey Licor.

Dot blot was conducted as previously described¹⁰³[\[103\]](#). In brief, an aliquot of isolated sEVs (0.5ug) was absorbed onto nitrocellulose membrane (Odyssey® P/N 926-31090) for 30min at room temperature and air dried. Next, membranes were blocked with PBS-3% BSA in PBS and incubated with primary antibodies (anti-CD63 1:500, anti-COLVI 1:200, anti-Alix, 1:500) in the absence or presence 0.2% (v/v) Tween-20 for 16h at the cold room (+4C) room temperature. Membranes were washed 3 times in PBS-3% BSA, and were incubated with Donkey Anti-Rabbit Secondary IRDye@800 CW (Li-COR, 926-32213) and Donkey Anti-Mouse Secondary IRDye@680 LT (Li-COR, 926-68022) (1:10,000), washed with PBS-0.2% tween20 and PBS and visualized and quantified using Odyssey ® CLx Infrared Imaging System.

Generation of 3D Extracellular Matrix

Generation of 3D matrices were conducted as previously described²⁴[\[24\]](#) with some modifications. In brief, plates or glass coverslides were covered with 0.2% gelatin in PBS for 1h at 37°C. Wells were washed with PBS and fixed with 1% glutaraldehyde in PBS for 30min at room temperature. Plates were washed with PBS and incubated with 1M ethanolamine for 30min at room temperature, then washed with PBS again. VSMCs (5×10^5 per well) were plated and cultured for 24h. Then the medium was replaced with medium containing 50µg/ml of ascorbic acid and cells were incubated for 9days. The medium was replaced every 48h. To extract matrix, cells were rinsed with PBS and incubated with pre-warmed extraction buffer (20mM NH₄OH containing 0.5%Triton X-100) for 3min until intact cells are not seen. Next, equal volume of PBS was added to extraction buffer and plates were incubated for 24h at +4°C. Plates were washed with PBS twice and kept with PBS containing 100 U/ml penicillin and 100 µg/ml streptomycin up to 2 weeks at +4°C.

sEVs isolation from 3D Extracellular Matrix

VSMCs were cultured on dishes pre-coated with 0.1% gelatin as described above, with minor modifications. Briefly, cells were maintained in DMEM supplemented with EV-free fetal bovine serum, 1% penicillin-streptomycin, and 501 µg/ml ascorbic acid. Cultures were maintained for 10 days, with media changes and collections performed every 2–3 days. At the end of the culture period, the medium was gently removed and collected as the “Media” fraction. Conditioned media was clarified by centrifugation at $2,500 \times g$ for 15 minutes to remove cellular debris; the resulting supernatant was retained for ExoView analysis and small EV (sEV) isolation. The cell monolayer was then rinsed with phosphate-buffered saline (PBS) and incubated with extraction buffer (0.5 M NaCl, 20 mM Tris, pH 7.5) for 4 hours at room temperature as previously described⁴². This NaCl-extracted ECM fraction, termed the “loosely-bound sEV fraction,” was collected and centrifuged at $2,500 \times g$ for 15 minutes. The supernatant was retained for ExoView analysis and sEV isolation. Following NaCl extraction, cells were treated with 0.1% sodium dodecyl sulfate and 25 mM ethylenediaminetetraacetic acid (EDTA), incubating at 37°C for 10 minutes to lyse any remaining cellular material. To extract the remaining ECM-associated fraction, the matrix was incubated with 4 M guanidine hydrochloride (GuHCl) in 50 mM sodium acetate buffer (pH 5.8) for 48 hours at room temperature. The GuHCl fraction, referred to as “strongly-bound sEV fraction” was collected and centrifuged at $2,500 \times g$ for 15 minutes at 4°C; the supernatant was retained for ExoView analysis. sEVs from all fractions were isolated by ultracentrifugation as described above.

ExoView analysis of ECM-derived sEV fractions

ECM-derived EV fractions (media, loosely bound and strongly trapped sEVs) were prepared as detailed above. For ExoView analysis, each fraction was diluted 1:2 in the manufacturer-supplied incubation solution. Leprechaun Exosome Human Tetraspanin chips (Unchained Labs, REF 251-1044) were used according to the manufacturer’s protocol.

Briefly, 35–501 µL of diluted sample (1:2–1:10) was applied to each chip well. Chips were incubated overnight at room temperature in a sealed humidity chamber to allow EV capture onto surface-immobilized antibodies. The following day, chips were washed per the supplied protocol to remove unbound material.

Secondary immunostaining was performed using fluorescently labeled antibodies provided in the Leprechaun Exosome Human Tetraspanin Kit (CD9, CD63, and CD81, Unchained Labs, LOT UL038460001F). After antibody incubation, chips were washed thoroughly as recommended and air-dried briefly.

Chips were scanned on an ExoView R200 platform (Unchained Labs). Imaging and data analysis were performed using ExoView Analyzer software, with spot intensity and particle counts determined for each tetraspanin (CD9, CD63, CD81) according to the manufacturer’s guidelines. Data were exported for downstream quantitative and statistical analysis.

Immunocytochemistry

VSMCs (10,000 cells per well) were plated onto the coverslips and incubated for 48h. Next, cells were treated with inhibitors in M199 supplemented with 2.5% exosome-free FBS and fixed using 3.7% paraformaldehyde for 15 minutes at +37°C. Cells were then washed with PBS, permeabilised with PBS-0.1% triton X-100 for 5 min at room temperature and washed with PBS again. Cells were blocked with PBS-3% BSA for 1 hour at room temperature and incubated with primary antibodies overnight at +4°C. Following washing 3 times with PBS-3% BSA cells were incubated with secondary fluorescently-labelled antibodies in the dark for 1 hour at room temperature, stained with DAPI for 5min and mounted onto 75-Superdex slides using gelvatol mounting media and analysed by spinning disk confocal microscopy as above (Nikon). Primary antibodies were used in

the following dilutions: CD63 1:500, Cortactin 1:500, Arp2C 1:500, Myo10 1:250, Vinculin 1:500, fibronectin 1:500, CD81 1:500, Rhodamine phalloidin 1:200 and secondary fluorescently labelled antibodies 1:200.

Focal adhesion turnover/Individual traction force

PDMS pillar (500 nm diameter, 1.3 μm height, 1 μm centre-to-centre) substrates were prepared as described previously¹⁰⁴. Briefly, CdSe/ZnS alloyed quantum dots (490nm, Sigma) were spun on the master 30s at 10,000rpm with a 150i spin processor (SPS), before the addition of PDMS. PDMS (Sylgard 184, Dow Corning) was mixed thoroughly with its curing agent (10:1), degassed, poured over the silicon master, placed upside-down on a plasma-treated coverslip-dish (Mattek), or coverslip 4-well dishes (Ibidi) and cured at 80°C for 2h. The mold was then removed and the pillars were incubated with fibronectin for 1h at 37°C, after which pillars were incubated with sEVs (10 $\mu\text{g}/\text{ml}$ in PBS) at 4°C overnight.

VSMC previously transfected with Paxillin-RFP by using electroporation as above were plated on the Pillars and imaged on a Nikon Eclipse Ti-E epifluorescent microscope with a 100x 1.4NA objective, Nikon DS-Qi2 Camera and a Solent Scientific chamber with temperature and CO₂ control. For calculation of pillar displacements, a perfect grid was assumed and deviations from the grid were calculated using reference pillars outside the cell. The traction stress was calculated for a 10 μm wide region at the cell edge that was enriched in paxillin staining, taking into account all pillar displacements above a ~ 20 nm noise level, which was calculated from pillars outside the cells.

The Stiffness of the pillars was calculated as described by Ghibaudo et al⁵² but taking into account substrate warping as described by Schoen et al (Eq1-5)⁵¹:

$$\text{Eq1: } k_{\text{bend}} = \frac{3}{64} \pi E \frac{D^4}{H^3},$$

$$\text{Eq2: } \text{corr} = \frac{\frac{16}{3} \left(\frac{L}{D}\right)^3}{\left(\frac{16}{3} \left(\frac{L}{D}\right)^3 + \frac{2 + 6 \nu L}{3 D} + 8 T_{\text{ult}}(\nu) \left(\frac{L}{D}\right)^2\right)},$$

$$\text{Eq3: } T_{\text{ult}}(\nu) = a \frac{(1 + \nu)}{2\pi} \left\{ 2(1 - \nu) + \left(1 - \frac{1}{4(1 - \nu)}\right) \right\}$$

$$\text{Eq4: } k = k_{\text{bend}} * \text{corr};$$

$$\text{Eq5: } E_{\text{eff}} = \frac{9k}{4\pi a};$$

Protein concentration

Protein concentration was determined by DC protein assay (BioRad).

Focal adhesion TIRF Imaging

μ -Slides (iBidi) were coated with FN (5 $\mu\text{g}/\text{mL}$) in PBS at 4°C overnight. Next, μ -Slides were gently washed once with cold PBS following incubation with EV or sEV at 4°C overnight. The following day μ -Slides were incubated with 0.1% BSA in PBS for 30 minutes at 37°C to block non-specific binding sites and washed with PBS three times and 250 μL of M199 media supplemented with 20% sEV-free FBS was added to each well. VSMCs were incubated in M199 media supplemented with 0.5% BSA overnight, detached with trypsin and resuspended to 80,000 cells/mL in M199 supplemented with 20% FBS. Cells (250 μL) were added to μ -Slides and were incubated for 30min at 37°C. VSMC were fixed with 4% PFA for 10 minutes at 37°C, washed with PBS three times and were permeabilised with PBS supplemented with 0.2% Triton X-100 for 10 minutes at 37°C. Next, VSMC were washed three times with PBS for 5 minutes and were incubated with 3% BSA in PBS for 1h at room temperature. VSMC were incubated with anti-vinculin antibody (1:400) in blocking buffer at 4°C overnight. Next, VSMC were washed with PBS for 5min and incubated with secondary

antibody (1:500, AlexaFluor488) diluted in blocking buffer for 1h at room temperature. Cells were incubated with CellMaskRed (ThermoFisherScientific) and DAPI (0.1 mg/mL) for 5min, washed and analysed by TIRF microscopy.

TIRF Microscopic images were collected on a Nikon Ti2 upright microscope with a TIRF illuminator (Nikon) attached using a 100x oil immersion (1.49 NA) objective lens. In each fluorescent channel the sample was excited with a laser beam at an angle indicated as follows (488nm at 62.3°, 561nm at 62.5° and 640nm at 65.5°). All emission light was filtered through a quad channel filter set (Chroma 89000) in the upper filter turret and a secondary bandpass emission filter in the lower turret position was used as follows for each channel (488nm excited: filter BP 525/50 - Chroma; 561nm excited: filter BP 595/50 - Chroma; 640nm excited: filter BP 700/75 - Chroma). Light was then passed to a Hamamatsu Orca-Flash 4v3.0 sCMOS camera and exposures for channels was set as follows (488nm: 100ms, 561nm: 100ms, 640nm:50ms) with 16bit channel images collected.

Acquisition was all controlled through NIS-Elements software (Nikon). Analysis of TIRF images was completed using NIS-Elements General Analysis Module. A general description of the process follows: Cell Area measurements were determined by use of an intensity threshold on the HCS CellMask Deep Red channel at an intensity of 100 AU (grey levels or arbitrary units) above the background to mask the cell. The Cell area, whole cell Vinculin and Far-Red intensity measurements calculated from this binary region. In addition, an inversion of the cell mask binary was generated to have a means to measure distance to edge of the cell.

Vinculin derived Foci were masked using an intensity-based threshold (Above 10000 AU) and a size minimum for objects was set to $0.5\mu\text{m}^2$ to eliminate smaller non foci objects from the binary mask. The foci mask was then limited to the cell mask regions to allow interpretation of the foci in the cell area. Minimum Distance to the outside of the cell for each foci was calculated by measurement of the nearest distance to the inverted non-cell mask. The average distance was then recorded for each image. For each foci intensity measurements were taken for the Vinculin and FarRed channel and the Mean intensity of the foci were recorded.

2D VSMC migration time lapse assay

24 well plates were incubated with FN (5 $\mu\text{g}/\text{ml}$) in PBS overnight at +4°C and treated with sEVs (10 $\mu\text{g}/\text{ml}$) diluted in PBS overnight at +4°C. VSMC (8,000 cells per well) were plated on the plate in M199 supplemented with 20% exosome-free FBS and incubated for 16h. Cells were washed twice with DMEM supplemented with 2.5% exosome-free FBS and incubated in this media for 2hrs +/-3'-OMS. Media was replaced for a fresh aliquot and cells were imaged using Opera Phenix High Content Screening System (PerkinElmer) every 6min20sec for 8h in a transmitted light channel. Quantification of cellular velocity and directionality was performed in Harmony 4.9 (Perkin Elmer) and the following filters were applied to the raw data: "Number of Timepoints" ≥ 5 , "First Timepoint" < 5 .

3D VSMC Invasion time lapse assay

Acquisition

Migration assay were completed using IBidi Chemotaxis μ -slides (Ibidi μ -slides #80326). VSMC were deprived overnight in M199 supplemented with 0.5% BSA, counted and mixed with Matrigel supplemented with/without sEV (5 μg total in 100 μL volume) to a final cell concentration of ca. 3×10^6 cells/ml and stained with Draq5 according to the manufacturer protocol. Slides were left for 30min at 37°C for gelation and chemoattractant-free medium (65 μL of M199 supplemented with 0.5% BSA) was filled through Filling Port E. Next, the empty reservoir was filled by injecting chemoattractant medium (65 μL of M199 supplemented with 20% FBS exosome-free through Filling

Port C. 3 slides were loaded and cells were imaged with the Opera Phenix (Perkin Elmer) high content imaging platform (10x Objective NA 0.3). Images were collected for digital phase contrast and a fluorescence channel for Draq5. Fluorescence was obtained by exciting with a 640nm laser and emission light was filtered with a 700/75 bandpass filter to a sCMOS camera. Images were recorded every 10min for 12h in several locations for each slide.

Migration tracking and analysis of tracking

Analysis of cell motion was completed on Draq5 images using Harmony Software (Perkin Elmer) with measurements of cell area, speed and direction were obtained for each cell/track and timepoint. Cells were not selected for tracking if they were in proximity to other cells of a distance of 35 μ m. The measurements and object data were exported and further analysis was performed. Data for each track included total length (timepoints), speed(μ m/s), Straightness, accumulated distance (μ m), lateral displacement (μ m) and vertical displacement (μ m).

Data analysis

This data was then analysed using an in-house script using Python library Pandas. Tracks were discarded if they were less than 3 time-points in length. Means and Standard Deviations were calculated for track length (timepoints), Speed (μ m/s), Track Straightness and accumulated distance. For each track the Parallel FMI (forward motion index) was calculated by taking a ratio of vertical displacement to accumulated distance. The means and standard deviations for these were calculated and reported for each condition similarly. Track Straightness is calculated as the ratio of total displacement over total track length. Accumulated distance is the total length of the track from the first to last time point of the track.

Clinical samples

Patients enrollment

The study was approved by the Local Ethical committee of the Research Institute for Complex Issues of Cardiovascular Diseases (Kemerovo, Russia, protocol number 20200212, dates of approval: 12 February 2020), and a written informed consent was provided by all study participants after receiving a full explanation of the study. The investigation was carried out in accordance with the Good Clinical Practice and the Declaration of Helsinki. Criteria of inclusion were: 1) performance of carotid endarterectomy due to chronic brain ischemia or ischemic stroke; 2) a signed written informed consent to be enrolled. A criterion of exclusion was incomplete investigation regardless of the reason; in this case, we enrolled another subject with similar age, gender and clinicopathological features who met the inclusion criteria. Cerebrovascular disease (chronic brain ischemia and ischemic stroke) as well as comorbid conditions (arterial hypertension, chronic heart failure, chronic obstructive pulmonary disease, asthma, chronic kidney disease, diabetes mellitus, overweight and obesity) were diagnosed and treated according to the respective guidelines of European Society of Cardiology, Global Initiative for Chronic Obstructive Lung Disease, Global Initiative for Asthma, Kidney Disease: Improving Global Outcomes, American Diabetes Association, and European Association for the Study of Obesity. eGFR was calculated according to the Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI) equation. Extracranial artery stenosis was assessed using the color duplex screening (Vivid 7 Dimension Ultrasound System, General Electric Healthcare). Data on age, gender, smoking status and pharmacological anamnesis were collected at the time of admission. In total, we enrolled 20 patients. The detailed characteristics of the study samples are presented in Table S1.

Sample collection and preparation

Carotid atherosclerotic plaques (n = 14) and adjacent intact arterial segments (n = 14) were pairwise excised during the carotid endarterectomy and divided into 3 segments each. The first segment was snap-frozen in the optimal cutting temperature compound (Tissue-Tek, 4583, Sakura) using a liquid nitrogen and was then sectioned on a cryostat (Microm HM525, 387779, Thermo Scientific). To ensure the proper immunofluorescence examination, we prepared 8 sections (7µm thickness), evenly distributed across the entire carotid artery segment, per slide. The second and third segments were homogenised in TRIzol Reagent (15596018, Thermo Fisher Scientific) for RNA extraction or in T-PER Tissue Protein Extraction Reagent (78510, Thermo Fisher Scientific) supplied with Halt protease and phosphatase inhibitor cocktail (78444, Thermo Fisher Scientific) for the total protein extraction according to the manufacturer's protocols. Quantification and quality control of the isolated RNA was performed employing Qubit 4 fluorometer (Q33238, Thermo Fisher Scientific), Qubit RNA BR assay kit (Q10210, Thermo Fisher Scientific), Qubit RNA IQ assay kit (Q33222, Thermo Fisher Scientific), Qubit RNA IQ standards for calibration (Q33235, Thermo Fisher Scientific) and Qubit assay tubes (Q32856, Thermo Fisher Scientific) according to the manufacturer's protocols. Quantification of total protein was conducted using BCA Protein Assay Kit (23227, Thermo Fisher Scientific) and Multiskan Sky microplate spectrophotometer (51119700DP, Thermo Fisher Scientific) in accordance with the manufacturer's protocol.

Immunofluorescence examination

Upon the sectioning, vascular tissues were dried at room temperature for 30min, fixed and permeabilised in ice-cold acetone for 10min, incubated in 1% bovine serum albumin (Cat. No. A2153, Sigma-Aldrich) for 1h to block non-specific protein binding, stained with unconjugated mouse anti-human CD81 (M38 clone, NBP1-44861, 1:100, Novus Biologicals) and rabbit anti-human fibronectin (F14 clone, ab45688, 1:250, Abcam) primary antibodies and incubated at 4°C for 16h. Sections were further treated with pre-adsorbed donkey anti-mouse Alexa Fluor 488-conjugated (ab150109, 1:500, Abcam) and donkey anti-rabbit Alexa Fluor 555-conjugated secondary antibodies (ab150062, 1:500, Abcam) and incubated for 1h at room temperature. Between all steps, washing was performed thrice with PBS (pH 7.4, P4417, Sigma-Aldrich). Nuclei were counterstained with 41,6-diamidino-2-phenylindole (DAPI) for 30min at room temperature (10µg/mL, D9542, Sigma-Aldrich). Coverslips were mounted with ProLong Gold Antifade (P36934, Thermo Fisher Scientific). Sections were examined by confocal laser scanning microscopy (LSM 700, Carl Zeiss). Colocalization analysis (n=12 images) was performed using the respective ImageJ (National Institutes of Health) plugins (Colocalisation Threshold and Coloc2). To evaluate the colocalization, we calculated Pearson's r above threshold (zero-zero pixels), thresholded Mander's split colocalisation coefficient (the proportion of signal in each channel that colocalizes with the other channel) for both (red and green) channels, percent volume colocalised with each channel, and intensity volume above threshold colocalised with each channel in both neointima and media.

Reverse transcription-quantitative polymerase chain reaction (RT-qPCR)

Reverse transcription was carried out utilising High Capacity cDNA Reverse Transcription Kit (4368814, Thermo Fisher Scientific). Gene expression was measured by RT-qPCR using the customised primers (500nmol/L each, Evrogen, Moscow, Russian Federation, Table S5), cDNA (20ng) and PowerUp SYBR Green Master Mix (A25778, Thermo Fisher Scientific) according to the manufacturer's protocol for $T_m \geq 60^\circ\text{C}$ (fast cycling mode). Technical replicates (n=3 per each sample collected from one vascular segment) were performed in all RT-qPCR experiments. The reaction was considered successful if its efficiency was 90-105% and R^2 was ≥ 0.98 . Quantification of the CD9, CD63, CD81, COL6A3, EDIL3, CSPG4, TGFBI, FN1, and MYADM mRNA levels in carotid atherosclerotic plaques (n=5) and adjacent intact arterial segments (n = 5) was performed by using the $2^{-\Delta\Delta C_t}$ method. Relative transcript levels were expressed as a value relative to the average of 3 housekeeping genes (ACTB, GAPDH, B2M).

Western blotting

Equal amounts of protein lysate (15µg per sample) of carotid atherosclerotic plaques (n=12), adjacent intact arterial segments (n=12), and plaque-derived sEVs (n=6) were mixed with NuPAGE lithium dodecyl sulfate sample buffer (NP0007, Thermo Fisher Scientific) at a 4:1 ratio and NuPAGE sample reducing agent (NP0009, Thermo Fisher Scientific) at a 10:1 ratio, denatured at 99°C for 5 minutes, and then loaded on a 1.5mm NuPAGE 4-12% Bis-Tris protein gel (NP0335BOX, Thermo Fisher Scientific). The 1:1 mixture of Novex Sharp pre-stained protein standard (LC5800, Thermo Fisher Scientific) and MagicMark XP Western protein standard (LC5602, Thermo Fisher Scientific) was loaded as a molecular weight marker. Proteins were separated by the sodium dodecyl sulphate-polyacrylamide gel electrophoresis (SDS-PAGE) at 150V for 2h using NuPAGE 2-(N-morpholino)ethanesulfonic acid SDS running buffer (NP0002, Thermo Fisher Scientific), NuPAGE Antioxidant (NP0005, Thermo Fisher Scientific), and XCell SureLock Mini-Cell vertical mini-protein gel electrophoresis system (EI0001, Thermo Fisher Scientific). Protein transfer was performed using polyvinylidene difluoride (PVDF) transfer stacks (IB24001, Thermo Fisher Scientific) and iBlot 2 Gel Transfer Device (IB21001, Thermo Fisher Scientific) according to the manufacturer's protocols using a standard transfer mode for 30-150 kDa proteins (P0 – 20 V for 1min, 23 V for 4min, and 25 V for 2 min). PVDF membranes were then incubated in iBind Flex Solution (SLF2020, Thermo Fisher Scientific) for 1h to prevent non-specific binding.

Blots were probed with rabbit primary antibodies to CD9, fibronectin or CD81, and GAPDH (loading control). Horseradish peroxidase-conjugated goat anti-rabbit (7074, Cell Signaling Technology) or goat anti-mouse (AP130P, Sigma-Aldrich) secondary antibodies were used at 1:200 and 1:1000 dilution, respectively. Incubation with the antibodies was performed using iBind Flex Solution Kit (SLF2020, Thermo Fisher Scientific), iBind Flex Cards (SLF2010, Thermo Fisher Scientific) and iBind Flex Western Device (SLF2000, Thermo Fisher Scientific) during 3h according to the manufacturer's protocols. Chemiluminescent detection was performed using SuperSignal West Pico PLUS chemiluminescent substrate (34580, Thermo Fisher Scientific) and C-DiGit blot scanner (3600-00, LI-COR Biosciences) in a high-sensitivity mode (12min scanning). Densitometry was performed using the ImageJ software (National Institutes of Health) using the standard algorithm (consecutive selection and plotting of the lanes with the measurement of the peak area) and subsequent adjustment to the loading control (GAPDH).

Statistics

Data were analysed using GraphPad Prism (version 8.4.3). CD63-beads assay, NanoView, adhesion assay, FA quantification were analysed by one-way ANOVA test. Two group CD63-bead assay was analysed by non-paired T-test. 2D migration multiple comparison data and VSMC invasion multiple comparison data were analysed using non-parametric Kruskal-Wallis test. Two group invasion data were analysed using non-parametric Kolmogorov-Smirnov test. Clinical data for two groups was analysed by non-paired T-test and multiple comparison was analysed by Mann-Whitney U-test. All analysis was conducted using PRISM software (GraphPad, San Diego, CA). Values of $P < 0.05$ were considered statistically significant.

Supplementary Figures

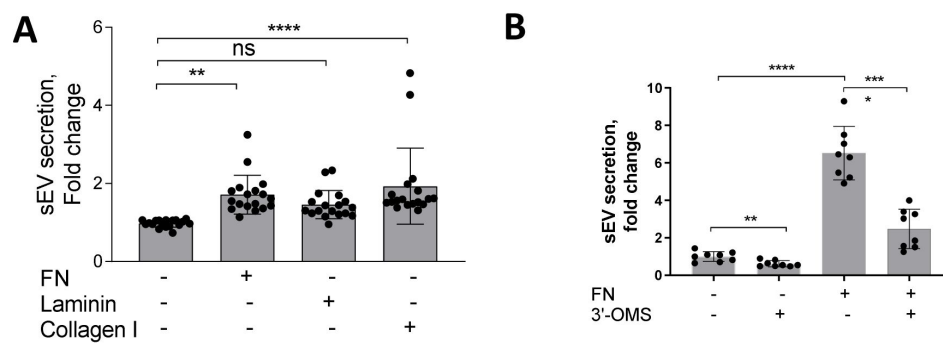


Figure S1.

FN matrix stimulates sEV secretion by VSMCs.

A, FN and collagen I but not laminin stimulate secretion of CD63-enriched sEVs. VSMCs were plated on the various matrices for 24h and sEV secretion was measured by CD63-beads assay. N=3, biological replicates, n=6 technical replicates. ANOVA **p<0.01 **B**, Inhibition of SMPD3 blocks sEV secretion by VSMC plated onto FN matrix. VSMCs were plated on non-coated or FN-coated plates for 24h and conditioned media was analysed by CD63-bead assay. N=2 biological replicates, n=4 technical replicates, ANOVA, ***, p<0.001, ****p<0.0001

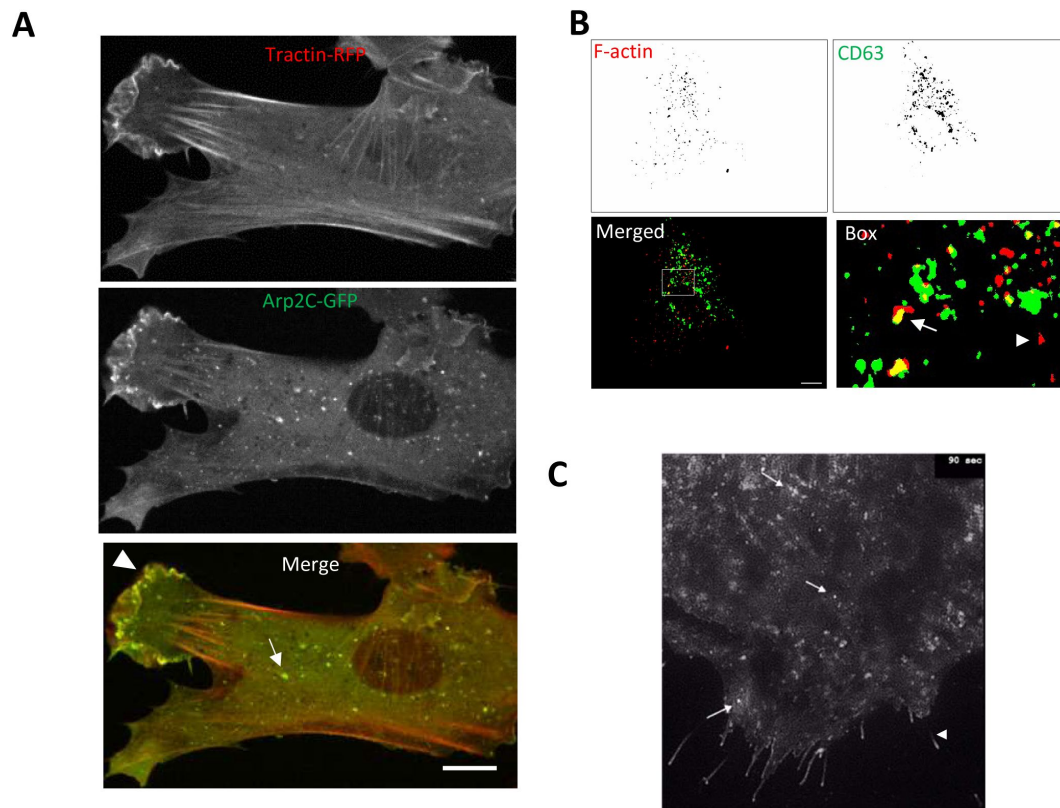


Figure S2.

A, Still images of a time-lapse showing that Arp2/3 and F-actin form tails in VSMC cytosol. VSMCs were co-transfected by ARPC2-GFP and F-tractin-RFP and cultured for 24h. Time-lapse video was captured using confocal spinning disk microscopy. Note, that Arp2/3 and F-actin are observed in lamellipodia but also detected in the cytosol with the unknown activity (*arrow*). Size bar, 10µm **B**, Analysis of F-actin/CD63 overlap using ImageJ in a fixed VSMCs. F-actin and CD63 images (**Fig 2D**) were converted to 8 bits and thresholded. Actin stress fiber structures were extracted from the image by using ParticleAnalysis plug-in (Size (pixel²) 10-infinity; Circularity 0.5-1) and overlap image created (merged). CD63 endosomes (pseudo coloured in green) were overlapped with F-actin spot-like structures (pseudo coloured in red). Note that endosomes partially overlapped with F-actin tail-like structures (arrows) though some F-actin spots and endosomes were not colocalised (arrowheads). **C**, sEV secretion detected with CD63-pHluorin. VSMCs were co-transfected with CD63-pHluorin and incubated for 24h. Time-lapse was captured using confocal spinning-disk microscopy. Arrows, typical “burst”-like appearance of sEV secretion at the cell-ECM interface. Arrows, an intense CD63-pHluorin staining along filopodia-like structures indicating that sEV release can occur in filopodia. Size bar, 10µm

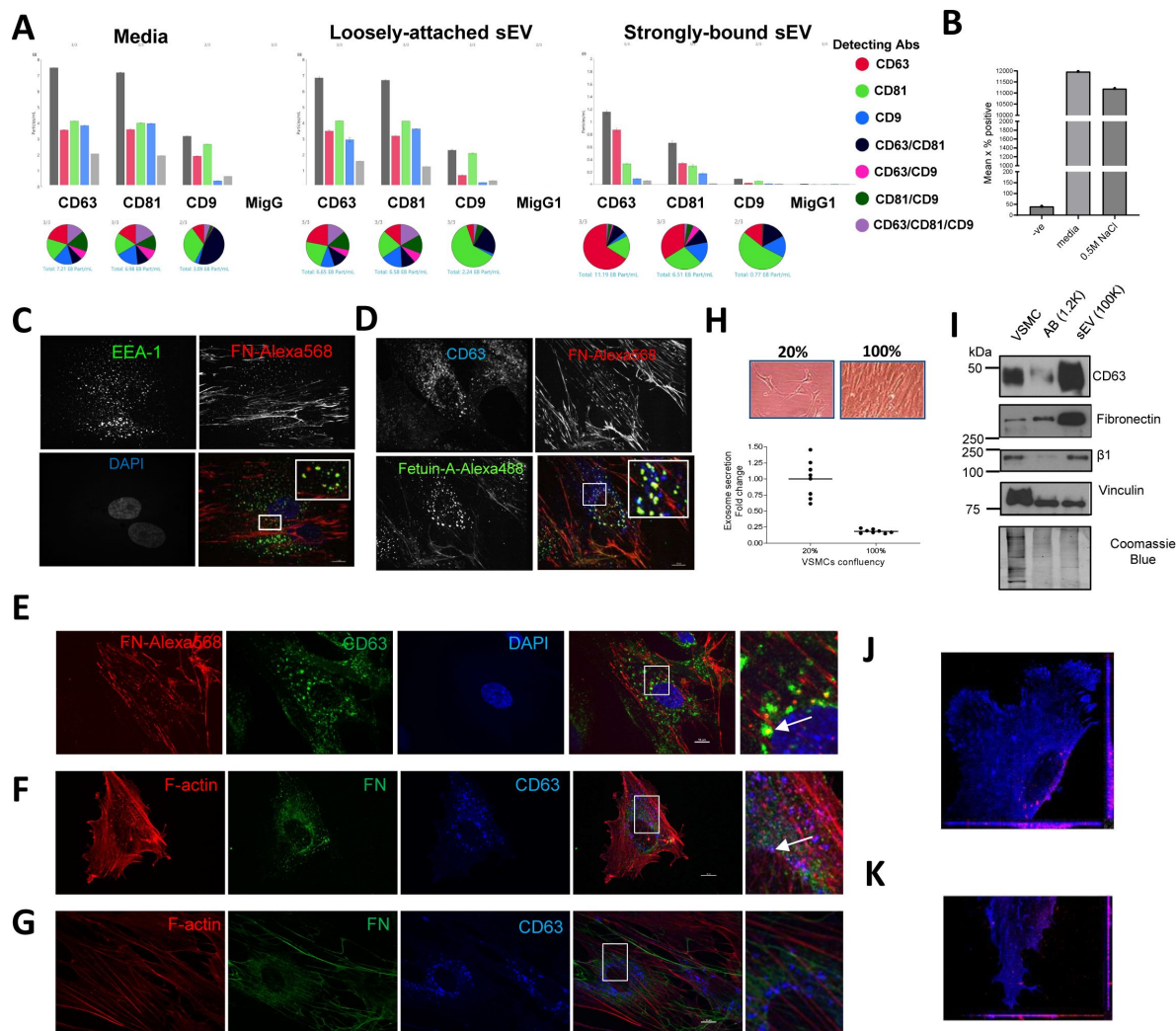


Figure S3.

SMPD3-dependent sEVs are trapped in ECM.

A, Characterisation of sEVs extracted from VSMC-derived media and ECMs fractions. sEVs were sequentially extracted and fractions were collected and analysed by using ExoView. X-axis – capturing spot antibodies (CD63, CD81, CD9 and control MigG1). **B**, CD63-bead capturing assay shows similar enrichment in CD63+/CD81+ sEVs in the conditioned media and ECM-associated sEVs extracted using 0.5M NaCl. VSMCs were plated onto gelatin-covered plates for 10 days and media was collected every 3d. sEVs were extracted in a mild (0.5M NaCl) conditions and media and 0.5M NaCl fractions were analysed by using CD63-beads capturing assay. **C, D**, Exogenously added FN-Alexa568 can be detected in the early endosomes and MVBs. VSMC were serum-deprived for 24h and then incubated with FN-Alexa568 for 30min (**C**) or 3h (**D**) in RPMI-0.5% BSA, thoroughly washed and fixed and stained for EEA-1 (**C**, green) or CD63 (**D**, blue). Size bar, 10µm, **E**, Exogenously added FN-Alexa568 can be detected in MVBs. VSMC were incubated with FN-Alexa568 for 3h and stained for CD63 which was visualised with anti-mouse-Alexa488 antibody. Size bar, 10µm, Note the partial colocalization of FN and CD63. **F, G**, FN can be detected in the CD63+ MVBs in a sparsely growing VSMCs. VSMC were plated at low (5,000 cells per well, **F**) or high density (20,000 cells per well, **G**) and were cultured for 24h in RPMI supplemented with 2.5% EV-free FBS, fixed and stained for F-actin, FN and CD63. **H**, VSMC plating density influences secretion of CD63+/CD81+ sEVs. Cells were plated at the different density and cultured for 24h and sEVs in conditioned media were measured by CD63-beads assay. N=1 biological replicates, n=8 technical replicates. **I**, FN is presented in sEV along with β1 integrin. **J**, VSMC were plated on the FN-coated dishes and Alexa568-labelled sEV were added to the cell media for 3h. Cells were fixed and stained for filopodia marker Myo10 (green) and vinculin (blue). Note perinuclear localisation of internalised sEVs. Size bar, 10µm. 3D projection. Myo10 staining channel is removed. **K**, VSMC were plated on the FN-coated dishes pre-coated with Alexa568-labelled sEV and incubated for 24h. Cell staining as in Fig S3J. Note even distribution of sEVs across the extracellular matrix and cell area.

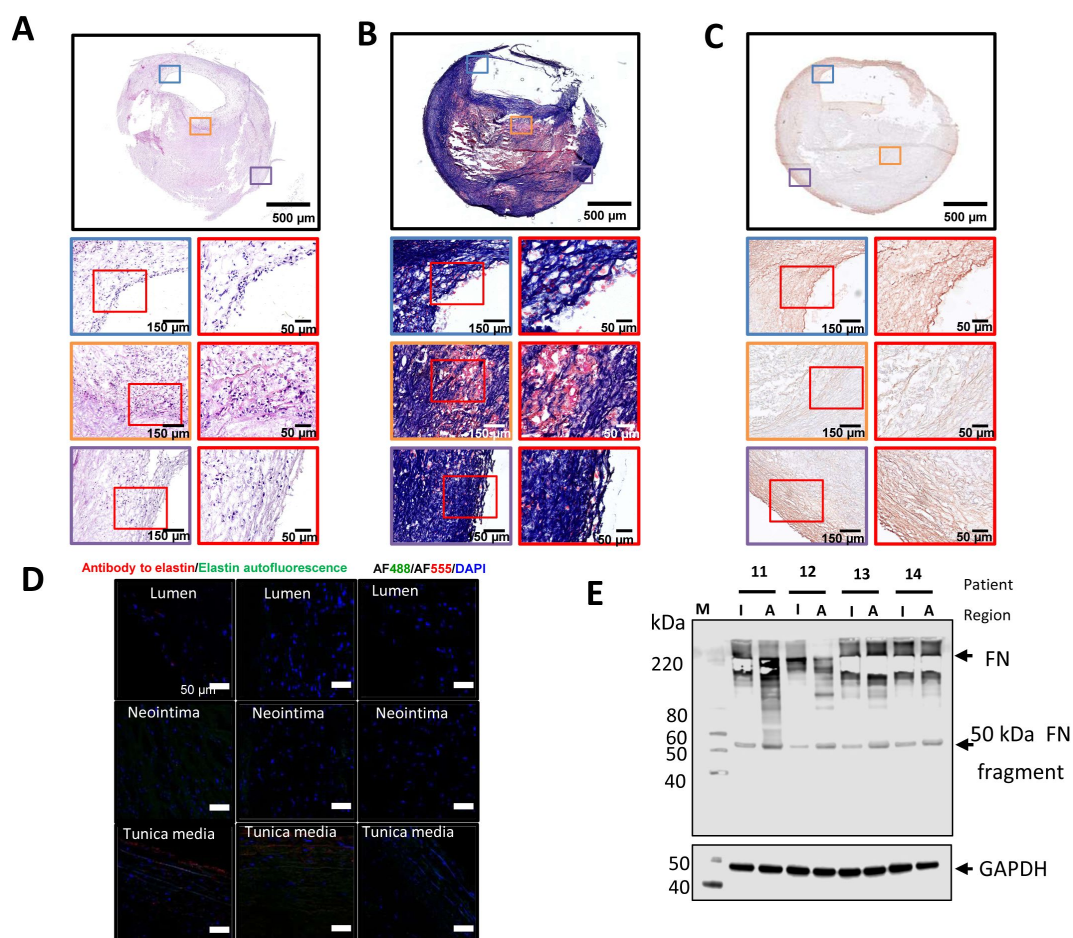
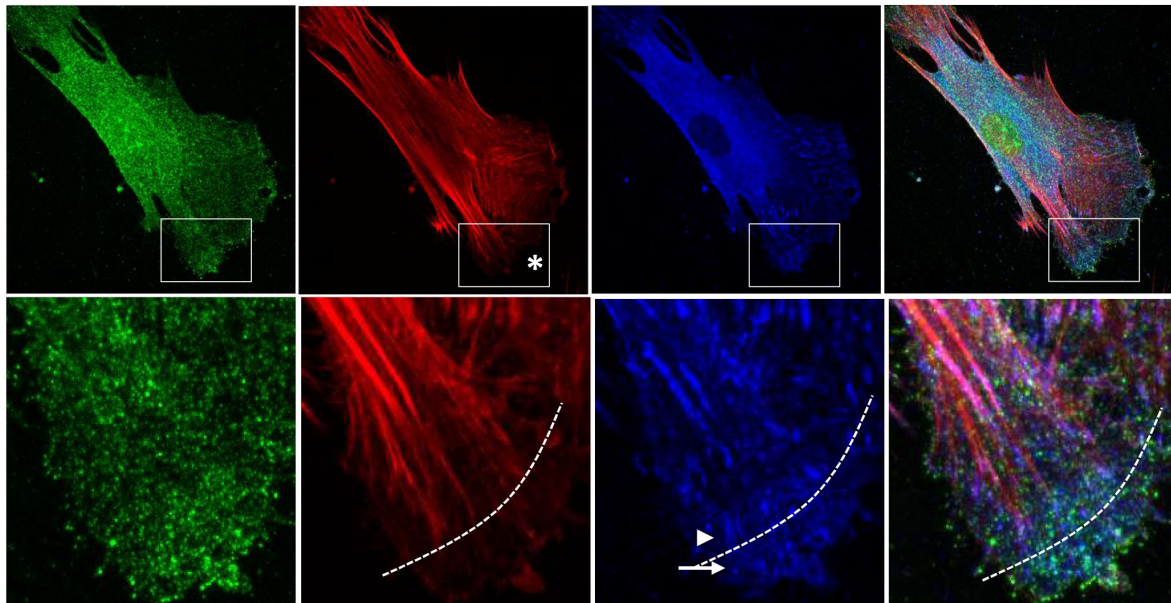


Figure S4.

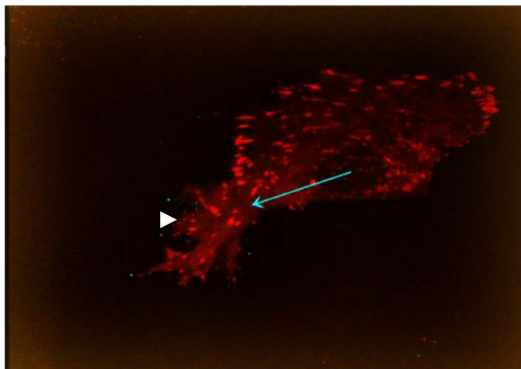
Fibronectin deposition in the atherosclerotic plaque is spatially associated with sEV marker CD81.

A. Staining of atherosclerotic plaques with haematoxylin and eosin. Unstable atheroma with an extracellular lipid or necrotic core and calcification (type IV according to the American Heart Association). Below are representative fields of view focusing on the lumen (blue contour), neointima (green contour), and tunica media (yellow contour). Red contours demarcate representative fields of view within these tissue compartments. Note multiple cells at the luminal side, numerous leukocytes in the neointima, and abundant cells in the tunica media. Overview images: magnification: $\times 1.8$, scale bar: 500 μ m; close-ups demarcated by blue, green, and gold contour: magnification: $\times 200$, scale bar, 150 μ m; close-ups demarcated by red contour: magnification: $\times 400$, scale bar: 50 μ m. **B.** Staining of atherosclerotic plaques with Masson's trichrome. Note the superficial erosions at the luminal side, abundant haemorrhages in the neointima, and sporadic haemorrhages in the tunica media. Overview images: magnification: $\times 1.8$, scale bar: 500 μ m; close-ups demarcated by blue, green, and gold contour: magnification: $\times 200$, scale bar: 150 μ m; close-ups demarcated by red contour: magnification: $\times 400$, scale bar, 50 μ m. **C.** Staining of atherosclerotic plaques with orcein. Note the moderate connective tissue staining at the luminal side, weak connective tissue staining in the atheroma with a confluent extracellular lipid core, and strong specific staining at the tunica media containing elastic fibres. Overview images: magnification: $\times 1.8$, scale bar: 500 μ m; close-ups demarcated by blue, green, and gold contour: magnification: $\times 200$, scale bar: 150 μ m; close-ups demarcated by red contour: magnification: $\times 400$, scale bar, 50 μ m. **D.** Demarcation of tunica media from the neointima using specific antibodies to elastin (red colour) and non-specific autofluorescence of elastic fibers (green colour). Nuclei are counterstained with 4',6-diamidino-2-phenylindole (DAPI, blue colour). Two representative images per each vascular compartment (luminal side, neointima, and tunica media) are provided. Negative controls (i.e., sections stained with species-specific pre-adsorbed fluorescent-labeled secondary antibodies and DAPI but without the respective antigen-specific primary antibody) are provided to the right from representative images. Magnification: $\times 200$, scale bar, 50 μ m. Note that elastin is expressed exclusively in the tunica media and is associated with elastic fibers. **E.** Expression of FN in atherosclerotic plaques. Atherosclerotic plaques (A) and adjacent intact arterial segments (I) were homogenised and analysed by Western blotting. FN is abundantly presented in both regions and 50 kDa FN fragment (Homandberg et al., 1992) intensity was used to quantify FN amount.

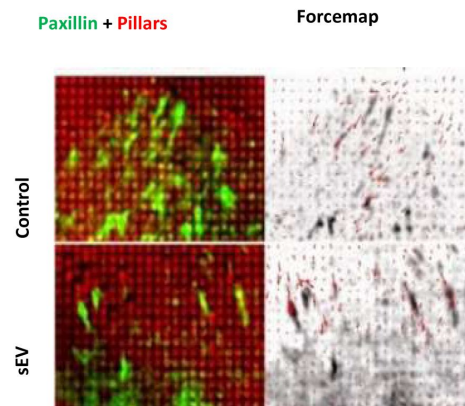
A



B



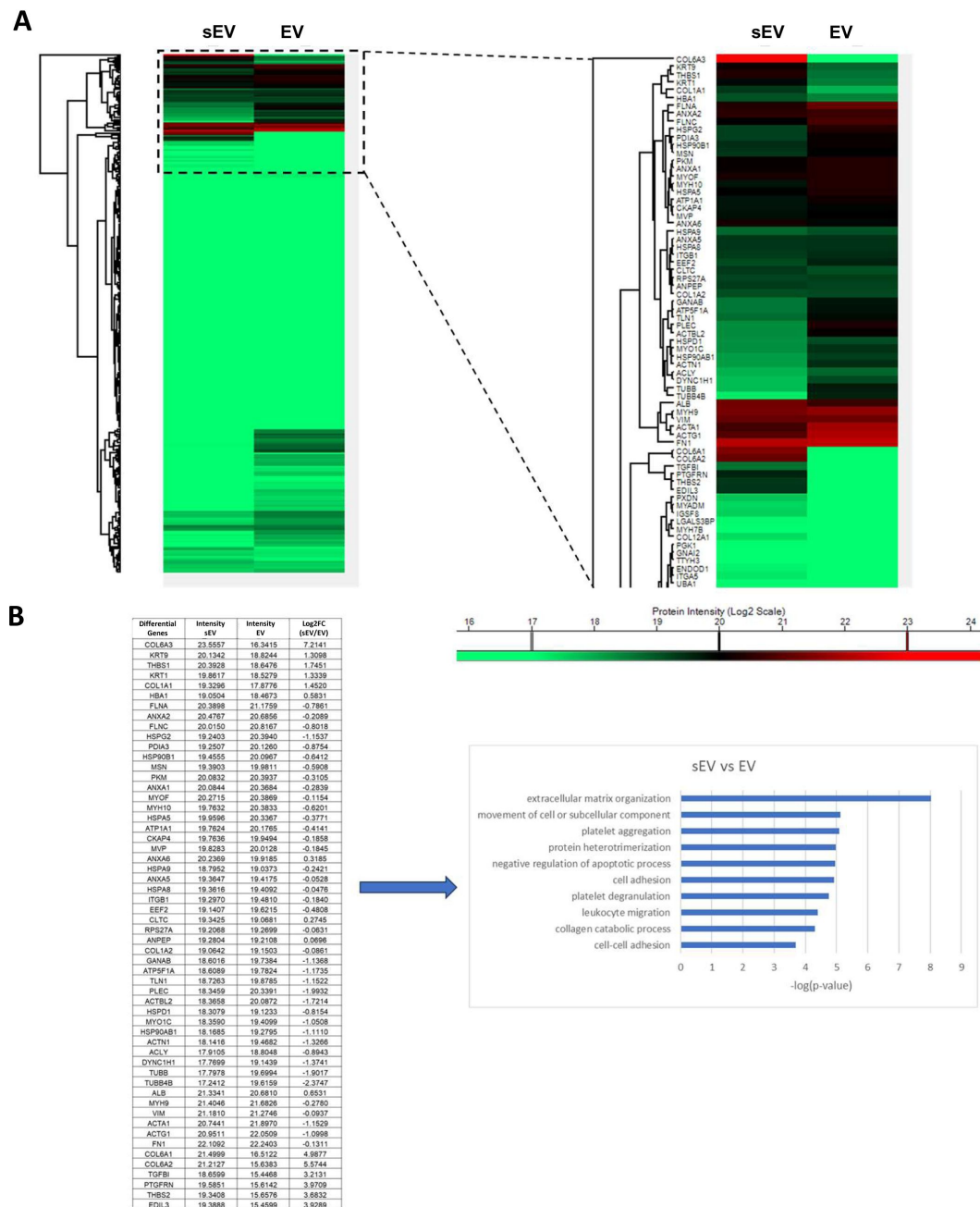
C



Supplementary Figure S5

sEVs regulate VSMC motility and invasion.

A, Centripetal FAs are linked to actin stress fibers. VSMCs were plated on FN-coated plate for 24h and cells were stained for Myo10 (green), CD63 (blue) and F-actin (phalloidin, red). Note the dot-like focal complexes in lamellipodium which are not associated with the contractile actin bundles (arrow) and an appearance of elongated FAs associated with the mature actin bundles (arrowhead). Dotted line, approximate position of the lamellipodium boundaries. Size bar, 10µm. **B**, Mature focal adhesion turnover is not affected by sEVs. VSMC were transfected with Paxillin-RFP and plated on the FN in the absence or presence of immobilised sEVs. Images were captured for 30min using confocal spinning disk microscopy. Note the appearance of the mature FAs in the lamellipodium (Arrowhead). Arrow, direction of the VSMC movement. Representative from N=4 biological replicates. **C**, sEV induces formation of FAs with the enhanced pulling force. VSMC transfected with Paxillin-RFP were plated on the PDMS pillars which were covered with FN and sEVs and pillar displacements were quantified. Representative image from N=2 biological replicates.



Supplementary Figure S6.

Proteomic analysis of VSMC-derived sEVs and EV.

VSMC-derived EVs and sEVs were isolated from cells and analyzed by protein mass-spectrometry. N=3 biological replicates. A. Clustered proteomic heatmap for EV and sEV. B. The GO term analysis was performed using the differentially expressed genes between the sEV and EV groups.

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Additional files

Supplementary Video S1 [↗](#)

Supplementary Video S2 [↗](#)

Supplementary Video S3 [↗](#)

Supplementary Video S4 [↗](#)

Table S1 [↗](#)

Table S2 [↗](#)

Table S3 [↗](#)

Table S4 [↗](#)

Table S5 [↗](#)

Table S6 [↗](#)

Table S7 [↗](#)

Table S8 [↗](#)

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Reviewer #1 (Public review):

In this revised submission from Kapustin et al., the authors have made significant changes to the manuscript. Namely, the authors have addressed several of the major issues with the original submission, providing a more concrete link between fibronectin and the secretion of extracellular vesicles. Additionally, the authors have moderated some of the conclusions to better suit the rigor of the experimental results and limitations of their approach. Generally, the findings convey an interesting cell autonomous pathway in which smooth muscle cells sense fibronectin, which canonically is a proinflammatory substrate with activating properties in many tissues. Fibronectin-mediated integrin signaling stimulates secretion of small extracellular vesicles containing collagen VI which is deposited into the surrounding extracellular matrix. Collagen VI itself gleaned from extracellular vesicle secretion seems to further alter smooth muscle cell morphodynamics. For this later finding, much of the mechanism behind collagen VI vesicle loading and secretion has yet to be worked out. The authors provide evidence of extracellular vesicles containing collagen VI trapped in fibronectin in atherosclerotic plaques providing a nice validation of their in vitro findings in a diseased human cohort. Some limitations do still exist in the manuscript in its current form such as the assessment of the vesicle origins, contents and their association with the actin cytoskeleton; however, the rigor and execution are much improved from the preceding version. Overall, the pathobiology underlying vascular smooth muscle remodeling in disease states is a critical area of research that warrants further exploration.

<https://doi.org/10.7554/eLife.90375.2.sa2>

Reviewer #2 (Public review):

The findings in the current manuscript are interesting and valuable contributions to the fields of vascular biology and extracellular vesicle-related mechanisms. They suggest a potential role for smooth muscle cell-derived extracellular vesicles in presenting Type VI collagen to cells to orchestrate their migration, with proposed relevance to aberrant smooth muscle cell movements in the progression of atherosclerotic lesions. A wide range of assays are utilized to test various aspects of this working model, with the resulting data being largely solid and supporting several of the interpretations articulated by the authors. The revised manuscript has adequately addressed key weaknesses.

The authors present data suggesting a working model in which vascular smooth muscle cells (vSMCs) are stimulated by fibronectin (FN) to generate small extracellular vesicles (sEVs) that

harbor Type VI Collagen (collagen VI). These collagen VI-associated sEVs are suggested to accumulate in the extracellular matrix (ECM) and influence cell migration and adhesion dynamics, potentially contributing to disease progression in atherosclerosis. Major strengths of this manuscript include robust imaging data and the inclusion of human-derived samples in their analysis. The authors also make a reasonable attempt to provide data to support the potential existence of these mechanistic connections, though some minor questions remain regarding data interpretation. The authors largely achieved their aims of finding evidence consistent with their interpretations, and they have presented logical support for their conclusions while acknowledging important limitations and caveats to their current study. This work will likely have a sustained impact on the field of sEV biology and potential intersections with vascular biology, including their methodology e.g., imaging approaches. As biologists continue to explore the role of sEVs in physiological and pathological processes, this work raises an interesting aspect that must be considered more broadly, and that is, what is the role of sEVs that are ECM-associated and not necessarily internalized by recipient cells? Are there discrete mechanisms that govern their role in maintaining and/or disrupting normal physiological processes? This manuscript makes an attempt to address these unresolved yet critical questions.

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Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public Review):

Summary

In this investigation Kapustin et al. demonstrate that vascular smooth muscle cells (VSMCs) exposed to the extracellular matrix fibronectin stimulates the release of small extracellular vesicles (sEVs). The authors provide experimental evidence that stimulation of the actin cytoskeleton boosts sEV secretion and posit that sEVs harbor both fibronectin and collagen IV protein themselves which also, in turn, alter cell migration parameters. It is well established that fibronectin is associated with increased cell migration and adherence; therefore, this association with VSMCs is not novel.

The reviewer is correct that FN has been associated with migration and adherence in previous studies. However we have extended these observations to show that the extracellular fibronectin matrix stimulates small extracellular vesicle (sEVs) secretion by modulating the actin cytoskeleton. We also showed that sEVs are trapped in the extracellular matrix and that by presenting collagen VI induce early focal adhesion formation, reduce excessive cellular spreading and guide cell invasion directionality through a 3D matrix. Hence, sEVs mediate cell-matrix cross talk and change cell behaviour in the context of fibronectin matrix. This is critically important for vasculature where regulated VSMC invasion is essential for repair with its deregulation leading to pathology.

The authors purport that sEV are largely born of filopodia origin; however, this data is not well executed and seems generally at odds with the presented data.

Our experimental data showed that CD63 MVs are associated with filopodia in fixed and live cells (Fig 2E, 2F and Video S1) and that inhibition of filopodia formation using the formin inhibitor, SMIFH2 reduced sEV secretion on FN (Fig 2B). However, we agree with the reviewer that further studies are required to connect sEV secretion to filopodia. To address

this we have provided further data analysis but also toned down our conclusions regarding this point: . Changes include:

(1) Title: Matrix-associated extracellular vesicles modulate smooth muscle cell adhesion and directionality by presenting collagen VI.

(2) Results, section title: 2. FN-induced sEV secretion is modulated by Arp2/3 and formin-dependent actin cytoskeleton remodelling

(3) Results, page 6 Line 27-44 and conclusion page 7, Ln 3 “Interestingly, CD63+ MVBs can be observed in filopodia-like structures suggesting that sEV secretion can also occur spatially via cellular protrusion-like filopodia but more studies are needed to confirm this hypothesis.”

(4) Discussion, page 12, line 19. “Curiously we observed CD63+ MVB transport toward the filopodia tips as well as inhibition of sEV-secretion with filopodia formation inhibitors suggesting that sEV secretion can be directly linked to filopodia but further studies are needed to define the contribution of this pathway to the overall sEV secretion by cells.”

Similarly, the effect of sEVs on parameters of cell migration has almost no magnitude of effect, making mechanism exploration somewhat nebulous.

VSMC are mesenchymal-type cells with a low migration rate and we agree that the changes in the motility are not of great magnitude even for the positive controls suggesting that this is a complex, multifactorial process for VSMCs. In our experiments we collected data from >5000 individual cells to measure the average speed and found that fibronectin matrix on its own increased VSMC speed from ~0.61 $\mu\text{m}/\text{min}$ to ~0.68 $\mu\text{m}/\text{min}$ (~12% raise) which was statistically significant (Fig 5A). Addition of a sEV inhibitor caused a modest but significant decrease in cellular speed. Interestingly, addition of ECM-associated sEVs did not influence cell speed in 2D or 3D assays. However in a 3D model we observed a 22% change in cell directionality (Fig 5G) and a 235% change in cell alignment index (FMI, Fig 5H) which we believe is very strong evidence that VSMC-derived sEVs are involved in a regulation of VSMC invasion directionality. These data are also in agreement with sEV effects in tumour cells (Sung et al., 2015) though this previous study did not identify the factor driving the directionality and we think our Collagen VI data extends significantly these previous observations.

Results, page 9: “Hence, ECM-associated sEVs have modest influence on VSMC speed but influence VSMC invasion directionality.”

Lastly, the proposed mechanism of VSMCs responding to, and depositing, ECM proteins via sEVs was not rigorously executed; again, making the conclusions challenging for the reader to interpret.

We appreciate the reviewer’s comment regarding the mechanistic aspects of VSMCs responding to and depositing ECM proteins via sEVs. In our revised manuscript, we have expanded the data demonstrating that sEVs can be retained within the extracellular matrix (see Figs 3A, 3B, S3A, S3B). Additionally, we show that collagen VI is present on the surface of sEVs, where it may modulate cell adhesion and influence the directionality of cell invasion (Fig 7E). Our results further indicate that both fibronectin (FN) and collagen VI can be recycled through multivesicular bodies (see Figs S3C, S3D, S3E–S3G). However, we acknowledge that the precise mechanisms governing the selective loading of ECM proteins onto sEVs, as well as the specific contributions of sEVs to overall ECM organization, remain to be fully elucidated and warrant further investigation. Based on our current evidence, we propose that collagen VI-loaded sEVs act primarily in a signaling capacity by modulating focal adhesion formation but are not directly involved in ECM structural remodeling.

Results, page 7: To quantify ECM-trapped sEVs we applied a modified protocol for the sequential extraction of extracellular proteins using salt buffer (0.5M NaCl) to release sEVs which are loosely-attached to ECM via ionic interactions, followed by 4M guanidine HCl buffer (GuHCl) treatment to solubilize strongly-bound sEVs (Fig S3A) [42]. We quantified total sEV and characterised the sEV tetraspanin profile in conditioned media, and the 0.5M NaCl and GuHCl fractions using ExoView. The total particle count showed that EVs are both loosely bound and strongly trapped within the ECM. sEV tetraspanin profiling showed differences between these 3 EV populations. While there was close similarity between the conditioned media and the 0.5M NaCl fraction with high abundance of CD63+/CD81+ sEVs as well as CD63+/CD81+/CD9+ in both fractions (Fig S3A). In contrast, the GuHCl fraction was particularly enriched with CD63+ and CD63+/CD81+ sEVs with very low abundance of CD9+ EVs (Fig S3A). The abundance of CD63+/CD81+ sEVs was confirmed independently by a CD63+ bead capture assay in the media and loosely bound fractions (Fig S3B).

Results, page 7: We previously found that the serum protein prothrombin binds to the sEV surface both in the media and MVB lumen showing it is recycled in sEVs and catalyses thrombogenesis being on the sEV surface⁴³. So we investigated whether FN can also be associated with sEV surface where it can be directly involved in sEV-cell cross-talk⁴³. We treated serum-deprived primary human aortic VSMCs with FN-Alexa568 and found that it was endocytosed and subsequently delivered to early and late endosomes together with fetuin A, another abundant serum protein that is a recycled sEV cargo and elevated in plaques (Figs S3C and S3D). CD63 visualisation with a different fluorophore (Alexa488) confirmed FN colocalization with CD63+ MVBs (Fig S3E). Next, we stained non-serum deprived VSMC cultured in normal growth media (RPMI supplemented with 20% FBS) with an anti-FN antibody and observed colocalization of CD63 and serum-derived FN. Co-localisation was reduced likely due to competitive bulk protein uptake by non-deprived cells (Fig S3F). Notably, when we compared FN distribution in sparsely growing VSMCs versus confluent cells we found that FN intracellular spots, as well as colocalization with CD63, completely disappeared in the confluent state (Fig S3F and S3G). This correlated with nearly complete loss of CD63+/CD81+ sEV secretion by the confluent cells indicating that confluence abrogates intracellular FN trafficking as well as sEV secretion by VSMCs (Fig S3H). Finally, FN could be co-purified with sEVs from VSMC conditioned media (Fig S3I) and detected on the surface of sEVs by flow cytometry confirming its loading and secretion via sEVs (Fig 3C).

Results: page 10 Collagen VI was the most abundant protein in VSMC-derived sEVs (Fig 7B, Table S7) and was previously implicated in the interaction with the proteoglycan NG2[53] and suppression of cell spreading on FN[54]. To confirm the presence of collagen VI in ECM-associated sEVs we analysed sEVs extracted from the 3D matrix using 0.5M NaCl treatment and showed that both collagen VI and FN are present (Fig 7D). Next, we analysed the distribution of collagen VI using dot-blot. Alix staining was bright only upon permeabilization of sEV indicating that it is preferentially a luminal protein (Fig 7E). On the contrary, CD63 staining was similar in both conditions showing that it is surface protein (Fig 7E). Interestingly, collagen VI staining revealed that 40% of the protein is located on the outside surface with 60% in the sEV lumen (Fig 7E).

Discussion page 12. “In fact, we observed that an extensive secretion of sEVs effectively ceased protrusion activity; also VSMCs acquired a rounded morphology when “hovering” over the FN matrix decorated with sEVs (data not shown). Hence, it will be interesting in future studies to investigate whether sEVs can stimulate Rho activity by presenting adhesion modulators—particularly collagen VI—on their surface, thereby guiding cell directionality during invasion..”

Discussion, page 14 “In summary, cooperative activation of integrin signalling and F-actin cytoskeleton pathways results in the secretion of sEVs which associate with the ECM and play

a signalling role by controlling FA formation and cell-ECM crosstalk. Further studies are needed to test these mechanisms across various cell types and ECM matrices.

Strengths

The authors provide a comprehensive battery of cytoskeletal experiments to test how fibronectin and sEVs impact both sEV release and vascular smooth muscle cell migratory activation.

We appreciate this comment reflecting our efforts to apply a range of orthogonal methods to show the role of the integrin/actin cytoskeleton in ECM-stimulated sEV secretion.

Weaknesses

Unfortunately, this article suffers from many weaknesses. First, the rigor of the experimental approach is low, which calls into question the merit of the conclusions. In this vein, there is a lack of proper controls or inclusion of experiments addressing alternative explanations for the phenotype or lack thereof.

We acknowledge this comment and agree that there was not sufficient evidence to conclude that sEV secretion occurs via filopodia despite the microscopy/inhibitory data so this claim has now been excluded from the study. However we believe that our experimental data does clearly show that FN stimulates the secretion of collagenVI-loaded sEVs which are trapped by the ECM and have the capacity to modulate VSMC adhesion and invasion directionality. To support this, we have now extended the dataset in the revised version:

- (1) In addition to the use of inhibitors and live cell analysis we have added quantitative data confirming that a large proportion of CD63+ endosomes are associated with F-actin/cortactin tails and this colocalization is increased upon the inhibition of sEV secretion with 3-OMS (Fig 2D, Fig S2B).
- (2) We developed a method to extract ECM-associated sEVs and quantified/characterized these using ExoView Assays further confirming significant sEV entrapment by the ECM (Figs 3B, S3A, S3B).
- (3) We extended the controls to confirm FN delivery to CD63+ endosomes and showed that FN recycling is stopped upon reaching cell confluence (Figs S3F, S3G and Fig S3H).
- (4) We included more intensive characterisation of human atherosclerotic plaque morphology (H&E, Masson's trichrome staining, Orcein, elastin fibers staining) to confirm predominant accumulation of sEV in the neointima (Figs S4A, S4B and S4C). We also excluded an endothelial origin for the CD81+ sEVs (Fig 4G).
- (5) We included individual cellular tracks to the 2D migration analysis to confirm the statistical significance and concluded that ECM-associated sEVs regulate cell invasion directionality but not the cell speed (Figs 5A and 5B).
- (6) We showed surface localisation of collagen VI on sEVs confirming that it can activate signalling pathways leading to early FA formation on the FN matrix (Figs 7D and 7E).
- (7) We included alternative explanations for some of our data in the discussion.

Reviewer #2 (Public Review):

Extracellular vesicles have recently gained significant attention across a wide variety of fields, and they have therefore been implicated in numerous physiological and pathophysiological processes. When such a discovery and an explosion of interest occur

in science, there is often much excitement and hope for answers to mechanisms that have remained elusive and poorly understood. Unfortunately, there is an equal amount of hype and overstatement that may also be put forth in the name of "impact", but this temptation must be avoided so that scientists and the broader public are not misled by overreaching interpretations and statements that lack rigorous and fully convincing evidence.

Thank you for your comment and we agree that investigating sEVs is particularly challenging due to their heterogeneity and nano-size, as well as complex biogenesis mechanisms. ECM-associated sEVs is a very new direction for the EV field but one that is particularly relevant to the vasculature where cells must invade through a thick ECM and where the accumulation of ECM-bound EVs is a unique and documented phenomenon. To further strengthen our conclusions we have included new data to support our statements but also excluded statements re: filopodia as the origin of sEVs, that are out of scope of our study and need to be investigated further.

The study presented by Kapustin et al. is certainly intriguing and timely, and it offers an interesting working hypothesis for the fields of extracellular vesicles and vascular biology to consider. The authors do a reasonable job at detecting these small extracellular vesicles, though some aspects of data presentation are missing such as full Western blots with accompanying size markers for the viewer to more fully appreciate that data and comparisons being made (see Figures 1 and 7).

We agree with the reviewer and have now included molecular weight markers (Fig 1F, 7C, 7D, S3I, S4E) and provided all original western blot scans (uncropped and unedited) to the eLife editor.

Much of the imaging data from cell-based experiments is strong and conducted with many cutting-edge tools and approaches. That said, the static images and the dynamic imaging fall short of being fully convincing that the small extracellular vesicles found in the neighboring extracellular matrix are indeed being deposited there via the smooth muscle cell filopodia. Many of the lines of evidence presented suggest that this could occur, but alternative hypotheses also exist that were not fully ruled out, such as the ECM-deposited vesicles were secreted more from the soma and/or the lamellipodia that are also emitted and retracted from the cells. In particular, the authors show very nice dynamic imaging (Supplementary Figure S2A and Supplemental Video S1) that is interpreted as "extracellular vesicles being released from the cell" and these are seen as "bursts" of fluorescent signal; however, none of these appear to occur in filopodia as they appear within the cell proper (a "burst" of signal vs. a more intense "streak" of signal), which would be a stronger and more consistent observation predicted by the working model proposed by the authors.

Our live and fixed cell microscope data as well as inhibitor analysis showed that sEV secretion can be associated with the filopodia. However we agree with the reviewer that the data generated using pHluoron GFP marker clearly indicate that the majority of sEVs are secreted from the cell soma toward the ECM:

To reflect this, we have added further changes:

- (1) Title: Matrix-associated extracellular vesicles modulate smooth muscle cell adhesion and directionality by presenting collagen VI.
- (2) Results, section title: 2. FN-induced sEV secretion is modulated by Arp2/3 and formin-dependent actin cytoskeleton remodelling

(3) Results, page 6 Line 27-36 “Formins and the Arp2/3 complex play a crucial role in the formation of filopodia, a cellular protrusion required for sensing the extracellular environment and cell-ECM interactions³⁶. To test whether MVBs can be delivered to filopodia, we stained VSMCs for Myosin-10 (Myo10)³⁷. We observed no difference between total filopodia number per cell on plastic or FN matrices ($n=18\pm8$ and $n=14\pm3$, respectively) however the presence of endogenous CD63+ MVBs along the Myo10-positive filopodia were observed in both conditions (Fig 2E, arrows). Filopodia have been implicated in sEV capture and delivery to endocytosis “hot-spots”³⁸, so next we examined the directionality of CD63+ MVB movement in filopodia by overexpressing Myo10-GFP and CD63-RFP in live VSMCs. Importantly, we observed anterograde MVB transport toward the filopodia tip (Fig 2F and Supplementary Video S2) indicative of MVB secretion”.

(4) Results, page 6, Ln 37-44 “We also attempted to visualise sEV release in filopodia using CD63-pHluorin where fluorescence is only observed upon the fusion of MVBs with the plasma membrane³⁹. Using total internal reflection fluorescence microscopy (TIRF) we observed the typical “burst”-like appearance of sEV secretion at the cell-ECM interface in full agreement with an earlier report showing MVB recruitment to invadopodia-like structures in tumor cells¹⁸ (Fig S2B and Supplementary Video S1). Although we also observed an intense CD63-pHluorin staining along filopodia-like structures we were not able to detect typical “burst”-like events to confirm sEV secretion in filopodia. (Fig S2C and Supplemental Video S1)”.

(5) Results, page 7 Ln 3 “Interestingly, CD63+ MVBs can be observed in filopodia-like structures suggesting that sEV secretion can also occur spatially via cellular protrusion-like filopodia but more studies are needed to confirm this hypothesis.”

(6) Discussion, page 12, line 19. “Curiously we observed CD63+ MVB transport toward the filopodia tips as well as inhibition of sEV-secretion with filopodia formation inhibitors suggesting that sEV secretion can be directly linked to filopodia but further studies are needed to define the contribution of this pathway to the overall sEV secretion by cells.”

Imaging of related human samples is certainly a strength of the paper, and the authors are commended for attempting to connect the findings from their cell culture experiments to an important clinical scenario. However, the marker selected for marking extracellular vesicles is CD81, which has been described as present on the endothelium of atherosclerotic plaques with a proposed role in the recruitment of monocytes into diseased arteries (Rohlena et al. Cardiovasc Res 2009). More data should address this potentially confounding interpretation of the signals presented in images within Figure 4.

We thank the reviewer for this insightful comment that the sEV marker CD81 can originate from endothelial cells in agreement with Rohlena et al., 2009. To address this we investigated the spatial overlap between CD81 and the endothelial marker, CD31. We observed very strong CD81 staining in the intact endothelial cell (intima) layer and occasional CD31 positive cells in the neointima. Importantly, quantification of colocalization confirmed that 80% of CD81 in the neointima does not overlap with CD31 excluding an endothelial origin of these sEVs. (Fig 4G). Moreover, we included complete morphological characterisation of the atherosclerotic plaques confirming that CD81 sEVs were primarily observed in the neointima where VSMCs constitute the cellular majority (Fig S4A, S4B, S4C and S4D).

On a conceptual level, the idea that the small extracellular vesicles contain Type VI Collagen, and this element of their cargo is modulating smooth muscle cell migration, is an intriguing aspect of the authors' working model. Nevertheless, the evidence supporting this potential mechanism does not quite fit together as presented. It is not entirely clear how the collagen VI within the vesicles is somehow accessed by the smooth muscle cell filopodia during migration. Are the vesicles lysed open once on the

extracellular matrix? If so, what is the proposed mechanism for that to occur? If not, how are the adhesion molecules on the smooth muscle cell surface engaging the collagen VI fibers that are contained within the vesicles? This aspect of the model does not quite fit together with the proposed mechanism and may be an interesting speculative interpretation, warranting further investigation, but it should not be considered a strong conclusion with sufficient convincing data supporting this idea.

We thank the reviewer for their insightful comments regarding the mechanism by which collagen VI associated with sEVs could modulate smooth muscle cell adhesion and migration. To clarify, our new data suggest that collagen VI is predominantly present on the surface of the sEVs, as evidenced by Fig 7E. This surface localization strongly implies that collagen VI can be directly accessed by cell surface adhesion receptors, without the need for vesicle lysis or opening. While we cannot entirely rule out all alternative mechanisms, we consider vesicle rupture or lysis within the extracellular matrix to be a highly unlikely route for collagen VI exposure, given the known stability of sEVs under physiological conditions. We have added these points to clarify:

(1) Results, page 10, Ln 45 “To confirm the presence of collagen VI in ECM-associated sEVs we analysed sEVs extracted from the 3D matrix using 0.5M NaCl treatment and showed that both collagen VI and FN are present (Fig 7D). Next, we analysed the distribution of collagen VI using dot-blot. Alix staining was bright only upon permeabilization of sEV indicating that it is preferentially a luminal protein (Fig 7E). On the contrary, CD63 staining was similar in both conditions showing that it is surface protein (Fig 7E). Interestingly, collagen VI staining revealed that 40% of the protein is located on the outside surface with 60% in the sEV lumen (Fig 7E).”

(2) Discussion, page 13, Ln 2 “Hence, it will be interesting in future studies to investigate whether sEVs can stimulate Rho activity by presenting adhesion modulators—particularly collagen VI—on their surface, thereby guiding cell directionality during invasion..”

(3) Discussion, page 14, Ln 30: In addition to collagen VI the unique adhesion cluster in VSMC-derived sEVs also includes EGF-like repeat and discoidin I-like domain-containing protein (EDIL3), transforming growth factor-beta-induced protein ig-h3 (TGFB1) and the lectin galactoside-binding soluble 3 binding protein (LGALS3BP) and these proteins are also directly implicated in activation of integrin signalling and cellular invasiveness⁸⁵⁻⁸⁷. Although we found that collagen VI plays the key role in sEV-induced early formation of FAs in VSMCs, it is tempting to speculate that the high sEV efficacy in stimulating FA formation is driven by cooperative action of this unique adhesion complex on the sEVs surface and targeting this novel sEV-dependent mechanism of VSMC invasion may open-up new therapeutic opportunities to modulate atherosclerotic plaque development or even to prevent undesired VSMC motility in restenosis. .

(4) Abstract Figure

On a technical level, some of the statistical analysis is not readily understood from the data presented. It is very much appreciated that the authors show many of the graphs with technical and biological replicate values in addition to the means and standard deviations (though this is not clearly stated in all figure legends). However, in figures such as Figure 5, there are bars shown and indicated to be different by statistical comparison (see panel B in Figure 5). It is not clear how the values for Group 1 (no FN, no 3-OMS, no sEV) are statistically different (denoted by three asterisks but no p value provided in the legend) than Group 3 (no FN, 3-OMS added, no sEV), when their means and standard deviations appear almost identical. If this is an oversight, this needs to be corrected. If this is truly the outcome, further explanation is warranted. A higher level of transparency in such instances would certainly go a long way in helping address the

current crisis of mistrust within the scientific community and at the interface with society at-large.

We thank the reviewer for their careful reading and important comments on the statistical analysis. We acknowledge that the technical and biological replicate data were not clearly reported in all figure legends and that the statistical approach for Figures 5A and 5B required clarification. In response, we have made several changes for greater transparency and rigor:

First, we have now explicitly included the numbers of biological replicates (N) and technical replicates (n) in all relevant figure legends for Figures 1–7. In addition, the number of individual cell tracks is now annotated for the migration/invasion analyses, along with the mean values for each dataset.

Upon review, we found that the original statistical analyses for Figures 5A and 5B were conducted using pooled averaged data. To address this, we have repeated the statistical tests using pooled individual cell track data, applying the Kruskal–Wallis test with Dunn’s multiple comparison correction. This more stringent approach revealed revised p-values, which are now indicated in Figures 5A and 5B.

With these corrections, we reconfirm our major findings: In the 2D model, fibronectin (FN) coating promotes VSMC velocity, while inhibition of sEV secretion with 3-OMS leads to reduced cell speed (Fig. 5A). Addition of sEVs to the ECM had no effect on VSMC speed at baseline but did rescue cell speed and distance in the presence of 3-OMS, consistent with EVs acting primarily on invasion directionality rather than speed in both 2D and 3D models (Fig. 5A, 5D). Furthermore, sEVs continue to significantly impact VSMC invasion directionality (Figs. 5G, 5H), in agreement with previous reports in tumor cells (Sung et al., 2015).

In summary, we have implemented the following revisions:

- (1) Figures 5A and 5B: Individual cell track data are now shown, and statistical analyses have been repeated using the Kruskal–Wallis test with Dunn’s multiple comparisons.
- (2) Figure legends and results sections: Numbers of biological and technical replicates, as well as individual data points, are now clearly stated.

Results, page 9, line 14: The text has been updated to clarify the statistical approach and major findings as described above.

We hope that these changes address the reviewer’s concerns and improve the transparency and reproducibility of our data presentation

Reviewer #1 (Recommendations For The Authors):

We are very thankful for the comprehensive review and comments which helped to improve our data.

Figure 1.

The authors clearly show that FN stimulation (immobilized or cell-derived) promotes sEV secretion via canonical integrin pathways. FN is a promigratory substrate, hence its extensive use as a cell adhesion aid; thus one could assume that simply plating on FN induces a pro-migratory phenotype (later data supports this notion). Does the addition of growth factors also increase sEV release? An endogenous function of FN is silencing of various GFs during clot formation. Also, FAK and SRC networks intersect with canonical RTK signaling in terms of promoting Rac1, CDC42 and other migration mediators. The reason I believe this is important is because the data could be interpreted in two ways: 1) FN induces pro-migration signaling and then sEVs are released, or visa versa, FN induces

sEV release and migration is initiated. GF supplementation in the absence of FN would clarify this relationship.

We thank the reviewer for this insightful comment regarding the possible role of growth factors (GFs) and the mechanistic relationship between FN stimulation, sEV secretion, and cell migration. We agree that FN is a well-established promoter of cell migration, and it is important to distinguish whether FN directly induces a pro-migratory phenotype or does so via sEV-mediated signaling.

Our data show that FN stimulation markedly increases VSMC motility, as reflected by enhanced cell speed (Fig. 5A), an increased number of focal adhesions (Fig. 6E), and facilitated centripetal movement of FAs (Fig. 6F). Interestingly, ECM-associated sEVs appear to play a complementary but distinct role: they do not significantly affect cell migration speed (Fig. 5A) but instead guide cell invasion directionality (Figs. 5G, 5H), reduce the number of FAs per cell (Fig. 6E), and promote early peripheral FA formation (Fig. 6F). In light of these findings, we have updated our graphical abstract to reflect the unique cross-talk mediated by sEVs between VSMCs and the ECM.

Regarding the influence of growth factors, we acknowledge that FN can bind and present different GFs, which could also contribute to changes in sEV secretion. Although our inhibition studies and integrin-blocking antibody results support a primary role for $\beta 1$ integrin activation and actin assembly in triggering sEV secretion, we cannot entirely exclude the possibility that FN-bound growth factors play a role in this process. We have now incorporated this point into the discussion to address the reviewer's suggestion.

Discussion, page 14, Ln 7 "Although our small inhibitors and integrin modulating antibody data clearly indicate that $\beta 1$ activation triggers sEV secretion via activation of actin assembly we cannot fully rule out that FN may also be modulating growth factor activity which in turn contributes to sEV secretion by VSMCs²³. Excessive collagen and elastin matrix breakdown in atheroma has been tightly linked to acute coronary events hence it will be interesting to study the possible link between sEV secretion and plaque stability as sEV-dependent invasion is also likely to influence the necessary ECM degradation induced by invading cells⁹⁶

Figure 2.

• The authors provide no evidence (or references) that SMIFH2 or CK666 halts filopodia extensions.

Thank you for this important note. We have included the corresponding references:

Results, page 5: "So next we tested the contribution of Arp2/3 and formins by using the small molecule inhibitors, CK666 and SMIFH2, respectively^{31, 32}".

• Is there an increase in filopodia density when plated on FN vs plastic? Similarly, if there are more filopodia present is that associated with more sEV? Please provide evidence in this regard.

We agree that connecting the number of filopodia with the secretion level of sEVs may be an important clue if sEV secretion can be driven by FN-induced filopodia formation. However, Myosin10 staining to quantify filopodia (Fig 2E) showed no difference between VSMCs plated on plastic versus FN matrix. Therefore, we agree with the reviewer that the filopodia contribution to sEV secretion needs to be investigated further. This idea is reflected in the following comments:

(1) Results, page 6, Ln 29 "We observed no difference between total filopodia number per cell on plastic or FN matrices ($n=18\pm 8$ and $n=14\pm 3$, respectively) however the presence of

endogenous CD63+ MVBs along the Myo10-positive filopodia were observed in both conditions (Fig 2E, arrows).

(2) Results, page 6, Ln 37 “We also attempted to visualise sEV release in filopodia using CD63-pHluorin where fluorescence is only observed upon the fusion of MVBs with the plasma membrane³⁹. Using total internal reflection fluorescence microscopy (TIRF) we observed the typical “burst”-like appearance of sEV secretion at the cell-ECM interface in full agreement with an earlier report showing MVB recruitment to invadopodia-like structures in tumor cells¹⁸ (Fig S2B and Supplementary Video S1). Although we also observed an intense CD63-pHluorin staining along filopodia-like structures we were not able to detect typical “burst”-like events to confirm sEV secretion in filopodia. (Fig S2C and Supplemental Video S1).”

(3) Discussion, page 12, Ln 15 : “Focal complexes either disassemble or mature into the elongated centripetally located FAs⁴⁸. In turn, these mature FAs anchor the ECM to actin stress fibres and the traction force generated by actomyosin-mediated contractility pulls the FAs rearward and the cell body forward^{12, 13}. Here we report that $\beta 1$ integrin activation triggers sEV release followed by sEV entrapment by the ECM. Curiously we observed CD63+ MVB transport toward the filopodia tips as well as inhibition of sEV-secretion with filopodia formation inhibitors suggesting that sEV secretion can be directly linked to filopodia but further studies are needed to define the contribution of this pathway to the overall sEV secretion by cells..”

As hinted above, this data could be interpreted in the light of generally inhibiting cell migration to blunt sEV shedding. Does cell confluence affect sEV release? If cells are cultured to 100% confluency this would limit filopodia formation regardless of ECM type. If sEV secretion remains elevated on FN in this culture condition it would suggest a lack of dependency on filopodia.

We thank the reviewer for this thoughtful suggestion regarding the influence of cell confluence on sEV release and filopodia formation. To directly address this hypothesis, we performed additional experiments comparing VSMCs cultured at low and high confluency. As described in the revised Results (page 7, line 39), we found that high cellular confluency reduced FN recycling, as indicated by the marked decrease in intracellular FN-positive spots and loss of colocalization with CD63 (Figs S3F, S3G). Importantly, this was accompanied by a significant reduction in CD63+/CD81+ sEV secretion by confluent cells (Fig S3H). These results suggest that VSMC confluence, which suppresses filopodia formation, nearly abolishes both intracellular FN trafficking and sEV secretion, even in the presence of FN. Thus, under our experimental conditions, sEV secretion by VSMCs appears to be closely linked to dynamic cell–matrix interactions and is dramatically reduced when these processes are limited by confluence:

(1) Results, page 7, Ln 39 : “Notably, when we compared FN distribution in sparsely growing VSMCs versus confluent cells we found that FN intracellular spots, as well as colocalization with CD63, completely disappeared in the confluent state (Fig S3F and S3G). This correlated with nearly complete loss of CD63+/CD81+ sEV secretion by the confluent cells indicating that confluence abrogates intracellular FN trafficking as well as sEV secretion by VSMCs (Fig S3H)..”

• Inhibition of branched actin polymerization has been shown to reduce both exocytic and endocytic activity. Thus, it is hard to interpret the results of Fig. 2B than anything more than a generalized effect of losing actin.

We thank the reviewer for this important point regarding the broad cellular functions of branched actin polymerization, and agree that generalized actin loss can influence both exocytic and endocytic pathways. To address this, we performed additional experiments and

analyses to better define the relationship between branched actin structures and sEV-related processes in VSMCs.

As described in the revised Results (page 6), we overexpressed ARPC2-GFP (an Arp2/3 subunit) together with F-tractin-RFP in VSMCs and carried out live-cell imaging. This approach revealed that Arp2/3 and F-actin organize into lamellipodial scaffolds at the cell cortex, as expected (Fig. S2A; Supplementary Video S2). Additionally, and more unexpectedly, we observed numerous Arp2/3- and F-actin-positive dynamic spots within the VSMC cytoplasm. These structures resemble actin comet tails seen in other systems, previously implicated in endosomal propulsion (Fig. S2A, arrow; Supplementary Video S2).

Quantitative analysis confirmed that a substantial fraction of these dynamic F-actin/cortactin spots colocalized with CD63+ endosomes (Fig. 2D), and that these structures are indeed branched actin tails based on cortactin immunostaining. Furthermore, inhibition of SMPD3 (with 3-OMS) induced enlarged cortactin/F-actin/CD63+ complexes, morphologically similar to invadopodia (Fig. 2D, arrowheads), supporting a functional link between actin branching and MVB dynamics.

To quantify the association, we calculated Manders' colocalization coefficients for F-actin tails and CD63+ endosomal structures in fixed VSMCs, observing that ~50% of F-actin tails were associated with ~13% of endosomes. Upon 3-OMS treatment, this overlap increased further (Fig. S2B).

Finally, using live-cell imaging (Fig 2C; Supplementary Video S4), we directly observed CD63+ MVBs being propelled through the cytoplasm by Arp2/3-driven actin tails, suggesting a mechanistic role for branched actin assembly in MVB intracellular transport, rather than a generalized effect of actin disruption alone.

We believe these combined data reinforce a more specific mechanistic role for Arp2/3-mediated branched actin in MVB/endosome transport and, consequently, in sEV secretion in VSMCs—over and above an indirect effect of global actin loss. We hope these additional experiments and quantitative analyses address the reviewer's concern and clarify the functional relevance of branched actin structures to sEV trafficking:

(1) Results, page 6, Ln 3 “As regulators of branched actin assembly, the Arp2/3 complex and cortactin are thought to contribute to sEV secretion in tumour cells by mediating MVB intracellular transport and plasma membrane docking[28, 33]. Therefore, we overexpressed the Arp2/3 subunit, ARPC2-GFP and the F-actin marker, F-tractin-RFP in VSMCs and performed live-cell imaging. As expected, Arp2/3 and F-actin bundles formed a distinct lamellipodia scaffold in the cellular cortex (Fig S2A and Supplementary Video S2). Unexpectedly, we also observed numerous Arp2/3/F-actin positive spots moving through the VSMC cytoplasm that resembled previously described endosome actin tails observed in *Xenopus* eggs[33] and parasite infected cells where actin comet tails propel parasites via filopodia to neighbouring cells[34, 35] (Fig S2A, arrow, and Supplementary Video S2). Analysis of the intracellular distribution of Arp2/3 and CD63-positive endosomes in VSMCs showed CD63-MVB propulsion by the F-actin tail in live cells (Fig 2C and Supplementary Video S4).”

(2) Results, New data Fig 2D, page 6, Ln 14. “we observed numerous F-actin spots in fixed VSMCs that were positive both for F-actin and cortactin indicating that these are branched-actin tails (Fig 2D). Moreover, cortactin/F-actin spots colocalised with CD63+ endosomes and addition of the SMPD3 inhibitor, 3-OMS, induced the appearance of enlarged doughnut-like cortactin/F-actin/CD63 complexes resembling invadopodia-like structures similar to those observed in tumour cells (Fig 2D, arrowheads)[18].”

(3) Results, New data Fig S2B, page 6, Ln 19 “To quantify CD63 overlap with the actin tail-like structures, we extracted round-shaped actin structures and calculated the thresholded Manders colocalization coefficient (Fig S2B). We observed overlap between F-actin tails and

CD63 as well as close proximity of these markers in fixed VSMCs (Fig S2B). Approximately 50% of the F-actin tails were associated with $\approx 13\%$ of all endosomes ($tM1=0.44\pm0.23$ and $tM2=0.13\pm0.06$, respectively, $N=3$). Addition of 3-OMS enhanced this overlap further ($tM1=0.75\pm0.18$ and $tM2=0.25\pm0.09$) suggesting that Arp2/3-driven branched F-actin tails are involved in CD63+ MVB intracellular transport in VSMCs”

• *In video 1 the author states (lines 8-9; pg6) "intense CD63 staining along filopodia" Although, there is some fluorescence (not strong) in these structures, there was no visible exocytic activity. This data is more suggestive that sEVs (marked by CD63) are not associated with filopodia. The following conclusion statement the authors make is overreaching given this result.*

We thank the reviewer for this careful observation and agree that the previous conclusion regarding sEV release from filopodia was overstated. In response, we have revised both the Results and Discussion sections to more accurately reflect the data..

(1) Results, page 6, Ln37 “We also attempted to visualise sEV release in filopodia using CD63-pHluorin where fluorescence is only observed upon the fusion of MVBs with the plasma membrane³⁹. Using total internal reflection fluorescence microscopy (TIRF) we observed the typical “burst”-like appearance of sEV secretion at the cell-ECM interface in full agreement with an earlier report showing MVB recruitment to invadopodia-like structures in tumor cells¹⁸ (Fig S2B and Supplementary Video S1). Although we also observed an intense CD63-pHluorin staining along filopodia-like structures we were not able to detect typical “burst”-like events to confirm sEV secretion in filopodia. (Fig S2C and Supplemental Video S1).”

(2) Discussion, page 12, Ln19 “Curiously we observed CD63+ MVB transport toward the filopodia tips as well as inhibition of sEV-secretion with filopodia formation inhibitors suggesting that sEV secretion can be directly linked to filopodia but further studies are needed to define the contribution of this pathway to the overall sEV secretion by cells.”

• *Fig 2D and video 2 are wholly unconvincing with regard to sEV secretion sites. The authors could use their CD63-pHluorin construct to count exocytic events in the filopodia vs the whole cell. Given the movie, I have a suspicion this would not be significant. The authors could also perform staining CD63 in non-permeabilized cells to capture and count exocytic events at the plasma membrane as well as their location between groups.*

We thank the reviewer for these constructive suggestions and their critical assessment of our current data regarding the sites of sEV secretion. We agree that our CD63-pHluorin approach clearly indicates sEV secretion events in the soma at the cell-ECM interface, while we did not observe comparable events in filopodia. Accordingly, we have clarified these points in the revised manuscript.

(1) Results, page 6, Ln37 “We also attempted to visualise sEV release in filopodia using CD63-pHluorin where fluorescence is only observed upon the fusion of MVBs with the plasma membrane³⁹. Using total internal reflection fluorescence microscopy (TIRF) we observed the typical “burst”-like appearance of sEV secretion at the cell-ECM interface in full agreement with an earlier report showing MVB recruitment to invadopodia-like structures in tumor cells¹⁸ (Fig S2B and Supplementary Video S1). Although we also observed an intense CD63-pHluorin staining along filopodia-like structures we were not able to detect typical “burst”-like events to confirm sEV secretion in filopodia. (Fig S2C and Supplemental Video S1).”

(2) Discussion, page 12, Ln19 “Curiously we observed CD63+ MVB transport toward the filopodia tips as well as inhibition of sEV-secretion with filopodia formation inhibitors suggesting that sEV secretion can be directly linked to filopodia but further studies are needed to define the contribution of this pathway to the overall sEV secretion by cells.”

• *Fig. 2E and video 4. Again, the conclusions drawn from this data are very strained. First, no co-localization quantification is presented on the proportion of CD63 vesicles with actin. Once again, the movie, if anything convinces the reader that 95-99% of all CD63 vesicles are not associated with actin; therefore, this is an unlikely mechanism of transport.*

We thank the reviewer for this valuable comment and for highlighting the need for quantitative co-localization analysis. In response, we developed a method to systematically quantify F-actin and CD63 co-localization in fixed VSMCs, as now presented in new Figures 2D and S2B. We acknowledge that the majority of CD63+ endosomes are not associated with F-actin, consistent with the reviewer's interpretation. However, our quantitative data now show that a specific subpopulation of MVBs appears to utilize this actin-based mechanism for transport. We believe this addresses the concern and more accurately reflects the prevalence and significance of the mechanism described.

(1) Results, page 6, Ln 19. "To quantify CD63 overlap with the actin tail-like structures, we extracted round-shaped actin structures and calculated the thresholded Manders colocalization coefficient (Fig S2B). We observed overlap between F-actin tails and CD63 as well as close proximity of these markers in fixed VSMCs (Fig S2B). Approximately 50% of the F-actin tails were associated with 13% of all endosomes ($tM1=0.44\pm0.23$ and $tM2=0.13\pm0.06$, respectively, $N=3$). Addition of 3-OMS enhanced this overlap further ($tM1=0.75\pm0.18$ and $tM2=0.25\pm0.09$) suggesting that Arp2/3-driven branched F-actin tails are involved in CD63+ MVB intracellular transport in VSMCs."

• *Are there perturbations that increase filopodia numbers? A gain of function experiment would be valuable here.*

We thank the reviewer for this important suggestion regarding the potential value of gain-of-function experiments to clarify filopodia's contribution to sEV release. In agreement with the reviewer's scepticism, we have removed statements linking filopodia to sEV release from both the title and abstract to avoid overinterpretation. At present, our understanding of filopodia biology and the lack of robust tools to selectively and substantially increase filopodia numbers in VSMCs prevent us from directly addressing this question through gain-of-function assays. We acknowledge that future studies using established methods—such as overexpression of filopodia-inducing proteins (e.g., mDia2 or fascin)—could provide insight into whether an increased number of filopodia affects sEV release. However, such experiments are beyond the scope of the current manuscript. We have made the following changes to clarify these points:

(1) Results, page 6, Ln37 "We also attempted to visualise sEV release in filopodia using CD63-pHluorin where fluorescence is only observed upon the fusion of MVBs with the plasma membrane³⁹. Using total internal reflection fluorescence microscopy (TIRF) we observed the typical "burst"-like appearance of sEV secretion at the cell-ECM interface in full agreement with an earlier report showing MVB recruitment to invadopodia-like structures in tumor cells¹⁸ (Fig S2B and Supplementary Video S1). Although we also observed an intense CD63-pHluorin staining along filopodia-like structures we were not able to detect typical "burst"-like events to confirm sEV secretion in filopodia. (Fig S2C and Supplemental Video S1)."

(2) Discussion, page 12, Ln19 "Curiously we observed CD63+ MVB transport toward the filopodia tips as well as inhibition of sEV-secretion with filopodia formation inhibitors suggesting that sEV secretion can be directly linked to filopodia but further studies are needed to define the contribution of this pathway to the overall sEV secretion by cells."

Figure 3

• Fig 3A. The CD63 staining is strongly associated with the entire plasma membrane. How are the authors distinguishing between normal membrane shedding and bona fide sEVs based on this staining alone (?)- this is insufficient as all membrane structures are seemingly positive. Additionally, there are very few sEVs in scrutinizing the provided images. For the "sEV secretion, fold change" graphs in previous figures, could the authors provide absolute values, or an indication of what these values are in absolute terms?

We thank the reviewer for raising this important point regarding the specificity of CD63 staining and the need to distinguish bona fide sEVs from membrane fragments or general membrane shedding. We agree that CD63 staining alone at the plasma membrane or in the extracellular matrix is not sufficient to unequivocally identify sEVs. To address this, we employed several complementary approaches to rigorously characterize ECM-associated sEVs:

First, using high-resolution iSIM imaging, we confirmed the association of CD63-positive particles specifically with the FN-rich matrix, and demonstrated that SMPD3 knockdown significantly reduced the number of CD63+ particles in the matrix (Fig. 3B; revised from Fig. S3A).

Second, by incubating FN matrices with purified and fluorescently labeled sEVs, we directly observed efficient entrapment of these labeled sEVs within the matrices (Fig. 3E), confirming that sEVs can interact with and be retained by the ECM.

Third, we developed and applied a sequential extraction protocol using mild salt buffer (0.5M NaCl) and strong denaturant (4M guanidine HCl) to selectively extract ECM-associated sEVs based on the strength of their association (see new Figs. S3A and S3B). Extracted vesicles were then characterized by ExoView analysis, which demonstrated a tetraspanin profile (CD63+/CD81+/CD9+) closely matching that of sEVs from conditioned media, providing evidence that these particles are true sEVs and not merely membrane debris. We also found that the more weakly bound (NaCl-extracted) fraction closely resembles media-derived sEVs, while the strongly bound (GuHCl-extracted) fraction is more enriched in CD63+ and CD63+/CD81+ sEVs but contains very few CD9+ vesicles, further supporting distinct extracellular vesicle subpopulations within the ECM.

In addition, the abundance of CD63+/CD81+ sEVs in both media and ECM-derived fractions was independently validated by CD63 bead-capture assay (Fig. S3B).

We hope these clarifications and the expanded data set address the reviewer's concerns about sEV identification and quantification in the extracellular matrix:

(1) Results, page 7, Ln 16. To quantify ECM-trapped sEVs we applied a modified protocol for the sequential extraction of extracellular proteins using salt buffer (0.5M NaCl) to release sEVs which are loosely-attached to ECM via ionic interactions, followed by 4M guanidine HCl buffer (GuHCl) treatment to solubilize strongly-bound sEVs (Fig S3A) 42. We quantified total sEV and characterised the sEV tetraspanin profile in conditioned media, and the 0.5M NaCl and GuHCl fractions using ExoView. The total particle count showed that EVs are both loosely bound and strongly trapped within the ECM. sEV tetraspanin profiling showed differences between these 3 EV populations. While there was close similarity between the conditioned media and the 0.5M NaCl fraction with high abundance of CD63+/CD81+ sEVs as well as CD63+/CD81+/CD9+ in both fractions (Fig S3A). In contrast, the GuHCl fraction was particularly enriched with CD63+ and CD63+/CD81+ sEVs with very low abundance of CD9+ EVs (Fig S3A). The abundance of CD63+/CD81+ sEVs was confirmed independently by a CD63+ bead capture assay in the media and loosely bound fractions (Fig S3B).

• A control of fig 3b would be helpful to parse out random uptake of extracellular debris versus targeted sEV internalization. It would be helpful if the authors added particles of similar size to that of the sEVs to test whether these structures are endocytosed/micropinocytosed at similar levels.

We thank the reviewer for this useful suggestion regarding the need for better controls to distinguish specific sEV uptake from nonspecific internalization of extracellular debris or similarly sized particles. As a comparison, in our study we analyzed the uptake of both sEVs and serum proteins such as fibronectin and fetuin-A (Figs S3C and S3D), and observed similar patterns of intracellular trafficking. However, we acknowledge that inert nanoparticles or beads of a similar size to sEVs could serve as potential controls to assess nonspecific micropinocytosis or endocytosis.

It is important to note, however, that the uptake of sEVs is strongly influenced by their surface protein composition and the so-called “protein corona.” Recent work from Prof. Khuloud T. Al-Jamal’s group underscores that exosome uptake mechanisms may be highly specific (Liam-Or et al., 2024), and studies from Mattias Belting’s lab have also shown the importance of heparan sulfate proteoglycans in exosome endocytosis (Cerezo-Magana et al., 2021). As a result, uptake comparisons with inert particles or beads may not fully recapitulate the specificity of sEV internalization, and distinct nanoparticle classes may rely on different uptake pathways.

Figure 4

• Fig. 4E,F,G. How are the authors determining the neointima and media compartments without ancillary staining for basement membrane or endothelial markers? Anatomic specific markers need to be incorporated here for the reader to evaluate the specificity of the FN and CD81 staining. It is also hard to understand the severity of the atherosclerotic lesion without a companion H&E cross section.

We thank the reviewer for highlighting the need for more rigorous characterization of atherosclerotic lesion architecture and anatomical compartments in our study. In response, we have incorporated additional histological analyses and now provide ancillary staining and companion images to enable clear identification of the neointima and medial compartments, as well as to assess lesion severity (see new Figs S4A–S4D):

(1)Results, page 8, Ln 28. . “To test if FN associates with sEV markers in atherosclerosis, we investigated the spatial association of FN with sEV markers using the sEV-specific marker CD81. Staining of atherosclerotic plaques with haematoxylin and eosin revealed well-defined regions with the neointima as well as tunica media layers formed by phenotypically transitioned or contractile VSMCs, respectively (Fig S4A). Masson’s trichrome staining of atherosclerotic plaques showed abundant haemorrhages in the neointima, and sporadic haemorrhages in the tunica media (Fig S4B). Staining of atherosclerotic plaques with orcein indicated weak connective tissue staining in the atheroma with a confluent extracellular lipid core, and strong specific staining at the tunica media containing elastic fibres which correlated well with the intact elastin fibrils in the tunica media (Figs S4C and S4D). Using this clear morphological demarcation, we found that FN accumulated both in the neointima and the tunica media where it was significantly colocalised with the sEV marker, CD81 (Fig. 4D, 4E and 4F). Notably CD81 and FN colocalization was particularly prominent in cell-free, matrix-rich plaque regions (Figs. 4E and 4F).”

• Figs s4c, S4d- proper controls are not provided. Again, a non-FN internalization control as well as a 4oC cold block negative control is required to interpret this data.

We thank the reviewer for this valuable suggestion. To enhance the rigor of our internalization assays, we have now included several additional controls using alternative treatments, fluorophore combinations, and internalization conditions:

a) We performed FN-Alexa568 uptake assays, followed by immunostaining for CD63 with a distinct fluorophore (Alexa488), to confirm the colocalization of internalized FN with CD63+ endosomal compartments in VSMCs (new Fig. S3E).

b) We also stained VSMCs, cultured under normal growth conditions, with an anti-FN antibody to visualize intracellular serum-derived FN and again observed colocalization with CD63 (new Figs. S3F and S3G). Notably, in cells grown to confluence, we observed a complete loss of intracellular FN staining and FN/CD63 colocalization, suggesting that FN recycling is prominent in sparse, motile cells, but not in confluent populations.

These additional controls strengthen our conclusions regarding FN internalization pathways and the conditions under which FN trafficking to the endosomal system occurs:

(1) Results, page 7, Ln 31 We treated serum-deprived primary human aortic VSMCs with FN-Alexa568 and found that it was endocytosed and subsequently delivered to early and late endosomes together with fetuin A, another abundant serum protein that is a recycled sEV cargo and elevated in plaques (Figs S3C and S3D). CD63 visualisation with a different fluorophore (Alexa488) confirmed FN colocalization with CD63+ MVBs (Fig S3E). Next, we stained non-serum deprived VSMC cultured in normal growth media (RPMI supplemented with 20% FBS) with an anti-FN antibody and observed colocalization of CD63 and serum-derived FN. Co-localisation was reduced likely due to competitive bulk protein uptake by non-deprived cells (Fig S3F). Notably, when we compared FN distribution in sparsely growing VSMCs versus confluent cells we found that FN intracellular spots, as well as colocalization with CD63, completely disappeared in the confluent state (Fig S3F and S3G)..

• *Can the authors please provide live and fixed imaging of FN and CD63-mediate filopodial secretion to amply support their conclusions.*

We have observed CD63 MVBs in both fixed (Fig 2E) and live VSMCs (Fig 2F) yet we agree that further studies are required to establish the contribution of filopodia to sEV secretion. Therefore, we have added the following changes:

(1) Results, page 6, Ln37 “We also attempted to visualise sEV release in filopodia using CD63-pHluorin where fluorescence is only observed upon the fusion of MVBs with the plasma membrane³⁹. Using total internal reflection fluorescence microscopy (TIRF) we observed the typical “burst”-like appearance of sEV secretion at the cell-ECM interface in full agreement with an earlier report showing MVB recruitment to invadopodia-like structures in tumor cells¹⁸ (Fig S2B and Supplementary Video S1). Although we also observed an intense CD63-pHluorin staining along filopodia-like structures we were not able to detect typical “burst”-like events to confirm sEV secretion in filopodia. (Fig S2C and Supplemental Video S1).”

(2) Discussion, page 12, Ln19 “Curiously we observed CD63+ MVB transport toward the filopodia tips as well as inhibition of sEV-secretion with filopodia formation inhibitors suggesting that sEV secretion can be directly linked to filopodia but further studies are needed to define the contribution of this pathway to the overall sEV secretion by cells.”.

Figure 5

• *Fig. 5A,B. The authors claim that sEV supplementation enhances VSMC migration speed and distance. The provided graphs show only a marginal increase in speed with sEV addition (A) but, concerning, there is a four-star significant difference between the FN*

condition compared with FN+sEV (B) while the means appear the same. How are these conditions statistically different? The statistics seem off for these comparisons.

We thank the reviewer for highlighting concerns regarding the statistical analysis in Figures 5A and 5B. In response, we have carefully re-examined our data and statistical approach to ensure accuracy and transparency.

First, we have now included all individual cell migration tracks in the data representation for these figures. The statistical tests were repeated using the Kruskal–Wallis test with Dunn’s multiple comparison correction across all groups. This more stringent analysis confirmed our key findings: fibronectin (FN) stimulates VSMC migration speed, while inhibition of sEV secretion (with 3-OMS) reduces cellular speed (Fig. 5A). Addition of exogenous ECM-associated sEVs modestly restored cell speed in the presence of 3-OMS, but had no effect on baseline migration speed in 2D or 3D models (Figs. 5A, 5D).

Regarding the four-star significance observed in the original Fig. 5B, the previous result reflected an analysis based on pooled group averages, which may have overstated marginal differences. The revised analysis, based on individual cell tracks, does not support a substantial difference between FN and FN+sEV groups. The revised p-values and comparisons are now provided directly on the figures and described in the figure legends. We also clearly report the numbers of biological replicates, technical replicates, and individual data points for every condition.

Further, the modest effect of ECM-associated sEVs on speed is consistent with our observation that sEVs influence invasion directionality rather than baseline migration velocity, in agreement with previous findings in tumor models (Sung et al., 2015).

The manuscript has been revised accordingly, with updates in:

- (1) Figures 5A and 5B: Individual cell track data are now shown, and statistical analyses have been repeated using the Kruskal–Wallis test with Dunn’s multiple comparisons.
- (2) Figure legends and results sections: Numbers of biological and technical replicates, as well as individual data points, are now clearly stated.
- (3) Results, page 9, line 14: “FN as a cargo in sEVs promotes FA formation in tumour cells and increases cell speed^{14, 15}. As we found that FN is loaded into VSMC-derived sEVs we hypothesized that ECM-entrapped sEVs can enhance cell migration by increasing cell adhesion and FA formation in the context of a FN-rich ECM. Therefore, we tested the effect of sEV deposition onto the FN matrix on VSMC migration in 2D and 3D models. We found that FN coating promoted VSMC velocity and inhibition of bulk sEV secretion with 3-OMS reduced VSMC speed in a 2D single-cell migration model (Figs. 5A, 5B) in agreement with previous studies using tumour cells^{14, 15}. However, addition of sEVs to the ECM had no effect on VSMC speed at baseline but rescued cell speed and distance in the presence of the sEV secretion inhibitor, 3-OMS suggesting the EVs are not primarily regulating cell speed (Figs 5A and 5B).”
- (4) Results, page 9, Ln 29 “Hence, ECM-associated sEVs have modest influence on VSMC speed but influence VSMC invasion directionality.”

We hope that these changes address the reviewer’s concerns and improve the transparency and reproducibility of our data presentation

• Fig d-h. Generally, the magnitude of the difference between the presented conditions are biologically insignificant. Several of the graphs show a four-star difference with means that appear equivalent with overlapping error bars. Do the authors conclude that a 0.1%, or less, effect between groups is biologically meaningful?

We thank the reviewer for drawing attention to the apparent mismatch between statistical significance and biological relevance in Figures 5d–h. In response, we have reanalyzed the data using individual cell tracks and more stringent non-parametric statistical tests, as described above. This reanalysis confirmed that the magnitude of differences in migration speed and related parameters between the groups is minimal and not biologically meaningful. Thus, we no longer claim that sEVs significantly affect VSMC migration speed under these conditions in either 2D or 3D assays. Our revised manuscript now accurately reflects this finding in both the Results and Discussion sections, and the updated figures and legends clarify the true extent of any differences observed.

Figure 6

• Generally, the author's logic for looking into adhesion, focal adhesion and traction forces is hard to follow. If there are sEV-mediated migration differences, then there would inexorably be focal adhesion alterations. However, the data indicates few differences brought on by sEVs, which speaks to the lack of migration differences presented in Fig. 5. Overall, the sEV migration phenotype has so little of an effect, to then search for a mechanism seems destined to not turn up anything significant.

We thank the reviewer for highlighting the importance of connecting the observed phenotypic effects of sEVs to the investigation of adhesion and focal adhesion mechanisms. While our revised analysis confirms that sEVs have little to no effect on VSMC migration speed or distance in 2D and 3D models, we did observe a robust effect of sEVs on the directionality of cell invasion (Figs. 5G and 5H). This prompted us to look more closely at pathways involved in cell guidance rather than bulk cell motility.

Our proteomic comparison between larger EVs (10K fraction) and sEVs (100K fraction) revealed a unique adhesion complex present specifically on the sEVs—comprising collagen VI, TGFBI, LGALS3BP, and EDIL3 (Figs. 7A–C)—each of which has previously been implicated in integrin signaling, cell adhesion, or invasion. Functional blocking and knockdown studies further identified collagen VI as a key mediator in the regulation of cell adhesion and invasion directionality influenced by sEVs (Figs. 7F and 7I).

In response to this mechanistic insight, we have modified the graphical abstract and discussion to clarify our approach:

We now explicitly state that our focus has shifted from analyzing baseline migration speed to mechanisms guiding invasion directionality, in line with our key phenotypic findings. We highlight that the unique adhesion cluster identified on sEVs—including collagen VI and its cooperative partners—provides a strong mechanistic rationale for examining focal adhesion dynamics and ECM interactions, even in the absence of changes in migration velocity. Discussion excerpts (pages 13–14) have been updated to reflect this rationale and to summarize the potential significance of these findings for vascular biology and disease.

We hope this clarifies the logic underlying our approach and justifies the mechanistic studies performed in this context:

(1) Discussion, page 13, Ln 2 “Hence, it will be interesting in future studies to investigate whether sEVs can stimulate Rho activity by presenting adhesion modulators—particularly collagen VI—on their surface, thereby guiding cell directionality during invasion.”

(2) Discussion, page 13, Ln 30 “In addition to collagen VI the unique adhesion cluster in VSMC-derived sEVs also includes EGF-like repeat and discoidin I-like domain-containing protein (EDIL3), transforming growth factor-beta-induced protein ig-h3 (TGFBI) and the lectin galactoside-binding soluble 3 binding protein (LGALS3BP) and these proteins are also directly implicated in activation of integrin signalling and cellular invasiveness^{85–87}. Although we

found that collagen VI plays the key role in sEV-induced early formation of FAs in VSMCs, it is tempting to speculate that the high sEV efficacy in stimulating FA formation is driven by cooperative action of this unique adhesion complex on the sEVs surface and targeting this novel sEV-dependent mechanism of VSMC invasion may open-up new therapeutic opportunities to modulate atherosclerotic plaque development or even to prevent undesired VSMC motility in restenosis”..

(3) Discussion, page 14, Ln 14 “In summary, cooperative activation of integrin signalling and F-actin cytoskeleton pathways results in the secretion of sEVs which associate with the ECM and play a signalling role by controlling FA formation and cell-ECM crosstalk. Further studies are needed to test these mechanisms across various cell types and ECM matrices.”.

Figure 7

• *The authors need to provide additional evidence Col IV is harbored in sEVs and not a contaminant of sEV isolation as VSMCs secrete a copious amount of this in culture. For instance, IHC of isolated sEVs stained for CD63 and Col IV as well as single cell staining of the same sort.*

We thank the reviewer for this important comment regarding the specificity of collagen VI detection in sEVs. To ensure that collagen VI is associated with bona fide sEVs—rather than being a contaminant resulting from high extracellular abundance—we performed a comparative analysis of vesicles isolated from the same conditioned media. Both proteomic mass spectrometry and western blotting revealed that collagen VI was exclusively present in the small EV (100K pellet) fraction and not in the larger EVs (10K pellet), as shown in Figs. 7B and 7C. Collagen VI was further identified in sEVs extracted from the ECM using our salt/guanidine protocol (new Fig. 7D).

Reviewer #2 (Recommendations For The Authors):

The authors have presented a nice collection of data with strong approaches to address their hypotheses. Nevertheless, an additional section within the Discussion would be welcome in addressing the potential limitations and important caveats to be considered alongside their study. These caveats and limitations could be reshaped by additional data supporting the ideas that: (1) small extracellular vesicles can be directly observed during their secretion from filopodia, (2) CD81 labeling in tissue can be interpreted clearly as extracellular vesicles and not the cell surface of other cell types (co-staining with an endothelial cell marker such as PECAM-1 perhaps), and (3) collagen VI within the vesicles is somehow accessed by adhesion molecules on the cell surface of migrating cells.

We thank the reviewer for these important suggestions and we have now added further studies and modified our conclusions to reflect the data more accurately:

(1) Results. Page 6, Ln37 “We also attempted to visualise sEV release in filopodia using CD63-pHluorin where fluorescence is only observed upon the fusion of MVBs with the plasma membrane³⁹. Using total internal reflection fluorescence microscopy (TIRF) we observed the typical “burst”-like appearance of sEV secretion at the cell-ECM interface in full agreement with an earlier report showing MVB recruitment to invadopodia-like structures in tumor cells¹⁸ (Fig S2B and Supplementary Video S1). Although we also observed an intense CD63-pHluorin staining along filopodia-like structures we were not able to detect typical “burst”-like events to confirm sEV secretion in filopodia. (Fig S2C and Supplemental Video S1)”..

(2) Discussion, page 12, Ln18: “Here we report that β 1 integrin activation triggers sEV release followed by sEV entrapment by the ECM. Curiously we observed CD63+ MVB transport toward the filopodia tips as well as inhibition of sEV-secretion with filopodia formation

inhibitors suggesting that sEV secretion can be directly linked to filopodia but further studies are needed to define the contribution of this pathway to the overall sEV secretion by cells”.

We quantified the colocalization of CD81 and CD31 to exclude the endothelial cell origin of sEVs and extended the characterisation of the atherosclerotic matrix as well as highlighting any limitations to interpretation ie re CD81 ECM localisation:

(1) Results, page 8, Ln 43 “An enhanced expression of CD81 by endothelial cells in early atheroma has been previously reported so to study the contribution of CD81+ sEVs derived from endothelial cells we investigated the localisation of CD31 and CD81. In agreement with a previous study, we found that the majority of CD31 colocalises with CD81 (Thresholded Mander's split colocalization coefficient 0.54 ± 0.11 , $N=6$) indicating that endothelial cells express CD81 (Fig 4G)45. However, only a minor fraction of total CD81 colocalised with CD31 (Thresholded Mander's split colocalization coefficient 0.24 ± 0.06 , $N=6$) confirming that the majority of CD81 in the neointima is originating from the most abundant VSMCs..

(2) Results, page 8, Ln 28: “To test if FN associates with sEV markers in atherosclerosis, we investigated the spatial association of FN with sEV markers using the sEV-specific marker CD81. Staining of atherosclerotic plaques with haematoxylin and eosin revealed well-defined regions with the neointima as well as tunica media layers formed by phenotypically transitioned or contractile VSMCs, respectively (Fig S4A). Masson's trichrome staining of atherosclerotic plaques showed abundant haemorrhages in the neointima, and sporadic haemorrhages in the tunica media (Fig S4B). Staining of atherosclerotic plaques with orcein indicated weak connective tissue staining in the atheroma with a confluent extracellular lipid core, and strong specific staining at the tunica media containing elastic fibres which correlated well with the intact elastin fibrils in the tunica media (Figs S4C and S4D). Using this clear morphological demarcation, we found that FN accumulated both in the neointima and the tunica media where it was significantly colocalised with the sEV marker, CD81 (Fig. 4D, 4E and 4F). Notably CD81 and FN colocalization was particularly prominent in cell-free, matrix-rich plaque regions (Figs. 4E and 4F). .”

We showed that collagen VI is presented on the surface of sEVs:

(1) Results, page 10, Ln43: “Collagen VI was the most abundant protein in VSMC-derived sEVs (Fig 7B, Table S7) and was previously implicated in the interaction with the proteoglycan NG253 and suppression of cell spreading on FN54. To confirm the presence of collagen VI in ECM-associated sEVs we analysed sEVs extracted from the 3D matrix using 0.5M NaCl treatment and showed that both collagen VI and FN are present (Fig 7D). Next, we analysed the distribution of collagen VI using dot-blot. Alix staining was bright only upon permeabilization of sEV indicating that it is preferentially a luminal protein (Fig 7E). On the contrary, CD63 staining was similar in both conditions showing that it is surface protein (Fig 7E). Interestingly, collagen VI staining revealed that 40% of the protein is located on the outside surface with 60% in the sEV lumen (Fig 7E)

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